

MEETING ABSTRACTS

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Proceedings of the 6th National Big Data Health Science Conference

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Introduction to the Proceedings of the 6th National Big Data Health Science Conference

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We are excited to present this proceeding from the 6th National Big Data Health Science Conference that took place February 13–14, 2025 in Columbia, SC. The conference, which is supported in part by NIH, centers on the critical questions and emerging methods in big data health science which are contributing to the present and future of healthcare.

The purpose of the conference is to foster multidisciplinary collaboration and serve as a venue for the exchange of new concepts, methods, and findings. The conference aims to bring together researchers from academia, government, and industry to share theoretical, methodological, and substantive knowledge about Big Data in biomedical science and public health.

This is the sixth year that the University of South Carolina Big Data Health Science Center (BDHSC) has hosted the National Big Data Health Science Conference. The theme of the 2025 conference was “Unlocking the Power of Big Data in Health: Transforming Data into Actionable Intelligence.”

The BDHSC is an interdisciplinary enterprise that conducts cutting-edge research and discovery, offers professional development and academic training, and provides service to the community and industry. The Center is comprised of five content cores including, Artificial Intelligence for Sensing and Diagnosis, Electronic Health Records, Genomics, Geospatial, and Social Media, and two functional hubs, Business/Entrepreneurship and Technology that collectively promote the utilization of Big Data Analytics in healthcare research, services improvement, and academic training. Since its establishment in 2019

through the USC Excellence Initiative, the BDHSC has made significant progress in enhancing collaborative Big Data health science research across South Carolina and beyond.

Our signature programs include this national conference and a national student case competition. Since the inaugural national conference in 2020, today's conference has grown by leaps and bounds. The 2025 conference received a record number of more than 150 abstract submissions. Over 300 participants joined the conference from 19 states across the United States and 12 countries around the world, including South Africa, Nigeria, and Australia.

Presenters discussed breakthroughs in Big Data science such as the utilization of Big Data and advanced data analytics to improve the precision diagnosis of infections and other clinical conditions (e.g., epilepsy), the use of geospatial data science for identifying and examining opioid overdose patterns in multimorbidity diseases like HIV and substance use disorder, the importance of balance between realism, security, and utility in synthetic data generation and application, and the ethical landscape to ensure fairness, transparency, accountability, privacy/data protection, human oversight on AI applied to healthcare. Besides 6 plenary speakers and 80 poster presentations, the conference organized 7 breakout scientific sessions with more than 40 oral presentations around the themes of “Big Data Analytics and Emerging Methodologies”, “Big Data for Public Health and Community Health (I&II)”, “Big Data for Biomedical Research”, “Big Data for Clinical Practice and Patient Care”, “Big Data Health Science Research: Findings from Africa”, and “Utilization of All of Us Research Platform for HIV Research”. In addition, the conference featured a NIH Trainee Session, a Career Development Luncheon, and a formal Networking Reception. During the NIH trainee session, trainees from four NIH/NIAID funded training programs (three R25 s, one T35) at USC presented their experiences in research training in the areas of Big Data and infectious diseases.

One mission of the conference is to provide a platform for students, researchers and healthcare professionals to interact and engage with data science industries. This year, the conference featured a pre-conference SAS Viya training workshop provided by SAS, Inc and a panel presentation on the AI agents and agentic workflows by two IBM innovators/engineers. The IBM presentation described agents as an “evolution of generative AI” capable of planning, multi-step reasoning, action-taking, and self-correction. Agentic use cases for healthcare



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like mammography screening and medical coding were discussed. Techniques to address challenges of AI, like hallucinations, bias, regulatory compliance, resource consumption, and privacy were provided. Industry advice on the adoption of AI agents, centered on impact on healthcare jobs and education, and the importance of continuous monitoring and evaluation.

The conference was preceded by a case competition featuring 39 teams totaling 115 students from 27 universities across the United States. This annual competition is designed to be a hands-on experience that tests undergraduate and graduate students' analytical, teamwork, communication, and presentation skills to further develop the talent pipeline in Big Data health science. Participating teams are given healthcare problems and datasets to solve a challenging problem using a data analytics approach. A panel of industry and academic experts judge the presentations based on teams' use of the full analytics process, from framing the problem to methodology selection, data use, model building, and innovation. This year, the teams were asked to identify the best performing antiretroviral treatment by patient cluster for people with HIV using synthetic data and advanced data analytics. The winning teams were recognized at the conference with cash prizes (\$5000 first prize; \$3000 second prize, \$2000 third prize, and \$500 for honorable mentions).

We want to thank all the organizing committee members, sponsors, volunteers, and all those who have worked tirelessly to make this event a reality. Your dedication and commitment to advancing the field of Big Data health science are truly commendable.

In closing, let us embrace the challenges and opportunities that lie ahead. May this (and future) conferences be a source of inspiration, knowledge, and collaboration that propels us toward a future where the fusion of Big Data and health science transforms lives and reshapes the healthcare landscape as we know it. We look forward to seeing more participants and more abstracts at the 7th National Big Data Health Science Conference that will take place February 12–13, 2026, in Columbia, SC.

Acknowledgement: Funding for this conference was made possible (in part) by NIH under the award R13LM014347 from the National Library of Medicine. The views expressed in written conference materials or publications and by speakers and moderators do not necessarily reflect the official policies of the Department of Health and Human Services; nor does mention by trade names, commercial practices, or organizations imply endorsement by the U.S. Government.

O1

Big Data Health Science Case Competition: Impact and Outcomes

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Introduction: Case competitions provide a form of experiential learning in higher education, allowing students to apply analytical skills learned in the classroom to real-world challenges. These competitions support the development of problem-solving, critical thinking, teamwork, communication skills, and social capital.^{1–3} Additionally, case competitions assist students in improving public speaking and presentation skills, which can enhance their self-confidence.

Big data case competitions offer students a platform to apply data analytics and data science skills to real-world problems.^{1–2} Students tackle challenging problems, work with real-world data, and present findings to judges. This experience prepares them for data science careers by requiring quick identification of key issues, effective data analysis, and presentation of solutions. Despite the time constraints, these competitions are rewarding, providing valuable skills and experiences.^{4–5} Literature shows increased motivation and pride among

participants, teamwork, and positive feedback from judges.⁴ Additionally, research indicates that case competitions offer a venue for diverse team collaboration.^{3–4}

This abstract examines the use of a healthcare data-specific case competition as an experiential learning tool, highlighting its potential to address real-world problems and presenting the resulting outcomes.

Innovation: As an integral part of the Annual National Big Data Health Science (BDHS) Conference, the Annual Big Data Health Science Case Competition offers dedicated teams of graduate and undergraduate students from the United States a distinctive opportunity to demonstrate their data science expertise by analyzing substantial healthcare datasets. Participants are tasked with addressing an authentic healthcare data challenge within a constrained timeframe, requiring them to analyze the problem, devise a data-driven solution, and present their conclusions to a national panel comprising academic and industry experts. This competitive format emulates the real-world challenges encountered by data scientists in professional environments, thereby providing participants with invaluable practical experience.^{1–2}

Process and Structure: The competition offers data science experience in healthcare, open to eligible undergraduate and graduate students. It has two rounds: preliminary and final. Teams are given the same problem and datasets and have 24 hours to solve it using a complete data science approach. National industry and academic experts evaluate the presentations based on analytics processes like problem framing, method selection, data use, model building, implementation strategy, and innovation. This process helps students grasp the full data science workflow.^{1–2} The competitive yet supportive environment and feedback from judges help identify areas for improvement and foster learning.^{3–6}

Outcomes: Over four years, 163 teams with 485 students from 55 U.S. universities have joined the BDHS case competition.^{7–8} Table 1 shows the number of institutions, teams, and students involved, while Fig. 1 depicts participants' degree levels. This high participation underscores the value of these competitions for data science learning and skill development.

Conclusion: Overall, case competitions are a valuable learning experience for students in higher education. More specifically, big data case competitions are a great way to learn and apply data science skills to real world healthcare problems. BDHSC case competitions also help students develop a strong foundation in data science applied to healthcare, including understanding the different types of data, how to collect and clean data, and how to use statistical and machine learning techniques to analyze data. They are a great way for students to improve their critical thinking, problem-solving, and teamwork skills.

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Table 1 (Abstract O1). Competing Teams in the Annual BDHSC Case Competition Across 2020–2025

Year	Participating Institutions	Teams	Students
2020	8	19	56
2021	14	22	64
2022	10	16	48
2023	15	37	111
2024	17	30	90
2025	27	39	116

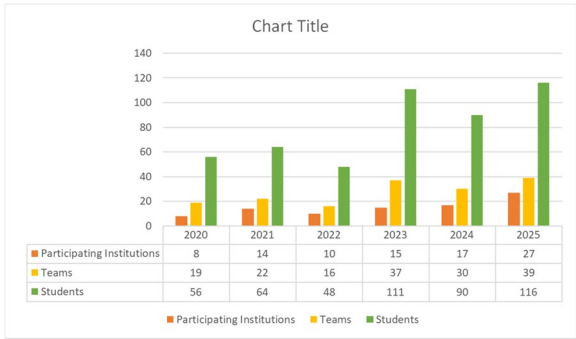


Fig. 1 (Abstract O1). The Annual Case Competition Distribution of Participating Students' Degrees

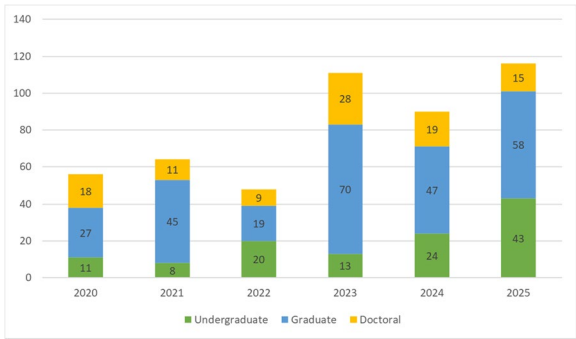


Fig. 2 (Abstract O1). See text for description

O2

Consistency of correct utilization of condoms with lubricants and associated factors among men who have sex with men in Ghana: protocol for evidence from integrated bio-behavioral surveillance survey

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Background: Human Immunodeficiency Virus (HIV) infection remains a public health challenge in sub-Saharan Africa. Ghanaian men who have sex with men (MSM) have a higher risk of contracting HIV compared to the general population. This is due to discrimination, stigmatization, and social isolation. Additionally, the unfriendly customary, cultural, and legal environments in most sub-Saharan African countries prevent them from having access to some of the basic HIV prevention needs and services such as condoms and lubricants. Although using condoms with lubricants correctly and consistently is very effective in reducing HIV transmission and acquisition, little is known regarding the scope and associated factors among the MSM population in Ghana.

Aim: The study aimed to determine the scope of consistent condom and lubricant use and associated factors among the MSM population in Ghana.

Methods: A retrospective analysis of the Ghana Men's Study II dataset was conducted, involving 4095 MSM. The data was imported into STATA (College Station, TX, USA, software version 17) to treat missing data and outliers, and for the analysis. Univariate and multivariable logistic regression analyses were conducted to determine the factors associated with consistent condom and lubricant use. All statistical analyses were performed at a 95% confidence interval, with significant differences at $P < 0.05$.

Results: The study included 4,095 MSM. The majority of the MSM (63.55%) were aged 18–24 years. Most were Christians (69.55%), and about half (52.92%) had at least a secondary level of education. The prevalence of consistent condom use for both men and women is 38.9%. Consistent condom use during penetrative anal sex with other males and females is 44.9% and 40.0% respectively. More than half of the MSM (55.1%) used lubricants consistently during anal sex. In multivariable analysis, being in the age category of 18–24 years (adjusted odds ratio (AOR): 1.19; 95% CI: 1.02–1.40, $p = 0.035$), having a secondary school education (AOR: 1.76; 95% CI: 1.20–2.60, $p = 0.004$), tertiary education (AOR: 2.24; 95% CI: 1.46–3.46, $p < 0.001$), informal employment (AOR: 1.37; 95% CI: 1.06–1.77, $p = 0.015$), sex worker (AOR: 3.42; 95% CI: 1.38–8.46, $p = 0.008$), being a Muslim (AOR: 0.56; 95% CI: 0.45–0.71, $p < 0.001$), having no religion (AOR: 0.66; 95% CI: 0.44–0.98, $p = 0.037$), and having a monthly income of at least Ghs 599/month (AOR: 0.62; 95% CI: 0.49–0.78, $p < 0.001$) were positively associated with consistent condom use. Among the main reasons for not using condoms consistently are trust in a sexual partner, partner's dislike for condoms, and reduction in sexual pleasure.

Conclusion: The study found a low prevalence of consistent condom use for all sexual partners during penetrative anal sex. It is therefore important to design with a predictive model to improve correct and consistent condom use with lubricants among MSM and other key population groups. This study contributes to knowledge and addresses a significant gap in understanding sexual health practices among the MSM population in Ghana to inform relevant implementations for improved sexual health practices in the future.

Keywords

Men who have sex with men; HIV; condoms; lubricants; anal sex; Ghana; sub-Saharan Africa

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O3

Association between physical functionality and cognitive performance in older U.S. adults: a national perspective from the 2011-2014 NHANES

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Introduction: Cognitive impairment, ranging from mild cognitive impairment to dementia, affects nearly 32% of U.S. older adults[1]. This prevalence is expected to surge 300% over the next three decades[2,3], posing significant economic and healthcare challenges. Physical and cognitive health decline often co-occur, suggesting a critical relationship[4,5]. This study hypothesizes physical limitations are associated with poor cognitive performance.

Objectives: To examine the association between physical functionality and cognitive performance among U.S. adults aged 60 and older using data from the 2011-2014 National Health and Nutrition Examination Survey (NHANES)

Method and Data Analysis: This cross-sectional study analyzed 2,149 participants who participated in two waves of the NHANES surveys 2011-2014, aged 60 and older, who had complete data on cognitive function and physical functionality. Physical functionality was assessed by measuring participants' ability to perform 18 physical activities across five domains (Table 1). Scores were categorized as no physical limitations, 1–3 physical limitations, and 4 or more physical limitations. Cognitive performance was evaluated using the Consortium to Establish a Registry for Alzheimer's Disease (CERAD) for Immediate and delayed recall (memory function assessment)[6], the Animal Fluency Test (AFT) for verbal fluency (executive function), and the Digit Symbol Substitution Test (DSST) for processing speed (executive function). Scores were treated as continuous variables. Covariates included age, sex, race, education, income, marital status, BMI, chronic conditions, and lifestyle factors. Multivariable linear regression models were used to examine the associations while adjusting for demographic, lifestyle, and health-related factors.

Results: Approximately 50% of participants reported one or more physical limitations, with 32.56% experiencing 1–3 limitations and 17.67% experiencing >3 limitations. Participants with >3 limitations showed significantly low cognitive performance ratings, particularly in measures of executive functioning, including processing speed (DSST: $\beta = -3.16$; 95% CI: $-5.67, -0.64$) and verbal fluency (AFT: $\beta = -1.00$; 95% CI: $-1.85, -0.16$). No significant associations were observed between physical limitations and immediate or delayed recall scores (CERAD) (Table 2).

Discussion and Conclusion: Results revealed that individuals with more than three physical limitations had poorer cognitive performance, particularly on tasks involving executive function (verbal fluency and processing speed). These findings align with previous longitudinal research indicating that declines in physical health disproportionately impact cognitive domains reliant on executive function and mental flexibility[7,8]. Interestingly, no significant relationship was found between physical limitations and recall memory, as measured by the CERAD Immediate and Delayed Recall tests. This supports the hypothesis that memory functions, particularly recall, may rely on neural pathways less affected by physical decline, such as the hippocampus, compared to the prefrontal cortex, which underpins executive function[8–10]. These findings highlight the importance of conducting further research that investigates the potential causal effect of physical limitations on poor executive cognitive performance. Interventions aimed at improving physical and cognitive health are beneficial to geriatric well-being.

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Table 1 (Abstract O3). Mean score on cognitive performance tests Stratified by Physical Functioning Limitations: NHANES 2011-2014^a

Cognitive performance score ^a	Total	No physical limitation	1–3 physical limitations	More than 3 physical limitations	P value ^b
Immediate recall	20.01 ± 0.24	20.42 ± 0.25	19.67 ± 0.36	19.50 ± 0.23	<.001
Delayed recall	6.37 ± 0.10	6.49 ± 0.10	6.30 ± 0.17	6.15 ± 0.18	<.001
Animal fluency	18.56 ± 0.20	19.40 ± 0.33	18.21 ± 0.26	16.83 ± 0.26	<.001
Digit symbol	54.07 ± 0.57	56.59 ± 0.66	53.32 ± 0.86	48.32 ± 1.01	<.001
Global score ^c	99.02 ± 0.92	102.92 ± 1.08	97.51 ± 1.38	90.82 ± 1.50	<.001

^a Values are mean ± SE

^b P-value from Least squares mean (LS-means) estimated for the categorical variable using weighted survey regression models

^c Sum of immediate recall, delayed recall, animal fluency and digit symbol

Table 2 (Abstract O3). Multivariate Analysis of the Association Between Physical Functioning Limitations and Cognitive Test Scores in NHANES 2011–2014

Physical functioning limitations	Immediate recall β (95% CI)	Delayed recall β (95% CI)	Animal fluency β (95% CI)	Digit symbol β (95% CI)	Global score β (95% CI)
Model 1^a					
No physical limitation	1 (Ref)	1 (Ref)	1 (Ref)	1 (Ref)	1 (Ref)
1–3 physical limitations	−0.39(−1.11, 0.33)	−0.02(−0.43, 0.39)	−0.52(−1.21, 0.16)	−1.24(−2.8, 0.35)	−2.17(−5.08, 0.73)
More than 3 physical limitations	−0.33(−0.91, 0.23)	−0.09(−0.48, 0.30)	−1.39(−2.1, −0.68)	−4.53(−6.93, −2.13)	−6.36(−9.72, −3.00)
Model 2^b					
No physical limitation	1 (Ref)	1 (Ref)	1 (Ref)	1 (Ref)	1 (Ref)
1–3 physical limitations	−0.38(−1.1, 0.33)	−0.03(−0.43, 0.37)	−0.51(−1.22, 0.19)	−1.20(−2.76, 0.36)	−2.13(−5.01, 0.74)
More than 3 physical limitations	−0.37(−0.95, 0.20)	−0.12(−0.51, 0.26)	−1.43(−2.20, −0.67)	−4.53(−6.77, −2.29)	−6.47(−9.71, −3.23)
Model 3^c					
No physical limitation	1 (Ref)	1 (Ref)	1 (Ref)	1 (Ref)	1 (Ref)
1–3 physical limitations	−0.37(−1.09, 0.33)	−0.00(−0.40, 0.39)	−0.32(−1.05, 40)	−0.84(−2.46, 0.78)	1.54(−4.4, 1.35)
More than 3 physical limitations	−0.23(−0.86, 0.40)	0.02(−0.38, 0.43)	−1.00(−1.85, −0.16)	−3.16(−5.67, −0.64)	−4.37(−8.03, −0.71)

^a Model 1: adjusted for age, gender, race, education level, marital status, family income
^b Model 2: adjusted for age, gender, race, education level, marital status, family income, smoking status, alcohol history, BMI
^c Model 2: adjusted for age, gender, race, education level, marital status, family income, smoking status, alcohol history, BMI, health insurance, overall health, chronic diseases

O4
Unhealthy lifestyle indicators increase hypertension in the southeastern United States

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Introduction: The six pillars of lifestyle medicine, stress management, physical activity, nutritious diet, avoidance of risky substances, quality sleep and social connections are health behaviors that have been found to be associated with lower hypertension [1, 2]. Inquiry into the combined impact of unhealthy lifestyle indicators on hypertension at the population level is limited.
Study Objective: To investigate the association between unhealthy lifestyle indicators and prevalence of adult hypertension in counties in the Southeastern United States (US).
Methods: Records from County Health Rankings and Roadmaps (2024) and from Places: Local Data for Better Health (2024) were merged to examine the relationship between unhealthy lifestyle measures and age-adjusted prevalence of adult hypertension within 1063 counties in the Southeastern United States. The percentage of adults in a county reporting mental distress, current smoking, obesity, physical inactivity, insufficient sleep, and living alone within each county represented six lifestyle measures that were combined to construct an unhealthy lifestyle summary index. Correlation (r) values and associated p-values <0.05 determined crude relationships and statistical significance between unhealthy lifestyle indicators, the summary lifestyle index and adult hypertension. Standardized beta coefficients (β) described associations adjusted for confounders, including county-level age, income, education, racial/ethnic composition, and rural status.

Results: Ecological cross-sectional analyses revealed strong correlations between unhealthy lifestyle measures and hypertension. Statistically significant positive correlations with hypertension (p-value <0.0001) were found for mental distress (r = 0.55), smoking (r = 0.64), obesity (r = 0.83), physical inactivity (r = 0.83), inadequate sleep (r = 0.67) and living alone (r = 0.36). A robust relationship was found between the unhealthy lifestyle summary indicator and increased hypertension (r = 0.72; p-value <0.0001). After adjusting for confounders, the positive association between the unhealthy lifestyle index and hypertension remained (β = 0.25; p-value <0.0001).
Discussion: Robust relationships between unhealthy lifestyle indicators and adult hypertension underscore the potential of lifestyle medicine for the prevention and control of hypertension at the community level. State differences in the associations between county-level lifestyle factors and hypertension also support targeted interventions for improving cardiovascular health within populations.

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O5
Robust text classification and LLM integration for personalized mental health support

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BMC Proceedings 2025, 19(13):O5

Introduction: Mental health conditions have become a leading cause of disability globally, yet stigma, financial barriers, and limited access to care hinder effective treatment[1]. Increasingly, many individuals turn to online platforms for support[2]. This study leverages social media to classify mental health conditions and provide personalized resources.
Study Objective: Utilizing the Reddit SuicideWatch and Mental Health Collection (SWMH) dataset [3], the study aims to use machine learning models to achieve multilabel classification of mental health conditions such as anxiety, bipolar disorder, depression, and suicidal ideation.
Methods: Predictive models were trained to classify posts from the SWMH dataset according to their relevance to Depression, Anxiety, Bipolar or SuicideWatch Reddit communities, in a multi-class classification problem. Prior to model training, text samples underwent text cleaning, label encoding, and feature extraction to prepare raw textual data for model input. Relevant inputs were obtained following Term Frequency-Inverse Document Frequency (TF-IDF) vectorization and tokenization. Posts from the r/offmychest community were excluded, ensuring that model training focused on mental health-specific posts and clearly delineated categories. Posts were processed separately into a training dataset and a test dataset, which was held out in training until the final evaluation. Multiple computationally efficient traditional models (Logistic Regression, SVM, XGBoost) and a lightweight transformer-based architecture (DistilBERT) were evaluated using accuracy, precision, recall, F1 score and the area under the receiver operating characteristic curve (AUROC). The two highest probability

classes were integrated into the LLM resource generation, providing contextual grounding.

Results: The DistilBERT classifier achieved the highest performance with an accuracy of 0.7613 (95% CI: 0.7524 - 0.7702) and an AUROC of 0.9161 (95% CI: 0.9116 - 0.9208). Compared to Logistic Regression (AUROC 0.8962) and XGBoost (AUROC 0.8962), DistilBERT showed superior discriminative capabilities. Its Precision of 0.7648 (95% CI: 0.7561 - 0.7737) and Recall of 0.7613 (95% CI: 0.7524 - 0.7702) underscored a balanced performance across multiple metrics (Table 1). Lastly, the DistilBERT classifier demonstrated strong identifying performance for individual mental health conditions, achieving AUROC values of 0.96 for anxiety, 0.95 for bipolar, 0.88 for depression, and 0.92 for suicide watch (Fig. 1).

These consistently high values made DistilBERT a suitable classifier for integrating Llama 3.1 8B Turbo, a Large Language Model (LLM), to generate personalized mental health resources.

Discussion: The pressing need to address the rising prevalence of mental health disorders in the United States and the global health community is accompanied by the issue of how to deliver the best resources to the many patients who are voicing their concerns outside a regulated healthcare visit. This study highlights the efficacy of combining efficient machine learning models and natural language processing to interpret patient concerns in unregulated settings like social media. Integrating LLM-generated resources offers a scalable solution to delivering mental health support to underserved populations.

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Table 1 (Abstract O). Comparative Performance Metrics of Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) Models for Predicting Mental Health Outcomes

Model	Accuracy	AUROC	F1 Score	Precision	Recall
DistilBERT	0.7613 (95% CI: 0.7524—0.7702)	0.9161 (95% CI: 0.9116—0.9208)	0.7621 (95% CI: 0.7533—0.7710)	0.7648 (95% CI: 0.7561—0.7737)	0.7613 (95% CI: 0.7524—0.7702)
Logistic Regression	0.7091 (95% CI: 0.7000—0.7187)	0.9026 (95% CI: 0.8980—0.9072)	0.7100 (95% CI: 0.7009—0.7196)	0.7128 (95% CI: 0.7037—0.7223)	0.7091 (95% CI: 0.7000—0.7187)
Random Forest	0.6836 (95% CI: 0.6744—0.6928)	0.8648 (95% CI: 0.8594—0.8708)	0.6852 (95% CI: 0.6761—0.6945)	0.6889 (95% CI: 0.6798—0.6982)	0.6836 (95% CI: 0.6744—0.6928)
XGBoost	0.7093 (95% CI: 0.7003—0.7182)	0.8962 (95% CI: 0.8913—0.9009)	0.7110 (95% CI: 0.7020—0.7197)	0.7206 (95% CI: 0.7121—0.7294)	0.7093 (95% CI: 0.7003—0.7182)

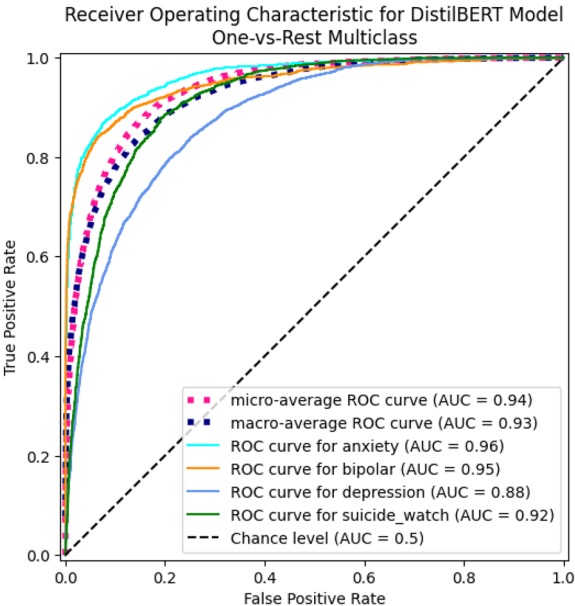


Fig. 1 (Abstract O). AUROC Curve for the Logistic Regression Model in Predicting Mental Health Outcomes

O6

Sexual orientation disparities in cervical cancer screening, HPV testing, and HPV vaccination rates: findings from nine years of CDC BRFSS data

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BMC Proceedings 2025, **19**(13):O5

Introduction: Sexual orientation-related differences in preventive care, especially concerning cervical cancer screening and HPV testing and vaccination, remain largely underexplored. Many sexual minority women face unique barriers to accessing these critical services, contributing to disparities in health outcomes. Identifying and addressing these differences in care is essential to advancing equitable preventative, detective, and interventional public health practices to mitigate disproportionately negative cancer-related outcomes in these groups [1,2].

Study Objectives: This study aimed to evaluate differences in HPV testing, HPV vaccination, and cervical cancer screening rates among heterosexual, lesbian or gay, and bisexual women in the United States. The goal was to identify potential disparities in preventive care practices across sexual orientations to inform targeted public health strategies [3].

Methods: We analyzed pooled data for 3,958,267 respondents from the CDC Behavioral Risk Factor Surveillance System (BRFSS) spanning 2014–2022 [1]. Self-reported outcomes included HPV testing, HPV vaccination, cervical cancer screening via Pap test, and overall cervical cancer screening. Outcomes were binary ("Yes" or "No"). Sexual orientation was categorized as heterosexual, lesbian or gay, and bisexual. Weighted Rao-Scott Chi-Square tests were used to account for the BRFSS complex survey design, and statistical significance was set at $p < 0.05$ [4].

Results: Significant disparities were found across sexual orientation groups. HPV testing was highest among bisexual women (58.8%) compared to heterosexual (47.9%) and lesbian or gay women (39.2%) ($p < 0.01$). HPV vaccination rates were similarly highest in bisexual women (35.2%), followed by heterosexual (23.2%) and lesbian or gay women (17.9%) ($p < 0.01$). Cervical cancer screening rates via Pap test were comparable for heterosexual (82.0%), lesbian or gay (80.3%), and bisexual (80.8%) women. However, overall cervical cancer screening was highest among heterosexual women (88.7%), with lesbian or gay (87.5%) and bisexual women (79.0%) trailing behind ($p < 0.01$) [1,4].

Discussion: This study highlights significant disparities in preventive care practices among women of differing sexual orientations. While bisexual women reported higher HPV testing and vaccination rates, their overall cervical cancer screening rates were lower. Conversely, heterosexual women had the highest overall screening rates, and lesbian or gay women reported the lowest rates of HPV testing and vaccination. These findings underscore the need for tailored public health interventions that address structural, cultural, and informational barriers faced by sexual minority women. Policy initiatives should prioritize equitable access to preventive care services and reduce disparities to improve health outcomes for these populations [2,5].

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07
Utilizing a multi-class classifier in predicting the severity and mortality of in-patients with COVID-19 in south Florida

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BMC Proceedings 2025, **19(13):07**

Background: The ongoing COVID-19 pandemic poses significant health and economic challenges in the United States, with new virulent strains continuously emerging. Predicting critical clinical events such as mortality, mechanical ventilation (MV) requirements, ICU admissions, and IMCU (Intermediate Care Unit) necessities for hospitalized patients with COVID-19 is a promising approach to alleviate these burdens. An AI-driven decision support system that can classify mortality and disease severities enables effective management of disease for patients with COVID-19. Utilizing this powerful classification tool, the current model can forecast treatment plans for critical clinical events by employing the essential features for the severity classifications.

Methods: We conducted a retrospective analysis on 5,371[1–4] COVID-19 patients from South Florida’s Memorial Healthcare System admitted between March 2020 and January 2021 and built a multi-class classifier using a Random Forest (RF) classification model trained on three sets of data: sociodemographic characteristics, comorbidities, and medication history collected during admission. The Venn diagram below (Fig. 1a) shows the distribution of patients and the overlap of mortality and disease severities. The mortality and disease severity were ranked as follows: mortality (3), MV (2), ICU (1), and IMCU (0), and the conditions were categorized as most to least severe, respectively. SMOTE (Synthetic Minority Oversampling Technique) was used to overcome the imbalance dataset and generate qualitative samples

that can aid the model in learning the decision boundary between mortality and different severity classes.

Results: The RF classifier achieved an accuracy of 94% in classifying the four classes. The confusion matrix revealed that the model correctly classified 3110 cases while misclassifying 184 cases. The Area Under the Curve (AUC: weighted average) achieved a high score of 99% for both cases: One-vs-One (OvO) and One-vs-Rest (OvR). The model interpretability (SHAP: SHapley Additive exPlanations) identifies the top features, diarrhea, age, BMI, diabetes, race, sex, early stages of chronic kidney disease, pneumonia, ACEIs/ARBs, ethnicity, and cardiac arrhythmias, in ranking order, that play a vital role in classifying severities (Fig. 1b). Using the waterfall plot, we identified the top thirteen features contributing to 90% of the model’s performance. Among them, five are from the patient’s sociodemographic, seven are from comorbidities, and one is from medication (ACEIs: Angiotensin Converting Enzyme Inhibitors and ARBs: Angiotensin Receptor Blockers) dataset. These results are comparable with other interpretability analyses, such as permutation importance and the Gini index.

Discussion: In contrast to our previous studies [1–2], which evaluated the prediction of individual disease outcomes, the current multi-class classifier was highly effective in predicting mortality and other severity conditions using a single model. The key features were identified and can help clinicians predict the severity of the illness.

Conclusion: The model shows reliability and fairness by efficiently capturing three sets of data. Healthcare professionals can use the valuable information the AI-driven decision support system provides. Rapid identification of high-risk individuals streamlines optimal care solutions when resources are limited and response time is minimal. The study illustrates how AI/ML can help healthcare workers categorize patients and facilitate the best care solutions in a time-sensitive, resource-constrained environment.

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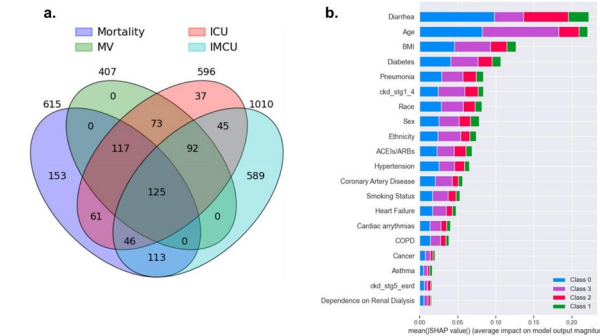


Fig. 1 (Abstract 07). a. (Venn diagram) shows the distribution of patients and the overlap of mortality and disease severities. b. (SHAP

interpretation) shows important features in ranking order (Y-axis) indicated by the SHAP values (x-axis). The classes are given as follows, 0: IMCU, 1: ICU, 2: MV, and 3: Mortality

O8

Designing QSARs of peptides and structural properties through mesoscale simulation model SIMFIM and Machine Learning

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BMC Proceedings 2025, 19(13):O8

Introduction: Bone Morphogenetic Proteins (BMPs) are clinically used in procedures for bone regeneration but they have a short half-life and often diffuse away from the target site, requiring higher doses that can lead to side effects such as bone overgrowth and tumor development (1). This has shifted the focus on peptides as alternate biomolecules to induce osteogenesis inspired by the bioactive domain of the BMP-2 protein known as the 20-mer amino acid sequence of the knuckle epitope (BMP-2 KEP) which has significantly different configurational properties like radius of gyration ($\langle R_g \rangle$) and end-to-end distance (EtE) in the free state as compared to when it is part of the protein (Fig. 1). Machine Learning (ML) can be used to predict possible 20-mer sequences with configurational properties closer to that of the native BMP-2 KEP.

Study Objectives: The objectives were to 1) generate data for machine learning through mesoscale simulation and 2) use the simulation-based database to train and build machine learning models for peptide configuration and shape.

Methods: A list of sequences was obtained by replacing amino acid residues in the BMP-2 KEP. Data on configurational properties of these sequences was generated using a mesoscale simulation model (SIMFIM) as we previously described (2). This data was then used to train and build different ML models to predict the $\langle R_g \rangle$ and $\langle EtE \rangle$ of the sequences (target output of the database). The sequences were represented by different amino acid descriptors such as z-scale and t-scale which were the input features of the database. The database was split into training and testing datasets where 80% of the data was used to train the models and 20% of the data was used to test the performance of the trained models using the R^2 score performance metric.

Results: The performance of the models improved as the size of the dataset increased from 50 to 100 to 200 datapoints. We observed that the neural network (NN) model performed the best when 200 datapoints were used and the normalized version of z-scale was used to describe the residues. The R^2 score of the NN model used to predict $\langle EtE \rangle$ of test peptides was 0.55 (Fig. 2).

Discussion: The $\langle R_g \rangle$ and $\langle EtE \rangle$ of the free BMP-2 KEP sequence were determined from SIMFIM to be 9.47 Å and 21.18 Å respectively. This indicated that the free BMP2-KEP peptide was relatively collapsed compared to $\langle R_g \rangle$ and $\langle EtE \rangle$ of the native BMP-2 KEP as part of the protein ($\langle R_g \rangle = 15.95$ Å and $\langle EtE \rangle = 46.90$ Å). In the database, the highest $\langle R_g \rangle$ and $\langle EtE \rangle$ values of the simulated sequences were observed to be 11.4 Å and 31.8 Å. The performance of these models could be further improved by increasing the number of datapoints using mesoscale simulation as evidenced by the increase in R^2 scores of models with larger training datasets.

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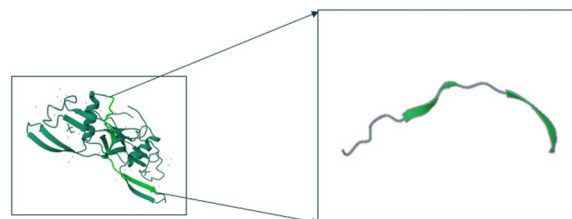


Fig. 1 (Abstract O8). BMP-2 protein (left) and the enlarged shape of the knuckle epitope sequence (right)

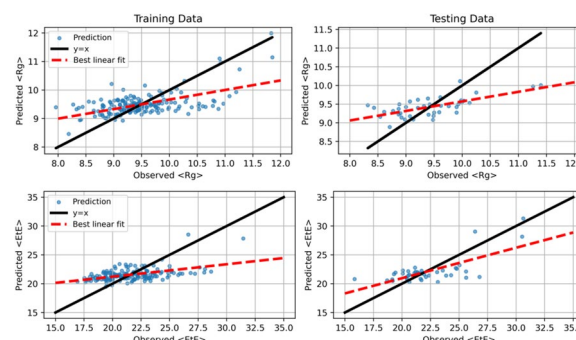


Fig. 2 (Abstract O8). Parity plot of NN models on training (top) and testing (bottom) datasets

O9

The application of machine learning algorithms to predict HIV testing among South African adult population: evidence from the 2017 population-based survey

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Background: Human immunodeficiency virus (HIV) testing is the cornerstone for HIV prevention and a pivotal step in realizing the Joint United Nations Program on HIV/AIDS (UNAIDS) goal of ending AIDS as a global epidemic by 2030 [1–4]. However, a significant portion of the South African adult population is still unaware of their status [5–7]. Hence, there is a need to identify key predictors of HIV testing among adults by using advanced statistical techniques like machine learning (ML) to reduce the spread of the virus.

Objective: This study aimed to determine predictors of HIV testing by applying supervised machine learning (SML) algorithms in the 2017 adult population-based survey in South Africa.

Methods: An SML technique was employed to predict HIV testing from the 2017 South African National HIV Prevalence, Incidence, Behavior, and Communication Survey (SABSSM) data [3]. The dataset was imported to R-Studio version 4.3.2 for cleaning, removing outliers, and analysis. Bivariate analysis was conducted to select important predictors (features) for the HIV testing predictive modeling. A logistic regression model was fitted to identify the predictors of HIV testing among adults in South Africa using the 2017 SASSM

dataset. The dataset was split into a training and testing set, and the outcome variable was based on the question in the SABSSM, which is "Ever had an HIV Test." This variable was a binary factor with levels "No" and "Yes" representing the test statuses. In order to obtain reliable performance estimates, we used a repeated 10-fold cross-validation with 100 repetitions.

Preliminary Results: The model underwent repeated 10-fold cross-validation to prevent overfitting and ensure accurate performance estimates. Metrics such as the area under the receiver operating characteristics curve (AUC-ROC), accuracy, sensitivity, and specificity were utilized to evaluate the model's classification ability. The final model was assessed based on its accuracy in distinguishing between individuals with and without HIV tests, and variable importance was calculated to highlight the most influential predictors. The results offer valuable insights into the factors influencing HIV testing status in the South African adult population. The AUC-ROC is 0.719, indicating that the model is good at distinguishing between the two groups, but there is still potential for improvement by using ML algorithms. Overall, our findings show that the logistic regression model looks promising in HIV testing predictive modeling.

Conclusion: This study contributes to knowledge and deepening understanding of factors linked to HIV testing beyond traditional methods. Our developed ML model shows promising results in HIV testing predictive modeling using nationwide survey data. Consequently, the findings will inform evidence-based policy recommendations to guide policymakers in formulating more effective and targeted public health approaches toward strengthening HIV testing.

Keywords: Machine learning; predictive modeling; HIV; testing; South Africa; support vector machines; random forest; supervised machine learning; decision trees.

Acknowledgments

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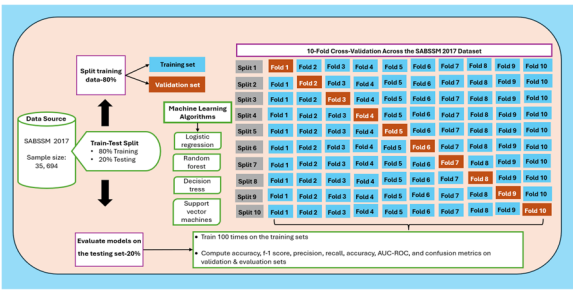


Fig. 1 (Abstract O9). Model training and evaluation with a tenfold validation technique across the 2017 SABSSBM dataset

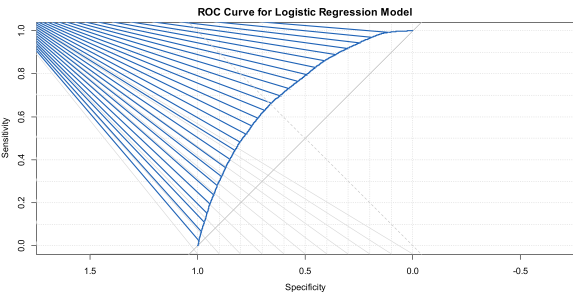


Fig. 2 (Abstract O9). Receiver operating curve for logistic regression model

O10 Exploring medical doctors' perspectives on generative AI tools: a Twitter-based sentiment and thematic analysis

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Introduction: Generative AI (GenAI) tools are transforming various aspects of medical practice, offering innovative applications in diagnostics, education, and workflow optimization, thus enhancing clinical decision-making and patient care (Ruan et al., 2024). Understanding medical doctors' perspectives on these technologies is crucial for guiding their adoption and addressing potential concerns related to trust, ethical considerations, and effectiveness (Javaid et al., 2023).

Study Objectives: This study aims to assess medical doctors' sentiments toward GenAI tools by analyzing their tweets. Using sentiment analysis, we categorize tweets by medical doctors about GenAI into positive, negative, and neutral, providing insights into the acceptance and perception of GenAI tools in medical practice. Also, we categorize the tweets into different themes.

Methodology: We used 34 keywords to retrieve Generative AI-related posts on X via Quid Monitor. The posts were then filtered based on user bios containing detailed 80 specialist titles to identify content authored or reposted by medical doctors about Generative AI. The dataset spans a 27-month period up to November 25, 2024.

The dataset is analyzed using the Pandas library to identify the most repetitive keywords. We also perform sentiment analysis with TextBlob and VADER to compute sentiment scores. Sentiments are classified as positive, negative, or neutral based on predefined thresholds. Finally, thematic analysis is conducted to identify the themes of the tweets.

Results: A dataset of 6,008 tweets from medical doctors' accounts was analyzed. Of these, 2,109 tweets (35.1%) mentioned ChatGPT, making

it the most frequently mentioned generative AI tool, followed by OpenAI (597 mentions, 9.9%) and Gemini (387 mentions, 6.4%). Sentiment analysis revealed 3,639 positive tweets (60.5%), 1,094 negative (18%), and 1,275 neutral (21.5%). Thematic analysis identified nine key themes: Medical Diagnostics and Research, Generative AI Applications in Healthcare, Privacy and Security Concerns, Ethical Considerations, AI's Impact on Society, Education and AI Literacy, Regulation and Policy, Social Media Misinformation, and Generative AI Applications in Healthcare. Fig. 1 summarizes the key themes identified in the dataset, using word clouds to depict the most relevant keywords for each theme.

Discussion: The analysis highlights valuable insights into medical professionals' perceptions of generative AI tools. Sentiment analysis revealed that a majority of tweets (60.5%) expressed positive sentiments, indicating an overall optimistic outlook toward the integration of AI in healthcare. However, the presence of negative (18%) and neutral (21.5%) tweets suggest nuanced perspectives, where concerns or uncertainties persist alongside enthusiasm.

Thematic analysis provided a deeper understanding of these perceptions, categorizing discussions into nine key themes. Topics such as *Medical Diagnostics and Research* and *Generative AI Applications in Healthcare* demonstrate the promising potential seen by professionals in leveraging AI for clinical and research advancements. On the other hand, themes like *Privacy and Security Concerns* and *Ethical Considerations* underscore apprehensions regarding data integrity and ethical dilemmas. Discussions on *Education and AI Literacy* and *Regulation and Policy Implications* reflect a need for structured frameworks to govern AI use responsibly. This thematic diversity emphasizes the importance of balanced AI implementation in healthcare.

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Fig. 1 (Abstract O10). Themes from Thematic Analysis

O11

Late seizure prediction after ischemic strokes: a pilot study using Machine learning and the TriNetX Research Network

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BMC Proceedings 2025, 19(13):O11

Background/Introduction: Late-seizures are frequent after ischemic stroke. Models designed to identify predict post-ischemic late-seizures have used limited variables and often assumed the potential association with post-stroke seizures so there may exist currently unidentified predicting variables. Combining big data from EHRs and using Machine learning could provide more accurate predictive tools.

Study objectives: To develop an artificial intelligence, machine learning prediction model for estimating the risk of seizures 1 year and 5 years after ischemic stroke (IS) using a large dataset from Electronic Health Records.

Methods: We identified patients with IS without a prior diagnosis of seizures from 2015 until inception (08/09/22) in the TriNetX Research Network, using the International Classification of Diseases, Tenth Revision (ICD-10) I63, excluding I63.6 (venous infarction). The outcome of interest was any ICD-10 diagnosis of seizures (G40/G41) at 1 year and 5 years following the index IS. We applied a conventional logistic regression, and a Light Gradient Boosted Machine (LGBM) algorithm to predict the risk of seizures at 1 year and 5 years. The performance of the model was assessed using the area under the receiver operating characteristics (AUROC), the area under the precision-recall curve (AUPRC), the F1 statistic, model accuracy, balanced accuracy, precision, and recall, with and without anti-seizure medication use in the models.

Results: We included 430,254 IS patients ≥ 18 years old without a prior diagnosis of seizures. Seizures were present in 18,502 (4.3%) and (5.3%) patients within 1 and 5 years after IS, respectively. At 1-year, the AUROC, AUPRC, F1 statistic, accuracy, balanced-accuracy, precision, and recall were respectively 0.7854 (standard error: 0.0038), 0.2426 (0.0048), 0.2299 (0.0034), 0.8236 (0.001), 0.7226 (0.0049), 0.1415 (0.0021), and 0.6122, (0.0095). Corresponding metrics at 5 years were 0.7607 (0.0031), 0.247 (0.0064), 0.2441 (0.0032), 0.8125 (0.0013), 0.7001 (0.0045), 0.155 (0.002) and 0.5745 (0.0095).

Discussion: Our model yielded similar performance as models that have used clinical and imaging granular variables such as the NIHSS, large artery occlusion, cortical location of the stroke, and early seizures. Our AUROC at 1 year and 5 years were, 79% and 76%, respectively. We did not have access to granular data but used claims data, suggesting that seizure prediction is possible using data routinely collected in clinical practice. Further, our model was developed using data from a diverse US population, suggesting that it could be generalized to this population.

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O12

A cubic spline estimation approach for time-varying mediation effect models and application to smoking cessation researchJingwei Li¹, Megan E. Piper², Donna Coffman¹¹Department of Psychology, University of South Carolina, Columbia, South Carolina, USA; ²Department of Medicine, University of Wisconsin - Madison, Madison, Wisconsin, USA**Correspondence:** Donna Coffman (dcoffman@mailbox.sc.edu)
BMC Proceedings 2025, 19(13):O12

Background: Developments in mobile and wearable device technology have enabled the collection of intensive longitudinal data such as ecological momentary assessment (EMA). EMA is particularly useful in health behavior change studies, for example, smoking cessation studies, in which data on variables such as craving, withdrawal symptoms, or stress, are measured in real-time contexts. Traditionally, mediation analysis typically examines the relations among an intervention, a time-invariant mediator, and a time-invariant outcome variable. Although there may be a total effect of the intervention on the outcome, there is a need to understand the process by which the intervention affects the outcome [1]. This indirect effect is frequently assumed to be time-invariant.

Methods: With improvements in data collection technology, it is possible to obtain repeated assessments over time resulting in intensive longitudinal data. This calls for an extension of traditional mediation analysis to incorporate time-varying variables and effects [2, 3, 5]. In this paper, we focus on estimation and inference for the time-varying mediation model, which allows mediation effects to vary as a function of time via cubic spline interpolation [4]. Covariates are added to the mediation model to show how to impact the mediation effects.

Results: A smoker's health research is studied to compare the spline estimation with local smoothing. More accurate results are obtained for the cubic spline interpolation when there are fewer time points. By adding covariates, more interesting results are shown in the mediation analysis.

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O13

Geographic variation and Type 2 Diabetes Mellitus (T2DM) in Kentucky adult Medicaid beneficiaries: increased prevalence T2DM and co-morbidities (AMI, retinopathy) but decreased renal failure rates from 2010 to 2018Bert Little, Michael Goldsby, Shaminul Shakib, Riten Mitra
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BMC Proceedings 2025, 19(13):O13

Introduction: Type 2 Diabetes Mellitus (T2DM) is a chronic metabolic disease that is increasing in prevalence worldwide. Kentucky is one of ten states that had increased diabetes prevalence of $\geq 25\%$ per the BRFSS surveys. Arkansas, Kentucky, and Nebraska reported the highest increases in prevalence of T2DM ($p < 0.01$) 2012 to 2022, more than 25% increased prevalence [1].

Study Objectives: The prevalence of T2DM was increased in the Kentucky Medicaid population 2010 to 2018, similar to the general population in the State. Geographic variation in T2DM prevalence estimates and the frequency of co-morbidities for Appalachia vs. non-Appalachia regions differ substantially. In this analysis, change in prevalence of five major co-morbidities (acute myocardial infarction, diabetic nephropathy, diabetic neuropathy, peripheral artery disease, diabetic retinopathy) with T2DM prevalence over time is analyzed.

Methods: Medicaid claims data for Kentucky beneficiaries was obtained from the Cabinet for Health and Family Services (CHFS). Diabetes was identified using the 250.xxx codes for ICD-9 and E11.xxx for ICD-10 coding. The cohort and longitudinal components of the data were extracted. The eight Kentucky Medicaid Regions were used to define geographic boundaries for the analysis. The area deprivation index (ADI) was used to quantify socio-economic factors using a simple bivariate linear regression.

Results: Certain T2DM co-morbidities increased over the study period (MI, PAD, retinopathy), one decreased (nephropathy), and one remained unchanged (neuropathy) (Fig. 1a). Poverty as indexed by the ADI was linearly related to the increased T2DM over time (Fig. 1b). Adult T2DM increased in Kentucky from 10.7% in 2010 to 14.5% in 2018 among Kentucky Medicaid beneficiaries [2], and the disease was highest in Appalachia (Fig. 2a). Myocardial infarction (MI) and peripheral artery disease (PAD) among Medicaid beneficiaries more than doubled between 2010 and 2018 and the highest frequency was in the Appalachia (east) region (Fig. 2b). End stage renal disease (ESRD or CKD5) decreased between 2010 and 2018, but the Appalachia region remained the most affected by CKD5. Diabetic retinopathy nearly doubled between 2010 and 2018, but no clear geographic pattern was apparent.

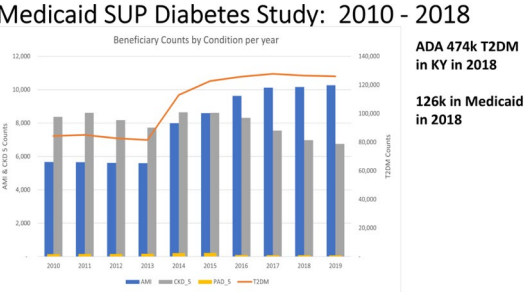
Discussion: These results indicate that poverty and geography are powerful determinants of T2DM and its co-morbidities. The ADI was most strongly related to MI. Low-income levels were associated with T2DM, ESRD, MI, and PAD but not with neuropathy or retinopathy. Lower education was associated with increased T2DM, ESRD, and MI prevalence, but not for neuropathy or retinopathy. PAD was associated with higher education, but may be explained by the association of higher education with sedentism, a known risk factor for PAD. Types of medical care were heterogeneously associated with T2DM and co-morbidities. It was clear that rural healthcare was less effective in preventing T2DM and its co-morbidities than suburban or urban dwelling Medicaid beneficiaries. One clear relationship was the protective effect between nurse practitioners and T2DM, ESRD, MI, and PAD. Obesity was highly associated with T2DM risk, but oddly was protective against ESRD, MI, PAD, neuropathy, and retinopathy. In summary, SDOH factors play a major role in T2DM and its co-morbidities, but the patterns are specific to the disease and geographic location.

Acknowledgements: We would like to acknowledge the Kentucky Cabinet for Health and Family Services for funding the study (State University Partnership) and for the Medicaid Administrative Claims data.

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(a) Diabetes comorbidities across time



(b) Area Deprivation Index and T2DM prevalence among adults 2010 to 2018

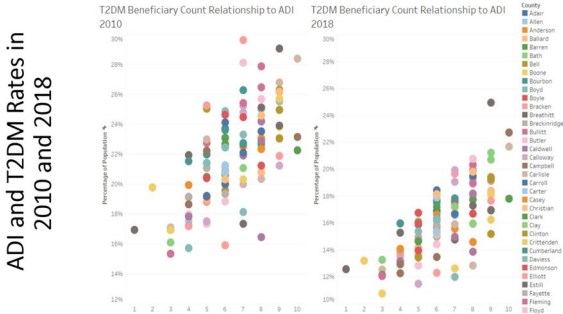
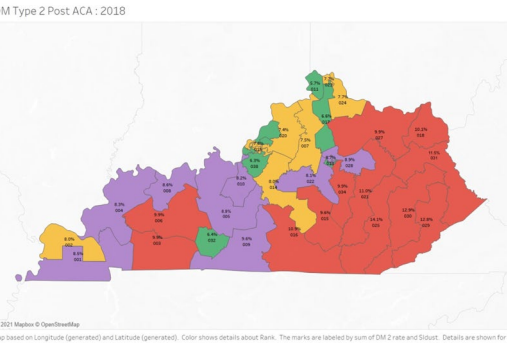


Fig. 1 (Abstract O13). Diabetes Prevalence over time. (a) Diabetes comorbidities across time. (b) Area Deprivation Index and T2DM prevalence among adults 2010 to 2018

(a) Geographic distribution of T2DM in Kentucky in 2018.



(b) Increased frequency of MI in Medicaid in Kentucky in 2010 and 2018



Fig. 2 (Abstract O13). Distribution of Diabetes across Kentucky. (a) Geographic distribution of T2DM in Kentucky in 2018. (b) Increased frequency of MI in Medicaid in Kentucky in 2010 and 2018

O14

Prevalence of Alzheimer's Disease and related dementias among transgender adults in South Carolina

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BMC Proceedings 2025, 19(13):O14

Introduction: Transgender individuals face distinct challenges that may increase their risk of Alzheimer's disease and Alzheimer's disease related dementias (AD/ADRD). The prevalence of AD/ADRD in transgender adults is not well known. Transgender adults experience more health disparities and psychosocial stressors that put them at higher risk for AD/ADRD. Approximately 17% of transgender adults report subjective cognitive decline (SCD), significantly higher than the 10.6% in cisgender adults. Transgender adults with SCD are 2.3 times more likely to report associated limitations and tend to be younger, more likely to be racial and ethnic minorities, uninsured, and have depression. In South Carolina (SC), over 122,699 individuals live with AD/ADRD as of 2023.

Objective: To analyze the prevalence of AD/ADRD and associated risk factors in transgender adults aged 18 and older in SC using the SC Integrated Data System.

Methods: We conducted a secondary analysis of de-identified all-payer hospital claims data from 2000 to 2022. Transgender adults, AD/ADRD diagnoses, and risk factors were identified using ICD-9 and ICD-10 codes. We used the 2010 Rural-Urban Commuting Area Codes to categorize areas as metropolitan or rural. Patient demographics and AD/ADRD risk factors were analyzed using descriptive statistics.

Results: Our analysis identified 1,160 unique individuals, stratified by age: 18–49 years (n=949, 81.8%) and ≥50 years (n=211, 18.2%). The overall AD/ADRD prevalence was 0.05% (n=53), with 0.01% (n=12) in the 18–49 group and 19% (n=41) in the ≥50 group. For ages 18–49, the highest prevalence of risk factors were depression (66%), smoking (54%), and hypertension (31%). For ages ≥50, the highest prevalence risk factors were hypertension (72%), hypercholesterolemia (67%), and diabetes (46%). Most individuals resided in urban areas, but a higher percentage of adults ≥50 (23%) lived in rural areas compared to the 18–49 group (15%).

Discussion: The 19% AD/ADRD prevalence in transgender adults ≥50 years old is notably higher than the general population, suggesting earlier onset or increased risk. This contrasts with a **previous** study showing lower prevalence rates in transgender populations. Our study found higher rates of depression, smoking, and hypertension in the younger cohort, unlike a **previous study** where these were more prevalent in older adults. The rural-urban disparity in the older age group may impact access to specialized healthcare services.

Conclusion: These findings highlight the need for targeted screening and healthcare strategies for transgender adults, particularly those in older age groups and rural areas. The high prevalence of modifiable risk factors in the younger cohort emphasizes the importance of early prevention efforts. Future research should focus on understanding factors contributing to earlier onset or higher prevalence of AD/ADRD in this population and developing tailored prevention and care strategies.

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O15

A mapping toolkit for built environment auditing based on street view imagery

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Introduction: Built environment auditing refers to the systematic documentation and assessment of physical characteristics and objects of the human-made surroundings in a particular area, such as measuring the sidewalk length and counting the traffic lights. One application of these audits is to assess the environmental impact on public health, such as environmental gerontology [1] and walkability impact on physical activities [2]. Publicly accessible street view imagery is a widely used data source [3] for built environment auditing. Object detection from images, a relatively mature technique, is the basis of built environment auditing. Because of the high overlap between adjacent street images, mapping the detected objects is needed for comprehensive counting without repetition and omission. Such mapping tools were underdeveloped, although the needed techniques were well studied [4], [5], [6], [7].

Study Objectives: This research focuses on creating a toolkit to automatically map the detected street objects. The toolkit receives the object pixels' positions and the image information of geographic location and orientation, and then returns the object's geographic location.

Methods: Traditional image-based mapping techniques usually use rigorous photogrammetric 3D reconstruction with accurate camera parameters or control points, which are not satisfied in the context of publicly available street view images. In addition, the built environment audit requires relatively lower object localization precision (e.g., meters). Therefore, the proposed toolkit adopts supplementary geometric information to map objects, including known object dimensions, depth data in the street view image metadata, and generated depthmaps. The main idea is to use various geometric relationships between objects and the street image camera to localize the objects.

Results: We developed a Python mapping toolkit named BEAT for street objects, providing three pipelines for various scenarios: 1) Mapping vertical objects with known dimensions using single-image, such as stop signs and house doors. The mean error is 2 meters for the case study of stop sign localization. 2) Mapping ground objects by converting the street view images to landcover images, e.g., sidewalk and road surface. In the test of road width measurement, the relative error is 10% (0.85 meters). 3) Mapping objects via triangulation based on stereo-images and depthmaps. The relative error is about 8% in the test of street tree diameter measurement. Overall, BEAT pipelines can provide localized objects for reliable quantitative built environment auditing.

Discussion: The researchers conducting built environment audits usually lack expertise in mapping objects from street view images. The practices using location sampling, e.g., every 100 meters along the street, can avoid mapping but bring uncertainties to the research. BEAT can provide mapped objects for comprehensive built environment auditing, filling the gap between the street view images and quantitative measures (e.g., length, area, and count) of street objects. In the future, we plan to simplify the operation and integrate object detection models, workflow control, and graphic user interface into the toolkit so that users can use BEAT to detect and map objects simultaneously, street by street, and conduct fully automatic auditing for large areas.

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O16

HIV Viral load testing and suppression from 2017-2022 using routine programme data in South Africa: a retrospective analysisEdith Phalane¹, Yegnanew Shiferaw^{1,2}, Refilwe Nancy Phaswana-Mafuya¹¹South African Medical Research Council/University of Johannesburg Pan African Centre for Epidemics Research Extramural Unit; ²Department of Statistics, Faculty of Sciences, University of Johannesburg**Correspondence:** Edith Phalane (edithp@uj.ac.za)*BMC Proceedings* 2025, **19**(13):O16

Introduction Viral load suppression is critical for the achievement of United Nations on HIV/AIDS Programmes (UNAIDS) 95-95-95 goals [1]. Sustained virologic suppression reduces the risk of onward HIV transmission [2,3]. Covid-19 related disruptions had the potential of impacting negatively on viral load testing and suppression in South Africa.

Study objective: To assess the variations in viral load testing and suppression rate before and after the Covid-19 pandemic period.

Methods: A retrospective analysis of aggregated routine HIV programme data from the National Department of Health, South Africa was analysed for the period January 2017 to December 2022. The National Department of Health, collects and captures routine programme data on the District Health Information System to monitor progress on HIV targets. A fixed effects model was used to assess the variations in outcome variables viral load testing and suppression (measured as <1000 copies/mL) rate before and after the Covid-19 pandemic period. The Covid-19 and time were the exploratory variables/predictors. The model has the advantage of accounting for unobserved individual differences specifically, the variations between provinces by eliminating time-invariant factors. The dataset used in this analysis is a balanced panel dataset (number of provinces=9, time points=24, total number of data points=3490512, aggregated rate n=216). This balance ensures a consistent structure in the design matrix, contributing to stable standard errors.

Results: The variation coefficient of Covid-19 for viral load testing rate was -3.60, $p < 0.001$, implying that during the Covid-19 pandemic, the average rate of viral load done decreased by -3.60 units per 1,000 individuals after controlling time effects. The rate of viral load testing decreased significantly during the Covid-19 period (Apr 2021 – Dec 2022). The fitted model yields an adjusted R-squared value of 0.21, which indicates that the model accounts for 21% of the variability in the dataset. The F-statistics are very significant ($p < 0.001$), indicating that the model accounts for a substantial portion of the variability in viral load testing. The coefficient for the Covid-19 of variation for viral load suppression rate was 1.88, $p < 0.001$. This implies that during the Covid-19 pandemic, the average viral load suppression rate increased

by 1.88 units per 1,000 individuals after controlling time effects. Viral load suppression rate decreased significantly during before (Mar 2018 – Jun 2019) and during the Covid-19 period (Jul 2020 – Dec 2021). The fitted model yields an adjusted R-squared value of 0.55, which indicates that the model accounts for 55% of the variability in the dataset. The F-statistics are very significant ($p < 0.001$), indicating that the model accounts for a substantial portion of the variability in the suppression rate.

Discussion: Although the number of viral load test done decreased during the Covid-19 period, the virological suppression (< 1000 copies/mL) was achieved Covid-19 mitigation measures such as multi-month drug dispensing strategy and appointment spacing might have positively affected ART adherence leading to viral load suppression [4]. The observed decrease in the number of viral load test done may partly be accounted for by the Covid-19 restrictions resulting in limited access to HIV services facilities.

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O17

Changes in HIV treatment outcomes from 2017–2022 in South Africa – retrospective analysis of aggregated routine programme data

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Introduction: The impacts of covid-19 on HIV service delivery need to be assessed for future pandemic preparedness [1,2]. The achievement of the HIV Initiation and retention in the HIV care continuum is crucial towards achieving the United Nations on HIV/AIDS Programmes (UNAIDS) 95-95-95 goals in the country and globally [3,4]. Interruption in antiretroviral therapy (ART) services not only increase the risk of HIV drug resistance [5], but also predisposes individuals to onward HIV transmission [6].

Study Objective: To quantify the changes in the HIV treatment outcomes over time using routine programme data in South Africa.

Methods: Provincial aggregated level longitudinal routine HIV data were analysed retrospectively. The data was secured from the National Department of Health, South Africa for the period January 2017 to December 2022. This data was collected and captured on the District Health Information System to monitor progress against national and international targets while allocating resources efficiently. A fixed effects model was used to assess the variations in HIV treatment indicators (ART initiation, ART adherence and loss to follow up) before and after the Covid-19 pandemic period. The Covid-19 and time were the exploratory variables. The model accounted for unobserved individual differences, specifically, the variations between provinces by eliminating time-invariant factors. The dataset used in this analysis is a balanced panel (number of provinces=9, time points=24, total number of data points=3490512, aggregated rate, n=216). This balance

ensures a consistent structure in the design matrix, contributing to stable standard errors.

Results: The coefficient for the Covid-19 variation for ART initiation was 14.7, $p < 0.001$. This implies that during the Covid-19 pandemic, the average ART initiation increased by 14.7 units per 1,000 individuals after controlling time effects. The ART initiation was significantly higher before the Covid-19 period (Mar 2018 – Nov 2019) and lower during the Covid-19 period (Jan 2020 – Jan 2022). Regarding ART adherence, the Covid-19 coefficient was 4.23, with an adjusted R-square value of 0.35 and F-statistic of 6.28; $p < 0.001$. This implies that when variable changed from t_0 to t_1 during the Covid-19 pandemic, the average ART adherence rate increased significantly by a rate of 4.23 per 1,000 individuals after controlling time effects. The ART loss to follow-up rate was significantly higher before the Covid-19 pandemic era (Jan 2018 – Dec 2019) and lower during the Covid-19 pandemic (Jan 2021 – Jan 2022). The Covid-19 coefficient of variation for the loss to follow-up rate was 14.7; $p < 0.001$. This implies that with any change in time during the Covid-19 pandemic, the average ART loss to follow-up rate increased significantly by a rate of 14.7.

Discussion: The ART initiation and adherence was higher during the Covid-19 pandemic period. Loss to follow up was significantly higher during the pandemic period. A plausible explanation for higher ART adherence may be the multi-month drug dispensing strategy that was implemented during the Covid-19 period [1,6] to promote continuity in HIV services during the lockdown period. The higher loss to follow up might be in part due to the movement restrictions, reassignment of some facilities to deal with Covid-19-related cases, and the multi-month drug dispensing intervention to ensure that clients on ART had continuous access to ART [5,6]. Timely and sustained pandemic response may have played a significant role in averting adverse Covid-19 effects and points towards the adaptability of the HIV programme in South Africa.

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O18

Characterizing the relationship between head anthropometry and brain material properties using MR imaging and elastography

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Introduction: Traumatic brain injury (TBI) risk depends on subject-specific anatomical variation, yet most computational models use the geometry of a 50th-percentile male head [1,2]. There is a critical need to make more inclusive predictions of TBI that account for geometric variation for individuals with varying brain size and shape [3].

Study Objectives: This study aims to advance subject-specific TBI modeling by investigating correlations between subject anthropometry (age, sex, head dimensions) and brain morphometry from magnetic resonance (MR) images [Aim 1] and between head anthropometry and brain material properties derived from MR elastography [Aim 2] using statistical models.

Methods: Neuroimaging data, including MR structural imaging and elastography, were sourced from a recently acquired open-source dataset [3]. The cohort contained 181 subjects (94 males, 87 females), aged 14–91, with T1- and T2-weighted images and associated data (age, weight, height). A standard protocol was developed to obtain manual measurements of the head size and shape dimensions in ITK-Snap. Brain dimensions were extracted from a binary brain mask using a custom code in MATLAB R2024b, and brain volume was obtained from a deep-learning segmentation [4]. Whole-brain MR elastography data, collected at 30, 50, and 70 Hz frequencies, was analyzed from a subset of 103 subjects (52 males, 51 females). Median and 95th-percentile material properties, including shear stiffness, damping ratio, storage modulus, and loss modulus, were extracted. Mathematical models (simple/stepwise linear regression, statistical shape analysis) were developed to predict brain morphometry and material properties from subject anthropometry.

Results: Simple linear regression indicated that head dimensions are moderate predictors of respective brain length, width, and depth (R^2 : 0.81, 0.65, 0.66, respectively), while stepwise linear regression lowered prediction error, modeling brain dimensions as specific combinations of head length, width or depth ($p < 0.05$). Gender-specific analysis revealed that males presented an 11.34 % larger average brain volume ($1316 \pm 110 \text{ cm}^3$) compared to females ($1182 \pm 114 \text{ cm}^3$), with larger head and brain dimensions on average, confirming existing literature [5]. Age-related trends showed a linear decline in brain volume with age of approximately 2.9 cm^3 per year. Analysis of the correlations between head size and MR elastography dataset will be presented at the BDHS Conference.

Discussion: The correlations between respective head and brain dimensions showed predictive potential, although not all variation was explained. Stepwise linear regression improved prediction accuracy, suggesting brain dimensions are influenced by specific combinations of head length, width and depth. Brain volume prediction improved when age and sex were included in the model. This emphasized the need to scale computational injury prediction models to account for subject-specific variation and demonstrated the power of big data in identifying patterns within large datasets and to make predictions representative of a wider population. Further analysis of brain MR elastography data, presented at the BDHS Conference, will quantify the relationship between head size/shape and mechanical properties to potentially inform future personalized TBI models.

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O19

Leveraging big data frameworks hive and spark for global diabetes prediction on AWS

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Introduction: The International Diabetes Federation (IDF) reports that 10.5% of adults globally have diabetes, with half undiagnosed [1]. Early identification using Machine Learning can help prevent serious conditions like cardiovascular disease and kidney failure [2, 3]. To address global early diabetes identification, this study uses open-source survey data to predict diabetes based on demographic, medical, and geographic factors.

Objectives: To develop an end-to-end pipeline on the Amazon Web Services (AWS) platform [4] that leverages Apache Hive and Apache Spark frameworks. This pipeline aims to reduce costs while efficiently storing and analyzing diabetes survey datasets from various locations, ultimately identifying patients with diabetes.

Methods: Publicly available survey data on diabetes is accessed for three locations India [5], China [6], and the United States [7], and stored in Amazon's scalable S3 storage bucket. We use Elastic MapReduce (EMR) to create a cluster configured to run open-source tools such as Apache Hive and Apache Spark on a general-purpose EC2 instance, M5.xlarge. This instance type offers a cost-effective solution while balancing various workloads. The end-to-end workflow is illustrated in Fig. 1.

Data Preprocessing in Hive: Apache Hive, built on Hadoop, utilizes MapReduce functions to process large datasets efficiently [8]. The preprocessing steps include adding new columns, categorizing the 'age' variable, and converting dichotomous responses into binary format. These transformations are performed using SQL-like queries in Hive to ensure the data is clean, structured, and ready for both the join operation and machine learning models in Spark. The preprocessed datasets are then stored in designated S3 output folders.

Machine Learning Analytics in Spark: AWS SageMaker's JupyterLab environment is used to set up a Spark session with the Application Name "Diabetes Prediction". Apache Spark SQL is utilized to perform a full outer join on the datasets, merging them based on common variables such as Diabetes Status, Age, Gender, BMI, Smoking, Alcohol, Blood Pressure, Country, and Year. The resulting dataset contains 258,931 rows and 9 columns. Apache Spark MLlib [9], Spark's scalable machine learning library, is used to build, train, and test models like Logistic Regression, Decision Tree, and Random Forest to predict diabetes outcomes. The results are then saved back to the S3 output folder. This pipeline seamlessly integrates data storage, processing, and analytics within AWS.

Results: Machine Learning models, including Logistic Regression, Decision Tree, and Random Forest, achieved an accuracy of 85.5%, 85.6%, and 85.6%, respectively, on the test data. Based on the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) values shown in Fig. 2, Logistic Regression and Random Forest (both 0.76) outperformed the Decision Tree model (0.71) in predicting diabetes. These findings highlight the efficiency of the implemented framework in processing large-scale data and achieving reliable predictive performance.

Discussion: Integrating Hive and Spark offers significant advantages for handling large-scale data while reducing costs. Hive's disk-based processing is efficient for dataset preprocessing, whereas Spark's in-memory capabilities overcome Hive's limitations. This combination successfully executes a full outer join on three large datasets and facilitates predictions using machine learning models. Future research could incorporate diabetes survey data from additional locations and explore deep learning techniques to further enhance model performance for global diabetes prediction. Overall, this study demonstrates the potential of integrating Hive and Spark on cloud platforms to streamline diabetes

prediction and analytics in healthcare, leveraging the AWS platform effectively.

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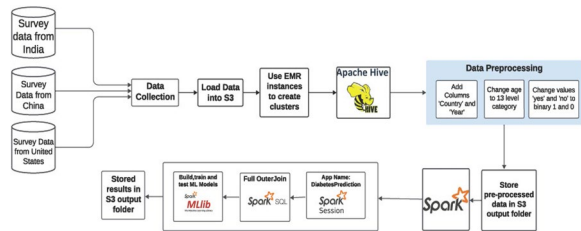


Fig. 1 (Abstract O19). Workflow representing the integration of Hive and Spark

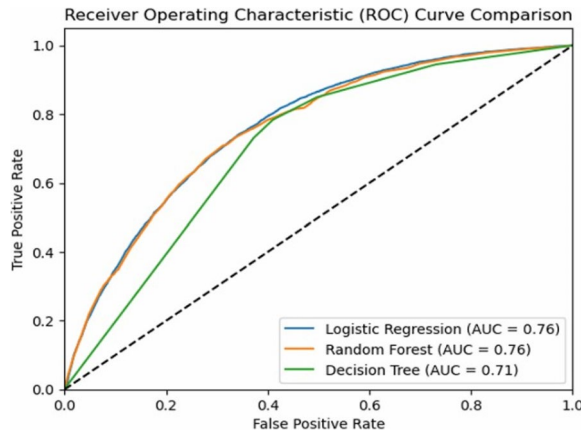


Fig. 2 (Abstract O19). AUC-ROC Curves Comparison

O20

COVID-19-related mortality: association with social determinants of health and comorbidities among hospitalized Medicaid beneficiaries in Kentucky: 2020-2021

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Introduction: The presence of pre-existing conditions, i.e., COPD, diabetes, OB, CKD, CHF, and A-Fib, and SDOH risk factors, i.e., older adults, males, non-Whites, and rural residents, increases the risk of adverse outcomes among hospitalized COVID-19 patients [1–3]. These studies have used data from either surveillance systems with limited clinical information or an individual health care provider. Furthermore, few studies have examined COVID-19-related outcomes such as hospitalization, mechanical ventilation, and in-hospital death among Medicaid beneficiaries [4]. Evidence on COVID-19-related in-hospital mortality for Medicaid beneficiaries’SDOH factors and comorbidities could aid states to reduce disease burden and healthcare expenditures.

Study Objectives: To assess risk factors (i.e., social determinants of health, comorbidities) associated with COVID-19-related mortality among hospitalized Medicaid non-pregnant adult beneficiaries in Kentucky.

Methods: This was an administrative claims-based retrospective cohort study of 28,128 COVID-19-positive Medicaid non-pregnant adult beneficiaries who were hospitalized during 2020–2021 in Kentucky. A Cox regression model was used to estimate the hazard ratios (HR) for COVID-19-related mortality, adjusted for age, sex, race, rural status, and comorbidities. A calculated mechanical ventilation severity score (0 [no mechanical ventilation], 1 [< 24 hrs.], 2 [24 hrs. to 96 hrs.], and 3 [96 hrs. plus]) was also included in the Cox regression model. Receiver Operating Characteristics (ROC) analysis, including area under the curve (AUC), was performed for Cox regression to evaluate model performance.

Results: Of the 28,128 COVID-19-positive Medicaid non-pregnant adult beneficiaries hospitalized, 2,356 (8%) were deceased based on the patient discharge status code. The risk of COVID-19-related mortality was significantly increased by the following predictors: age (ref: 18–44): 45–54 (adjusted hazard ratio (aHR): 1.28, 95% confidence interval (CI) = 1.12–1.46), 55–64 (aHR: 1.61, 95% CI = 1.42–1.83); male (aHR: 1.10, 95% CI = 1.01–1.19); white (ref: non-White) (aHR: 1.18, 95% CI = 1.04–1.33); rural (ref: urban) (aHR: 1.15, 95% CI = 1.02–1.31); COPD (aHR: 1.12, 95% CI = 1.03–1.23); CKD (aHR: 1.22, 95% CI = 1.10–1.36); A-Fib (aHR: 1.21, 95% CI = 1.09–1.36); mechanical ventilation severity score (aHR: 1.64, 95% CI = 1.58–1.70) (Table 1). ROC analysis for COVID-19 mortality with survival probability from the Cox regression model produced a significant AUC value of.834 (standard error:.005, 95% CI =.825–.844) (Fig. 1). The Cox regression model for COVID-19 mortality performed well in classifying the groups, significantly better than by chance.

Discussion: During the COVID-19 pandemic (2020–2021), older adults (aged 45–64), males, and rural residents among COVID-19-positive Medicaid non-pregnant beneficiaries were at significantly higher risk of in-hospital mortality in Kentucky. Pre-existing conditions like COPD, CKD, and A-Fib also increase the risk of in-hospital mortality, which coincides with other studies [5–7]. Evidence from current and other studies indicates increased mortality risk among COVID-19-positive mechanically ventilated patients [8, 9]. Improper use of mechanical ventilation can increase the risk of in-hospital mortality [10]. Addressing patients with these SDOH characteristics and comorbidities in primary care settings can reduce future healthcare costs and improve patient care. Advanced training in proper ventilation monitoring can also reduce in-hospital mortality.

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Table 1 (Abstract O20). Unadjusted and Adjusted Hazard Ratio (HR) of COVID-19 Mortality Among Hospitalized COVID-19 Positive Non-Pregnant Adult Medicaid Beneficiaries in Kentucky: 2020–2021

	No. of COVID-19 Mortality/Total No. (%)	Unadjusted Model			Adjusted Model		
		Unadjusted HR	95% CI		Adjusted HR	95% CI	
			Lower	Upper		Lower	Upper
Age							
18–44	329/7875 (4)	Ref	-	-	-	-	-
45–54	606/7868 (8)	1.46	1.28	1.67	<.001	1.28	1.12
55–64	1421/7285 (11)	1.94	1.72	2.19	<.001	1.61	1.42
Sex							
Female	978/13072 (7)	Ref	-	-	-	-	-
Male	1378/15056 (9)	1.11	1.03	1.21	.001	1.10	1.01
Race							
Non-White	315/4374 (7)	Ref	-	-	-	-	-
White	2041/23754 (9)	1.23	1.10	1.39	<.001	1.18	1.04
Rural Status							
Urban	2080/25353 (8)	Ref	-	-	-	-	-
Rural	276/2775 (10)	1.27	1.12	1.44	<.001	1.15	1.02
*Comorbidities (Ref: No)							
COPD	696/4909 (14)	1.36	1.25	1.49	<.001	1.12	1.03
Diabetes	1082/10285 (11)	1.21	1.11	1.31	<.001	1.06	0.98
OW/OB	1096/10932 (10)	1.15	1.06	1.25	<.001	1.07	0.98
CKD	517/3534 (15)	1.30	1.18	1.44	<.001	1.22	1.10
CHF	493/3130 (16)	1.29	1.17	1.43	<.001	1.07	0.96
A-Fib	410/1996 (21)	1.62	1.45	1.80	<.001	1.21	1.09
*Mechanical Ventilation Severity Score							
0 (No Mechanical Ventilation)	277/22851 (1)	1.67	1.61	1.73	<.001	1.64	1.58
1 (< 24 h)	307/71202 (26)						
2 (24 h to 96 h)	393/1146 (34)						
3 (96 h, plus)	1379/2929 (47)						

	No. of COVID-19 Mortality/Total No. (%)	Unadjusted Model			Adjusted Model		
		Unadjusted HR	95% CI		Adjusted HR	95% CI	
			Lower	Upper		Lower	Upper
0 (No Mechanical Ventilation)	277/22851 (1)	1.67	1.61	1.73	<.001	1.64	1.58
1 (< 24 h)	307/71202 (26)						
2 (24 h to 96 h)	393/1146 (34)						
3 (96 h, plus)	1379/2929 (47)						

¹ An unadjusted Cox regression model was performed between COVID-19 mortality and the respective independent variable

² An adjusted Cox regression model was performed between COVID-19 mortality and the respective independent variable, controlling for the rest of the predictors

³ Acronyms for Comorbidities: Chronic obstructive pulmonary disease (COPD), overweight/obesity (OW/OB), chronic kidney disease (CKD), chronic heart failure (CHF), and atrial fibrillation (A-Fib)

⁴ Mechanical Ventilation Severity Score was treated as a continuous variable in the Cox regression model

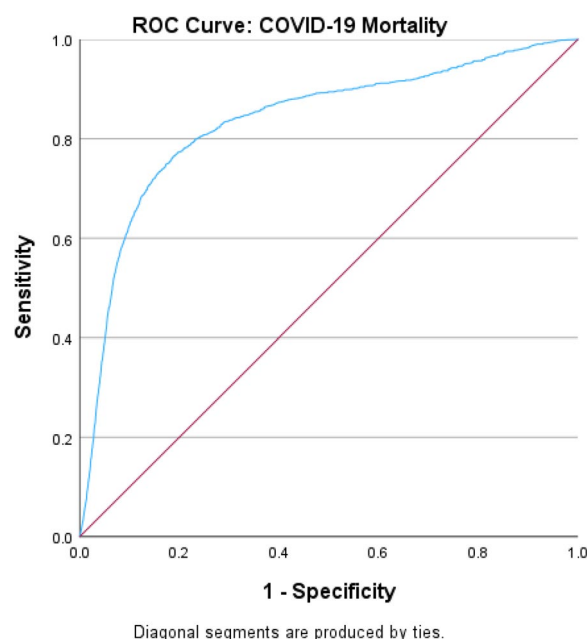


Fig. 1 (Abstract O20). ROC Curve (Cox regression model)

O21

Modeling precise district-level HIV prevalence among individuals aged 15 years and above in South Africa using area-level Poisson mixed model

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Background: South Africa has one of the highest HIV prevalence rates globally, with significant geographic and demographic variations. While household-based HIV surveys provide estimates at national and provincial levels, it is important to have estimates at district municipality levels to understand the HIV burden better nationally.

Objective: The primary aim of this study is to estimate HIV prevalence across the 52 districts of South Africa, disaggregating the data by four age groups and gender. This will be accomplished using information from the 2017 South African National HIV Prevalence, Incidence, Behavior, and Communication Survey (SABSSM) in conjunction with the 2022 South African population census datasets.

Methods: The study used area-level Poisson mixed models to estimate HIV prevalence at the district municipality level, especially in areas with limited or no available data. The HIV prevalence at the district municipality level was estimated using the empirical best predictor (EBP) and plug-in estimator. These estimates were derived using the method of moments, which is a common approach for fitting generalized linear models.

Results: The findings indicate that HIV prevalence varies across district municipalities in South Africa and differs by age group (15–24, 25–34, 35–49, and 50+) as well as by gender (women and men). For instance, in District Municipality DC44 in the Eastern Cape, the HIV prevalence rates are exceptionally high. Among females, the prevalence rate stands at 30.75%, with a 95% confidence interval ranging from 30.68 to 30.83. For the age group of 15 to 24 years, the prevalence rate is 14.32% (95% CI: 14.28 to 14.37). In the 25 to 34 age group, the prevalence rate rises to 27.85% (95% CI: 27.77 to 27.93), and among those aged 35 to 49 years, the rate is 39.48% (95% CI: 39.36 to 39.59). These municipalities are classified as high-priority areas for targeted HIV intervention efforts.

Discussion and policy implications: The area-level Poisson mixed model provides robust estimates for districts with limited data, facilitating a more nuanced analysis of HIV prevalence. By integrating data from the 2022 population census, we enhance the accuracy of our estimates, which is vital for effective resource allocation. This analysis underscores the necessity for targeted interventions in high-prevalence districts and among vulnerable groups, particularly young women and middle-aged adults. Notably, women tend to exhibit higher prevalence rates than men, especially in the 25–34 and 35–49 age brackets, revealing critical opportunities for focused prevention and treatment strategies. These insights can assist policymakers in directing resources more efficiently, ensuring that those most in need receive the support they require. Addressing age and gender disparities in HIV prevalence is essential for fostering health equity and improving overall health outcomes.

022

Severe maternal morbidity and postpartum readmissions through one year

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Introduction: Little is known about the indicators for postpartum readmission (PPR), a preventable adverse outcome. PPR is strongly associated with severe maternal morbidity (SMM); yet its association beyond the early postpartum period and potential racial/ethnic variations remains poorly explored.

Study Objective: This study aimed to identify the primary diagnostic indicators for PPR, investigate the relationship between SMM at birth and the risk of PPR within one-year post-discharge, and assess whether this association differed across racial/ethnic groups.

Methods: A statewide retrospective cohort study using vital records linked to all-payer hospital records data for mothers giving birth from January 2018 through December 2021 in South Carolina. All birthing individuals were followed up using January 2018–December 2022 data. SMM was defined by the Centers for Disease Control and Prevention's 21 indicators measured up to 42 days after birth. PPR was identified using *International Classification of Disease, Tenth Revision, Clinical Modification (ICD–10-CM)* diagnosis and procedure codes for delivery-related diagnosis and procedures. The indicators for PPR were characterized as primary and associated diagnoses. We measured the occurrence PPR at four cumulative timepoints: within 42, 90, 180 and 365 days after the initial birth hospitalization discharge day (hereafter, post-discharge). Separate multivariable logistic regression models were conducted for each time point. We report odds ratios (OR) with 95% confidence intervals (CI) of PPR at each timepoint, sequentially adjusting for prenatal and clinical factors.

Results: Of 166,969 study mothers, SMM prevalence was 1.4%, and the prevalence of PPR increased from 1.5% within 42 days post-discharge to 3.3% within 365 days post-discharge. Among mothers with SMM, 36.9% had a PPR, over 13-fold of those without SMM (2.8%). Compared to all other race/ethnicities, NH Black mothers had the highest rates of PPR overall (4.4%). The most common indicator for PPR was an infection related diagnosis (62%). After adjustment for potential confounders, compared to mothers without SMM, the odds for PPR throughout 1 year post-discharge are as follows: up to 42 days OR: 37.8 (CI: 34.0, 42.0), up to 90 days OR: 30.6 (CI: 27.7, 33.9), up to 180 days OR: 24.0 (21.8, 26.5), and up to 365 days OR: 15.6 (CI: 14.1, 17.1). Using NH White as the reference group, the odds of PPR for NH Black mothers after adjustment 1.1 (1.0, 1.2). In Hispanic mothers, the odds of PPR were 0.8 (0.6, 0.9). In considering the interaction between SMM and race/ethnicity, the OR for NH Black with SMM was 1.2 (0.9, 1.4), and the OR for Hispanic mothers with SMM was 0.8 (0.5, 1.3).

Discussion: These findings emphasize the long-term burden of SMM on maternal health, underscoring the importance of sustained postpartum care and surveillance and the need for extended postpartum care models that go beyond the traditional 6-week visit. The persistent disparities in PPR rates for NH Black mothers call for tailored interventions to address systemic inequities in maternal healthcare. While the SMM and race/ethnicity interaction findings were not statistically significant, they highlight potential nuances in how SMM and race/ethnicity jointly influence PPR risk, warranting further investigation.

023

A digital twin for increasing adherence to dietary guidelines with personalized, AI-driven, long duration meal recommendations

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BMC Proceedings 2025, 19(13):023

Introduction: Longitudinal studies [7] have shown that poor dietary adherence and imbalanced nutrition choices detrimentally impact physical and mental health challenges [3, 9, 10]. Studies show that prioritizing convenience over nutrition [8] can also lead to suboptimal lifestyle choices and health outcomes [4]. To address this, we propose a data-driven meal recommendation system that emphasizes long-term, personalized meal plans to balance convenience and nutrition [1].

Study Objectives: This work aims to develop and evaluate various meal recommendation methods designed to deliver personalized meal plans over extended time horizons.

Methods: For this problem, three meal recommendation methods were considered and evaluated: a random selection method (M0), a sequential cycling method (M1), and a reinforcement learning (RL) approach using boosted bandits (M2). M0 is a baseline approach which creates a randomized meal plan, M1 systematically cycles

through each food item within the dataset and forms a meal, and M2 leverages using a boosted bandit algorithm [2] to drive context-aware recommendations by incorporating user preferences into the meal plans (i.e., yield maximize user satisfaction while also promoting a balanced and nutritious diet). The methods were selected to cover a range from baseline to more sophisticated algorithms. M2 was also adapted from a different work on AI-assisted group recommendation [5, 6], where it was shown to effectively address N user constraints within a different context (teaming).

Results: The three methods were evaluated across different time frames (1, 3, and 5 days) and varying dietary preferences (e.g., allergens, diet types) with 24 simulated users. The experiments measured method performance across three metrics: user constraints (uc), duplicate meals (dm), and meal coverage (mc).

Fig. 1 shows that M2, being the most informed, outperformed the others in user constraints and meal coverage metrics. When users had fewer negative preferences, M0 and M1 performed comparably to M2. M1 scored perfectly on duplicate meal prevention, while M2 struggled due to favoring fewer items. However, M2 demonstrated superior performance in the combined metrics uc_dm_mc.

The initial study shows that boosted bandit algorithms (M2) are promising meal recommendation methods, as they show precision across different user preference configurations. More details can be found in our prototype work [1].

Discussion:

Simulated users do not adequately represent real-world scenarios, so we plan to evaluate our system using real-world user studies. We are currently building a web application to integrate M2 in the UI, allowing real users to test the system's effectiveness. Fig. 2 shows a mockup of the proposed user interface (UI), where a user can construct a profile, receive meal plans according to provided inputs, and view various insights from those plans.

It presents a promising approach to the challenge of balancing convenience and nutrition in meal planning, particularly for public health. The ability to provide personalized, health-conscious meal plans can increase adherence to dietary guidelines and improve long-term health outcomes. Future work includes scaling to larger datasets, accommodating complex dietary needs, and integrating them into health interventions.

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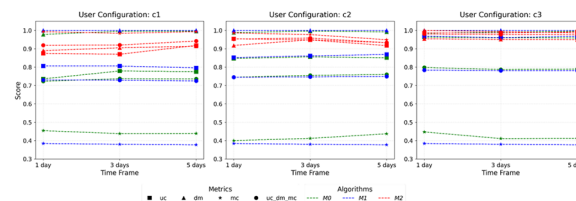


Fig. 1 (Abstract O23). Evaluation of Recommendation Methods

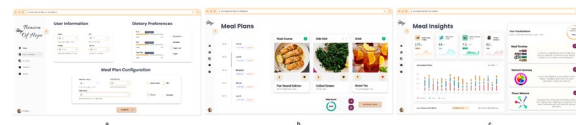


Fig. 2 (Abstract O23). Prototype UI

O24

Using machine learning tools to predict awareness of lung cancer screening among U.S. adults

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Introduction: Lung cancer is the leading cause of U.S. cancer-related deaths. Although lung cancer screening (LCS), which uses low-dose computed tomography (LDCT), can reduce mortality, such screening is severely underutilized. Physician–patient discussions play an important role in increasing LCS awareness and adherence; however, few studies have focused on factors related to LCS awareness.

Study Objectives: This study aimed to (1) identify factors associated with awareness of LCS using LDCT screening and (2) predict awareness of LCS standards using machine learning (ML) approaches.

Methods: Data from 2,620 individuals, whose smoking status (current smoker, former smoker, or non-smoker) was known and were aged 50–80 years old, were selected from the 2022 Health Information National Trends Survey (HINTS 6). LCS awareness was defined by the question “At any time in the past year, did a doctor or other health professional talk with you about having an LDCT scan to check for lung cancer?” Initial analysis included 76 variables. The Boruta algorithm and the least absolute shrinkage and selection operator (LASSO) were used to perform feature selection with awareness of LCS. Seven ML tools—including support vector machines (SVMs), logistic regression (LR), naïve Bayes, random forest (RF), k-nearest neighbor (KNN), gradient boosting machine (GBM), and extreme gradient boosting (XGBoost)—were applied to develop ML models.

Results: 26.3% of current smokers and 10.3% of former smokers reported that they had discussed LDCT screening with a health provider in the past year. The Boruta algorithm and LASSO selected 13 identical variables. Based on a grid search, the radial basis function (RBF) kernel in SVM provided the best model with an accuracy of 0.989, sensitivity of 0.996, specificity of 0.981, and AUC of 0.997. Multiple LR further confirmed 8 of 13 variables as positively associated with awareness.

Discussion: This is the first study using ML tools to predict awareness of LCS among adults in the United States. Our study provides evidence that ML techniques are helpful in predicting awareness of LCS based on survey data. Furthermore, feature selection is essentially the process in ML to pick key informative and relevant features. Lung disease,

smoking status, and beliefs about cancer causation and prevention are linked to the likelihood that a patient has discussed LCS with a health professional. Clinical practice changes are needed to educate physicians about LCS guidelines, engage more patients in screening discussions, and remove barriers to ordering and performing screening tests.

Acknowledgments: The authors would like to thank the NCI for providing the data from the 2022 Health Information National Trends Survey.

O25

Mining social media posts: An AI-powered approach to exploring home health care workers' experiences and concerns

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BMC Proceedings 2025, 19(13):O25

Background: Home health care is a fast-growing sector [1]. Despite its significance, home health care workers (HCWs) experience high turnover rates [2], yet they have historically been underrepresented in research [3, 4]. Traditional studies on HCWs predominantly rely on interviews and surveys, while insights from HCWs' posts on social media have been largely overlooked. This study aims to mine social media posts with objectives: (1) to identify major topics that HCWs are concerned about; (2) to analyze sentiments associated with these topics; and (3) to assess the effectiveness of combining AI-based text analysis with computer-assisted human coding in retrieving insights from social media data.

Materials and methods: We selected the online community "r/HomeHealthAide" on Reddit as the research context and used a web scraping tool Apify to extract all 545 posts on the subreddit. To analyze these posts, we employed Idiomatic, an AI-powered text analysis tool [5], to identify major topics and sentiments. Because AI-powered text mining tools have limitations, we supplemented it with computer-assisted human coding, in which we followed the procedure outlined by [6] and employed NVivo 12 to assist in the coding and analyses.

Results: The results of AI-powered text analysis show that the discussion on workplace safety and support dominates the online community (85% of the messages). The general tone of the posts is negative (24/100). Around half of the posts (52.1%) are negative and few posts are positive (0.4%). Results of computer-assisted human coding demonstrate that in about 59% posts, HCWs are asking questions or seeking advice from others. Most of the questions are about careers (e.g., job certificates) and clients (e.g., how to serve certain types of clients). The second largest category is shared experience, in which negative posts dominate while few posts are positive.

Conclusions: HCWs' experiences are generally negative while positive emotions are rare, which raises alarms for HCW mental health. HCWs are concerned about hazardous work environments, unfriendly clients and lack of support from their agencies.

This study contributes to the literature on HCWs in two important ways. First, this study identified a new data source – HCW's social media posts. Compared with survey data, social media posts are more likely to reveal HCW's true feelings and experiences since they are self-initiated in their natural environment. Second, this study demonstrated that employing AI-powered text mining supplemented by computer-assisted human coding is an effective way to generate insights from social media.

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O26

Telemental health service utilization and mental health emergency room visits among children in foster care in a Southeastern state: evidence from statewide linked administrative data

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BMC Proceedings 2025, 19(13):O26

Introduction: Children in foster care are disproportionately affected by mental health issues and are more likely to seek care in emergency rooms (ERs) for mental health-related concerns. This trend is partially driven by barriers to accessing timely mental health services, including transportation challenges, scheduling difficulties, lack of sufficient mental health providers and multiple parties' coordination. Telemental health (TMH) services can mitigate some of these barriers, providing more accessible and timely care to children in foster care. This, in turn, may reduce the likelihood of ER visits for mental health issues. The present study aims to examine the relationship between TMH service utilization and mental health ER visits among children in foster care in a southeastern state.

Methods: We analyzed statewide integrated administrative data, which included foster care and service records from the Department of Social Services, mental health service data from the Department of Mental Health, and Medicaid claims data from the Department of Health and Human Services. The dataset included a cohort of children who entered foster care for the first time between 2018 and 2019 and remained in care for at least 30 days. The dependent variable was the number of mental health-related ER visits within a 12-month period after the child entered foster care. The key independent variable was TMH utilization (used versus not used). Other control variables included region (rural versus urban), child age (continuous, by year), race/ethnicity (White, Black, Other), sex (female versus male), type of maltreatment (e.g., physical abuse, neglect, abandonment, sexual abuse, medical issues, threat of harm, other), mental health diagnoses (e.g., ADHD, anxiety/depression, adjustment disorder, conduct disorder, PTSD, and other mental health disorders), and placement instability rate. Mental health diagnoses were identified using ICD-10 codes. Placement instability was measured by the total number of placements a child experienced during their time in care, divided by the cumulative number of days spent in foster care over a 12-month period, with higher numbers indicating greater placement instability. We used SAS 9.4 to conduct descriptive statistics, bivariate analyses, and negative binomial regression.

Results: Foster children who used TMH services had higher rates of ER visits (17.88%) than children using only in-person mental health services (9.85%). After adjusting for geographic, sociodemographic, and clinical factors, using TMH service compared to in-person mental health service was not significantly associated with mental health ER visits (IRR [incidence rate ratio] =1.51, p<0.139). However, some other significant predictors were identified, including children's anxiety/depression (IRR=1.95, p <0.038), ADHD (IRR=2.02, p <0.018), other mental health diagnoses (IRR=1.14, p<0.017), child older age

(IRR=1.14, $p=0.001$), and placement instability rate (IRR=1.06, $p < 0.001$).

Conclusions: The findings suggest that mental health conditions, older age, and placement instability increase the likelihood of mental health-related ER visits among children in foster care. However, the use of TMH services was not significantly associated with a reduction in mental health ER visits. Future research should investigate the specific types of TMH services provided to children in foster care, as well as the modalities of service delivery, to better understand the relationship between TMH use and mental health ER visits.

027

Do neighborhood-level stressors and resilience factors impact alcohol use among sexual minority groups: insights from the NIH All of Us Research Project

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BMC Proceedings 2025, **19(13):O27**

Background: Sexual minority populations (SMPs) report a disproportionately high prevalence of alcohol use, often attributed to coping with bi/homonegativity and systemic inequities across various social domains[1–3]. This study aims to explore alcohol use patterns and associated neighborhood-related factors among SMPs using data from the NIH All of Us dataset[3–6]. By identifying mechanisms through which neighborhood factors affect alcohol use, this research seeks to inform the development of culturally tailored interventions to reduce alcohol misuse disparities and promote the well-being of SMPs.

Method: Alcohol use was assessed using the AUDIT-C (Alcohol Use Disorders Identification Test - Consumption) scale among 9,454 gay, 15,284 bisexual, 5,267 lesbian, and 349,748 straight participants. We used Chi-square and ANOVA to explore demographic characteristics and alcohol use patterns among these groups. Logistic regression models were then employed to examine associations between alcohol use and neighborhood- and individual-level factors. Finally, we conducted predictive modeling to identify key factors related to alcohol use among SMPs [7]. We used multiple imputation strategies to handle missing values during the analyses.

Results: Our analyses revealed that gay participants had the highest AUDIT-C scores (mean = 3.60, SD = 2.27), followed by bisexuals (mean = 3.35, standard deviation [SD] = 2.21), other SMPs (mean = 3.18, SD = 2.19), lesbians (mean = 3.04, SD = 2.08), and straight individuals (mean = 3.05, SD = 2.06). Bivariate analyses of alcohol use and neighborhood-level factors revealed statistically significant associations (Fig. 1). The full logistic regression revealed that alcohol use was positively associated with neighborhood disorder (adjusted Odds Ratio [aOR] = 1.08, 95% CI = 1.04, 1.12), residential density (aOR = 1.06, 95% CI = 1.03, 1.09), housing insecurity (aOR = 1.24, 95% CI = 1.14, 1.36) while accounting for demographics. Interestingly, food insecurity (aOR = 0.74, 95% CI = 0.71, 0.78) was negatively associated with alcohol use (Table 1). The prediction models also indicated that neighborhood disorder and housing insecurity play a negative role in alcohol use.

Discussion: Consistent with existing research, our study highlights significant differences in alcohol use across SMPs and emphasizes the influence of neighborhood-level stressors such as disorder and housing insecurity [8, 9]. These findings underscore the need for tailored interventions addressing both the social and environmental determinants of alcohol use in SMPs to mitigate the negative impacts of alcohol consumption on their health and well-being.

Acknowledgements: We acknowledge the support of the University of Rochester's Research Support Grant for this study.

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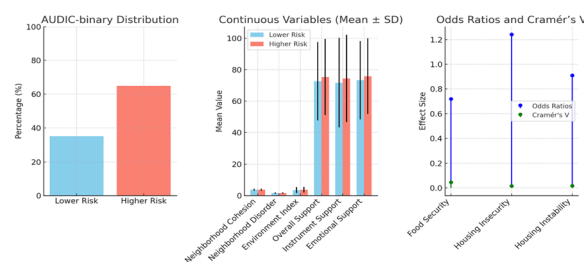


Fig. 1 (Abstract O27). Bivariate Analyses of Alcohol Use and Neighborhood-Level Factors

Table 1 (Abstract O27). Stepwise Logistic Regression Models for Alcohol Use and Neighborhood-Level Characteristics

[illegible]

Notes: * Model 1–9 control all demographic information (e.g., age, income, education, race/ethnicity, sexual orientation)

028

Identifying delayed human response to external risks: an econometric analysis of mobility change during a pandemic

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BMC Proceedings 2025, **19(13):O28**

Background: A major methodological challenge in estimating human behavior in response to changes in environmental risks is that societal response is not immediate; it often involves a delay with an unknown duration and shape. Econometric methods, such as Autoregressive Distributed Lag (ARDL), offer flexible approaches for estimating such delayed effects empirically, but results often lack theoretical clarity and do not fit into simulation modeling frameworks used in policy analysis and long-term forecasting.

Study Objectives: This study aims to develop a method that bridges empirical econometric rigor and theory-based simulation modeling. Specifically, it focuses on quantifying delayed human behavioral responses to external risks, using COVID-19 mobility data as a case study. The method is designed to offer more interpretable and applicable results for policy-making and long-term forecasting while addressing methodological challenges related to delayed societal responses.

Method: In this paper we propose to bridge the two, leveraging the empirical rigor of ARDL, and translating those estimates to theoretically transparent Erlang lag distributions with a length and shape parameter. We demonstrate the method using an extensive dataset from the recent COVID-19 pandemic, and estimating how weekly deaths, with a delay, impacted county-level mobility in the USA.

Results: Results offer stronger empirical linkages between cause and effect than some intuitive alternatives. We find 1st to 3rd order Erlang delays with lags of 2 to 18 weeks provide a theoretically cogent summary of how mobility responses succeeded deaths during the first year and half of the pandemic.

Discussion: This study successfully introduces a hybrid ARDL-Erlang model to estimate delays in human behavior in response to health risks and can be used as a guideline and an example of identifying delayed human behavior in other settings. Our findings highlight the significant variability in both the total impact of COVID-19 deaths on mobility and the timing of responses across states. These results suggest the importance of tailored public health interventions depending on state-specific response patterns. By offering a more interpretable method, this approach enhances the practicality of estimating delayed effects, contributing to both econometric and epidemiological modeling. Future research can expand on this method to explore other societal responses to disruptions.

Keywords: Human behavior, causality, delay, COVID-19

O29

Effective differentiated service delivery: pioneering community-led strategies for optimal continuity-in-treatment and viral suppression in North-Eastern Nigeria

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BMC Proceedings 2025, 19(13):O29

Background: Gombe State, Nigeria, faces significant challenges in sustaining people living with HIV (PLHIV) on treatment and achieving viral suppression. By 2018, the state had the second-highest HIV prevalence (1.2%) in the North-East geopolitical zone, with treatment saturation at 71% (13,719/19,328) and viral load (VL) coverage at 38% (7,345/19,328). Recognizing these gaps, we initiated tailored differentiated service delivery (DSD) models in Gombe to enhance client-centered care, optimize retention, and ensure long-term sustainability. This study evaluates the outcomes of two community-led DSD models implemented between 2018 and 2023.

Methodology: A retrospective cohort analysis was conducted on 5,587 PLHIV who transitioned to two DSD models from July to September 2018 across 24 health facilities in Gombe State. The models included: Healthcare Worker-Led Community ART Groups (HCW-CAGs) – where Healthcare workers delivered antiretroviral therapy (ART) to decentralized community hubs; and PLHIV-Led Community ART Groups (PLHIV-CAGs) – where stable PLHIV managed ART distribution within peer-supported groups. Eligibility required at least one year on ART, good adherence, and signed consent for their chosen model. Baseline and follow-up data were abstracted from DSD registers, and descriptive statistics were used to analyze retention, viral load coverage and viral suppression trends over four years.

Results: Of 5,587 PLHIV 29% (n=5587/19328) of active clients devolved, 4,494 enrolled in HCW-CAG, and 1,093 in PLHIV-CAG. For the HCW-CAG, retention was consistently 100% throughout the analysis period. Viral load coverage increased from 71% (3,190/4,494) in 2018 to 98% (4399/4489) in 2023, while suppression improved from 86% (n=3864/4494) in 2018 to 99% (n=4355/4399) in 2023 (Fig. 1). Two deaths and four relocations were recorded during the period.

For the PLHIV-CAG, retention remained at 100% (1,093/1,093) discounting three deaths and nine relocations. Viral load coverage rose from 62% (671/1,093) to 99% (1,060/1,071), while suppression improved from 93% (625/671) to 98% (987/1,007) (Fig. 2).

Conclusion: Both community-led DSD models showed remarkable improvements in retention, viral load coverage and viral suppression, highlighting the potential benefits of aligning services with patient preferences. These tailored approaches leveraged existing community structures and enhanced patient engagement which may be useful as scalable and sustainable interventions in resource-limited settings toward achieving HIV epidemic control.

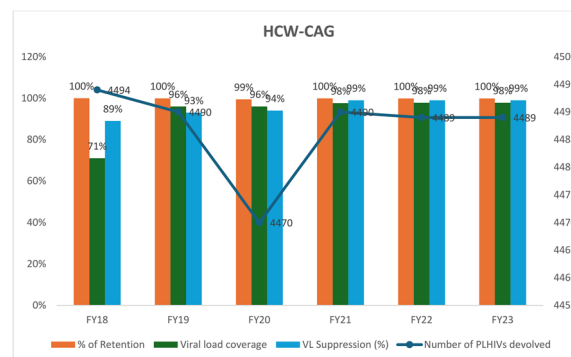


Fig. 1 (Abstract O29). Healthcare Worker Led Community Art Group

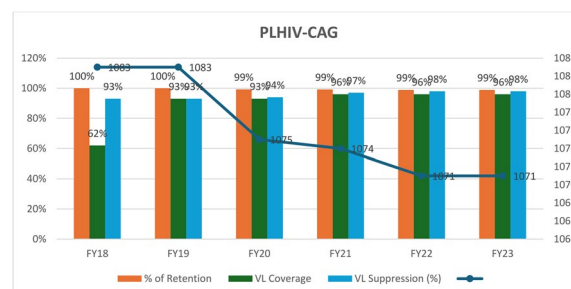


Fig. 2 (Abstract O29). PLHIV Led Community Art Group

Fig. 1 (Abstract P1). Health facility distribution in Iraq, 2014–2020

P2

Decoding colon cancer: unveiling the genetic impact for tailored treatments

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BMC Proceedings 2025, 19(13):P2

Introduction: The diversity of colon cancer treatments allows the potential personalization of regimens for individual patients, such as existing targeted therapies for mutations in *BRAF*, *KRAS*, and *PIK3 CA*. While studies have typically analyzed patient responses to colon cancer therapies based on genetic mutations only in a handful of genes, our research takes a comprehensive approach by examining the interplay between a substantially larger set of mutated genes and treatment options.

Study Objectives: This project aims to explore the relationships between multiple colon cancer mutations, treatment plans, and responses to determine how genetic mutations influence the outcomes to certain regimens.

Methods: Clinical data for 181 colon cancer patients was obtained from the Prisma Health Cancer Institute's Biorepository derived from EPIC[®]. The mutation status of 50 cancer-critical genes was determined with through multiplex PCR using Ion Ampliseq Cancer Hotspot panel v2 by Precision Genetics and USC Functional Genomics Core. Multiple rounds of treatment were further subdivided, with a single data observation corresponding to the applied treatment regimen and response for each round. Data analysis was performed using Microsoft Excel to assess the frequencies of mutations and patient demographics. Additional analysis is being performed using R software and machine learning techniques to identify relationships between different treatments, mutation status, and patient outcomes.

Results: The most frequent mutations in the colon cancer patients were identified in the genes *TP53* (54.1%), *KRAS* (37.6%), *APC* (33.7%), *KDR* (29.8%), and *PIK3 CA* (25.4%). A greater proportion of mutations in *BRAF*, *KDR*, and *KRAS* was observed in female patients compared to male patients. The majority of colon cancer cases were classified as adenocarcinoma (94.0%), with the remaining being mucinous adenocarcinoma (3.8%), mucinous (1.1%), sarcoma (<1%), or spindle cell (<1%). The most prominent stages of colon cancer were IIA (20.8%), IIIB (18.0%), and I (12.6%). FOLFOX (fluorouracil, leucovorin, oxaplatin) alone was utilized in 23.9% of the colon cancer therapies, capecitabine in 14.9%, FOLFOX + bevacizumab in 11.7%, FOLFIRI (fluorouracil, irinotecan, leucovorin) in 7.2%, and the remaining treatment regimens being comprised of various combinations of these therapies alongside other chemotherapeutic drugs.

Discussion: Our preliminary analysis shows the variety of genes mutated in the colon cancer dataset, highlighting the predominance of *TP53*, *KRAS*, and *APC*. This supports the frequencies of mutated genes observed in previous studies, reinforcing the dataset's quality and potential for ongoing analysis. The higher proportion of *BRAF* mutations among the female population is also consistent with patterns reported in prior studies. As we continue our research, we aim to identify correlations between mutated genes, treatments, and patient outcomes in an effort to pave the way for improving therapeutic strategies.

P3

Utilization of the key population unique identifier codes in monitoring HIV treatment cascade in Sub-Saharan Africa

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BMC Proceedings 2025, 19(13):P3

Introduction: There is a gap in tracking key populations (KPs) data on the HIV treatment cascade in Sub-Saharan Africa (SSA) [1,2]. Published and grey literature on HIV treatment cascade among key populations using unique identifier codes remains sparse, highlighting the need for a routine data management system. This system should track KPs across the HIV care continuum for a targeted response. Key populations include female sex workers (FSW), men who have sex with men (MSM), transgender (TG) populations, people who use drugs (PWUD), and people in prisons and other closed settings.

Study Objective: To assess the utility of the unique identification code (UIC) in measuring HIV treatment outcomes among KPs within routine data management systems in SSA.

Methods: A systematic review on the utility of the unique identification code (UIC) in measuring HIV treatment outcomes such as HIV testing, linkage to care, ART adherence and viral load suppression among KPs within routine data systems in SSA was conducted. The review was done following the Preferred Reporting Items for Systematic Reviews and Meta-analyses statement (PRISMA) and registered with the International Prospective Register of Systematic Reviews (PROSPERO) registration number: registration number: CRD42023437904. A comprehensive literature search was conducted using a range of databases, websites, reports and references of screened eligible sources.

Results: Of the 297 articles, only 12 published and unpublished documents were eligible for inclusion (Fig. 1). The utilization of UICs for disaggregation of HIV indicators for key populations was reported only for eight SSA countries, namely Zambia, Sierra Leone, Burkina Faso, Togo, Mali, Malawi, Ghana, and Kenya. The UICs were implemented between the years 2009 and 2017. Only, two types of UICs were identified, namely alphanumeric code (n=5) and biometric fingerprint (n=3). Out of the seven countries, only 2 studies (Ghana and Mali) reported the utility of the UIC in tracking the HIV treatment outcomes for key populations such as individuals reached with HIV services, HIV testing, HIV positivity rates, and linkage to care. Out of 12 publications, 2 countries (Malawi, Ghana implemented UIC or biometrics for general populations however information on treatment outcomes was not reported. Whereas Sierra Leone has also implemented UIC for both general and key populations, data on HIV treatment outcomes for these target populations were not available. Eight countries utilized UIC for KPs. Two countries (Ghana and Mali) were able to disaggregate by HIV treatment outcomes while Ghana was able to disaggregate by type of Key Population.

Discussion: The findings from the systematic review support the development of unique identifiers to measure the HIV treatment cascade. Currently, the utilization of UIC in Ghana and Mali allows for accurate monitoring of indicators along the treatment cascade. The lack of widespread use of unique identifiers in SSA by countries presents a challenge to the evolution of existing systems, whether electronic or paper based [3]. It is important for the UICs developed for general population to include KPs in tracking HIV treatment outcomes. Unique identifiers play a pivotal role in linking records across systems. Adopting an individual-level person-centered approach to resolve data duplication is critical for tracking progress towards the UNAIDS 95-95-95 targets [4]. The utilization of UIC for KP HIV treatment cascade outcomes must be included in national and global reporting metric and strategic reports to improve treatment cascade.

Acknowledgements

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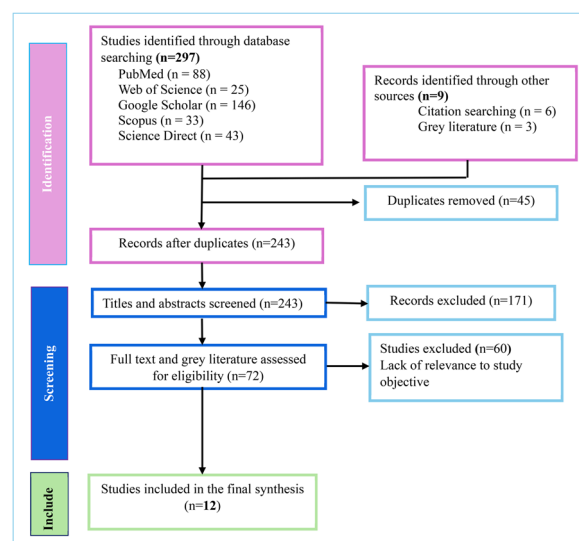


Fig. 1 (Abstract P2). PRISMA flow diagram for the review process

P4

HIV testing and associated factors among men who have sex with men in Ghana: a protocol paper

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BMC Proceedings 2025, 19(13):P4

Background: Human Immunodeficiency Virus/Acquired Immunodeficiency Syndrome (HIV/AIDS) remains a major global public health issue, disproportionately affecting men who have sex with men (MSM), a population at higher risk of infection. Despite advances in HIV prevention and treatment, HIV testing uptake among MSM remains insufficient, particularly in sub-Saharan Africa (SSA) and Ghana.

Objective: This study aims to understand the socio-demographic, behavioural, and structural factors influencing HIV testing rates. Additionally, interventions aimed at improving HIV testing among MSM need to be identified, assessed, and their challenges and successes evaluated.

Methods: A mixed-methods study will examine factors influencing HIV testing among MSM in Ghana using data from the 2017 Ghana Men's Study, which surveyed 4,095 MSM across 10 regions. Socio-demographic, HIV risk, prevention knowledge, perceptions risky sexual behaviour and facility types data will be analyzed. In-depth interviews will be conducted with a purposive sample of 15 stakeholders to provide insights into interventions to improve HIV testing. The study will integrate qualitative and quantitative findings to develop a framework for enhancing HIV testing among MSM, informing future

healthcare strategies. Statistical analysis will use Stata version 18, and qualitative data will be analyzed with Atlas.ti.

Expected results: The study is expected to reveal that socio-demographic factors such as age, education, and socioeconomic status significantly influence HIV testing uptake among MSM in Ghana. Barriers like stigma, discrimination, and limited access to services are likely contributors to low testing rates. The qualitative interviews with stakeholders are expected to highlight successful interventions, such as mobile and community-based testing, that have improved testing rates. These findings will contribute to a framework for future interventions and policy recommendations aimed at increasing HIV testing and reducing transmission in Ghana and beyond.

Discussion

This study protocol aims to understand HIV testing behaviours among MSM in Ghana, addressing ongoing challenges despite progress in HIV prevention and treatment. It examines how socio-demographic, behavioural, and structural factors such as age, education, sexual practices, stigma, and discrimination affect testing uptake and access to care. Findings will inform evidence-based policy recommendations and interventions, including community engagement and healthcare provider training. The study also explores new testing technologies and socio-cultural factors, contributing to improved public health outcomes. It aligns with SDG 3 (ending AIDS) and SDG 10 (reducing inequalities), guiding inclusive policies for better health outcomes for MSM in Ghana.

This study aims to understand HIV testing behaviors among MSM in Ghana using qualitative and quantitative methods to develop a framework for improving HIV testing services. It will explore factors affecting testing uptake, identify barriers, and examine preferences for interventions.

Acknowledgment

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P5

Implementation of a secure cloud enclave for CMS regulated datasets

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BMC Proceedings 2025, 19(13):P5

Issues: As a research university, the University of South Carolina (USC) is required to make data available, accessible, discoverable, and usable. Public and private needs for sharing and collaboration on large volumes of health data are growing and faculty researchers at R1 Universities are looking for internal and external partners in IT to enable their research at scale while navigating shifting logistical and data privacy concerns. This project enabled research by designing and building a secure computing environment at USC to provide resources for large-scale processing of Regulated Research Data. One of these projects with secure processing needs is the The South Carolina Center for Effectiveness Research in Orthopaedics (CERortho) at Arnold School of Public Health (ASPH) in partnership with the USC Research Computing group.

Project: The CERortho scientists undertake research that involves PHI data from Prisma Health, in cooperation with both SC based and geographically distant collaborators. After the loss of their primary on-premises hardware solution the group reached out to RC for help designing a new environment. CERortho required both moderate computing power to complete their research analyses as well as strict

adherence to compliance and security measures for the PHI they are analyzing. The USC Research Computing (RC) group designed and deployed a secure computing environment that meets these rigorous computing and security requirements as a cloud resource (Fig. 1) to allow researchers to securely perform their computational workloads. Further work is being done to investigate the potential for integration into any future on-premises secure computing solutions at USC for greater cost-savings.

The cloud solution is more fault tolerant than on-premises solutions and can offer greater hardware scaling opportunities through the enormous capacity of cloud infrastructure. At the same time, the center's scientists ensure that the PHI they possess is being protected by a solution that meets all compliance and security requirements as stipulated by CMS ARS.

Lessons learned: During the initial portion of the project, it was quickly discovered that while there was a strong desire by RC to eventually lift CERortho's workloads to existing SecureHPC environments, it was not the best decision currently. The center needed a solution that could get them back to operational status quickly, avoid additional cumbersome CUI security controls that would be present in SecureHPC, and was very similar to their previous solution using windows graphical environments.

While it was initially obvious that there would be significant work dedicated to the administrative and governance tasks necessary for compliance requirements it quickly became clear that these tasks were tightly integrated with each step of the technical process. CMS offers a comprehensive list of security controls outlined in their ARS framework that mapped very well to freely available CIS windows server security benchmarks. Creating and documenting processes such as configuration management, change management, and security processes were time-consuming pieces of the project but were mandatory for achieving compliance.

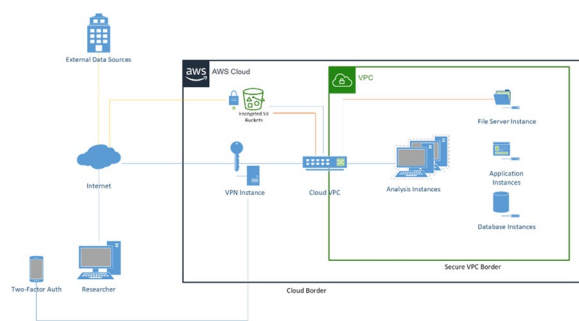


Fig. 1 (Abstract P5). Cloud (AWS) architectural diagram for secure computing environment for CERortho

P6

Breast cancer's genetic map: revealing racial disparities with genomic tools

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Introduction: Breast cancer affects 13% of U.S. women, with African American women disproportionately impacted by aggressive triple-negative breast cancer. This study analyzes 194 breast cancer specimens using big data techniques to identify racial disparities in

mutation profiles across 50 oncogenesis-related genes, leveraging genomic, clinical, and demographic data for deeper insights.

Study Objectives: This study aims to analyze racial disparities in breast cancer mutation profiles using large-scale genomic and clinical datasets. By identifying differences in mutation frequencies, molecular subtypes, and amino acid substitutions, the research seeks to advance personalized treatments, uncover molecular mechanisms driving disparities, and promote equitable care through inclusive genomic research.

Methods: The study uses data from Prisma Health Cancer Center Biorepository (Greenville, SC, USA) collected between 2012 and 2021, with 194 breast cancer specimens (161 White, 33 African American), screened for mutations by Precision Genetics. Analyzed genes include 10 tumor suppressors, 33 oncogenes, and 7 context-dependent genes. Patient data from the Prisma Health Cancer Institute database and EPIC[®], including demographic details such as age, race, sex, and smoking status, along with tumor characteristics like tissue site, histology, grade, and TNM staging. Our analysis assessed racial differences in mutation frequency, molecular-subtype, and specific amino acid substitutions in each gene.

Results: This study analyzed 194 breast cancer samples (161 White, 33 African American) for mutations in 50 cancer-associated genes. PIK3 CA (29%), KDR (21%), and TP53 (12%) were the most frequently mutated genes overall. Significant racial differences emerged, notably in JAK3 mutations, which were more prevalent in African Americans (24%) than Whites (2%, $p < 0.001$). Subtype-specific mutations included PIK3 CA p.His1047 Arg in Whites (30%) and p.Ile391Met in African Americans (79%). TNBC was more common in African Americans (27% vs. 11%, $p < 0.05$), who also exhibited higher PIK3 CA and JAK3 mutations. Total mutation rates were similar (82% African American, 81% White).

Discussion: The findings reveal critical racial differences in breast cancer mutation profiles, particularly in the prevalence of JAK3 and PIK3 CA mutations, and highlight the disproportionate burden of TNBC among African Americans. These disparities may contribute to the poorer outcomes observed in African American women and underscore the need for molecularly targeted treatments. Identifying specific amino acid substitutions, such as PIK3 CA p.Ile391Met in African Americans, provides insights into the distinct genetic mechanisms that may drive these differences. Additionally, the similarity in overall mutation rates suggests that disparities are driven more by specific genetic patterns than by overall mutational burden. The results emphasize the importance of inclusive genomic research to inform equitable, personalized cancer therapies. Addressing these disparities requires a commitment to enrolling underrepresented populations in clinical trials and integrating genetic findings into treatment protocols to improve outcomes for all breast cancer patients.

P7

Unlocking nutritional insights: a big data analysis of social media's role in public health

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Introduction: Social media shapes public nutrition perceptions, serving as both an educational tool and a source of misinformation [1]. While engaging content can enhance health communication, it may also spread unverified claims. Understanding engagement patterns is key to addressing this issue.

Study Objectives: This study analyzes engagement metrics, sentiment trends, audience demographics, and themes of nutrition-related posts to assess whether popular content aligns with evidence-based nutrition or contributes to misinformation. The goal is to improve strategies for disseminating accurate nutrition information.

Methods: We consulted nutrition literature to identify eight nutrition-related categories and their corresponding subcategories (Table 1). Using Quid Monitor and these subcategories as keywords, we retrieved nutrition-related posts in English from X, YouTube, Reddit, forums and blogs. The dataset covers a 27-month period, ending on December 6, 2024.

Results: A total of 201,981,643 posts were retrieved, with potential impressions of 46,556.1 billion. The top platforms were X (133,490,535 posts), Reddit (14,722,439 posts), and YouTube (11,461,608 posts), with additional data from blogs, news, reviews, and comments. The net sentiment was a positive 72 (Figure1). Age distribution showed that 12% of posts were from users under 18, 16% from 18–24, 17% from 25–34, 15% from 35–44, and 13% from 45–54. The most liked and commented post,"Homemade Milk Chocolate,"received 1,119,876 likes and 5,690 comments, while the most viewed post,"Food Lover vs. Ball Lover,"had 29,565,424 views. Popular posts included humor, DIY recipes, cultural cuisine, product promotions, and motivational content.

Discussion: Social media presents both opportunities and risks for nutrition communication. The high engagement found in this study reflects public interest, but viral trends, DIY recipes, and product promotions often lack scientific accuracy. The high positive sentiment suggests trust in social media content, raising concerns about misinformation acceptance. Younger audiences dominate discussions, highlighting the need for targeted digital health communication. The popularity of cultural cuisine offers a chance for inclusive, evidence-based messaging. Future research should assess misinformation spread and explore collaborations between health professionals, influencers, and platforms to promote credible nutrition content.

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Table 1 (Abstract P7). Category and keywords

Category	Keywords
Diet Types	Keto, Vegan, Paleo, Intermittent Fasting, Low Carb, Plant Based, Gluten Free, Sugar Free, Dairy Free, High Protein, Whole30, Low Fat, Balanced Diet, Mediterranean Diet
Healthy Eating and Nutrition	Healthy Food, Healthy Eating, Nutrition, Nutrition Tips, Nutritious, Eating Well, Eat Clean, Superfoods, Organic, Clean Eating, Real Food, Mindful Eating, Food for Thought
Food Sourcing and Sustainability	Farm to Table, Local Produce, Seasonal Eating, Farm Fresh, Eat Seasonal, Eco-Friendly Food, Sustainable Eating, Zero Waste Cooking, Natural Food
Meal Preparation and Ideas	Breakfast Ideas, Lunch Ideas, Dinner Ideas, Healthy Snacks, Smoothie Recipes, Meal Prep, Quick Meals, Family Dinners, Comfort Food, One-Pot Meals, Low Calorie Meals
Food Aesthetics and Plating	Plate It Perfect, Food Plating, Food Styling, Food Inspiration
Cultural and Global Cuisine	Asian Cuisine, Indian Food, Mexican Food, Italian Food, Soul Food, Comfort Cuisine, Global Eats
Lifestyle and Emotional Connection	Foodie, Food Is Life, Food Lover, Good Eats, Cravings, Food Adventure, Food Cravings, Eat Happy, Healthy Indulgence, Food Goals
Festive and Holiday Food	Holiday Eats, Festive Foods, Healthy Holiday Recipes

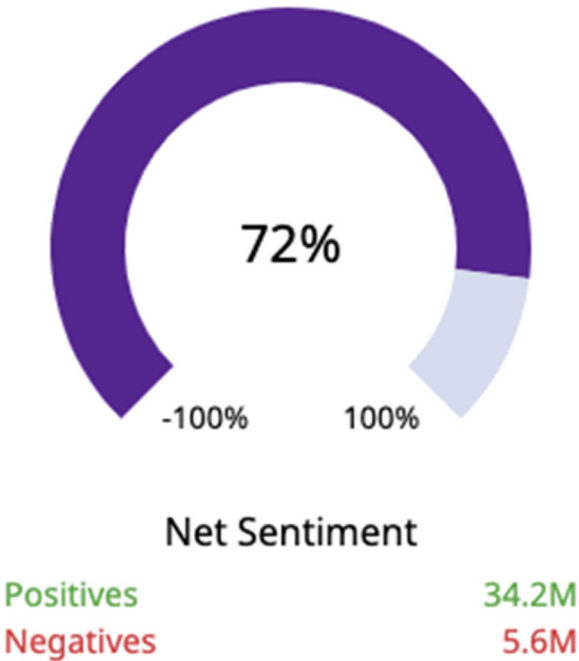


Fig. 1 (Abstract P7). Net Sentiment Analysis

P8
Birthing Black: The Impact of Perceived Discrimination on the Birthing Experiences of Black Women

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Background: Embedded in our healthcare system lies a level of cultural incompetency that is compounded by the historical siege of the autonomy of the Black body through bias and institutional racism. The result of these systems has enabled the continued increase in the Black maternal mortality rate in the United States, with perceived discrimination playing an imminent role in the birthing experience. Previous research has explored the effect of discrimination on perinatal outcomes of pregnancy and its potential impact on Black maternal outcomes.

Purpose: While the perinatal outcome in its entirety is significant, this research specifically seeks to evoke a deeper understanding of the perceived discrimination of Black women and its impact on them during labor. In addition, this study examines perinatal outcomes (during) by exploring the type of care Black women receive and the conditions that play a role in shaping their birthing experiences.

Method: This mixed-methods study explored perceived discrimination during labor among four African American women (ages 22–41). The quantitative component used the Discrimination in Medical Settings Scale to assess experiences of being ignored, dismissed, or treated with less courtesy. Participants answered survey questions measuring perceived bias in medical interactions. The qualitative component involved a narrative analysis, where participants responded to three open-ended questions about their birthing experiences. These interviews provided deeper insight into how discrimination shaped their labor and postpartum health. Combining both methods allowed for a comprehensive understanding of the systemic challenges affecting Black maternal care and outcomes.

Conclusion: Quantitative data revealed that 75% of participants experienced less courtesy, poor service, and feeling unheard, while 50% reported disrespect from medical staff. Qualitative analysis identified four major themes: feeling unseen, being rushed into labor, stereotypes about pain management, and post-pregnancy chronic conditions. These findings highlight the urgent need for culturally competent care in Black maternal health.

Keywords: Black birthing experiences, Black maternal mortality, Perceived discrimination

P9

Shingles vaccine and the risk of dementia in a cohort of people with HIV using large-scale data analytics

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BMC Proceedings 2025, 19(13):P9

Introduction: Dementia is a major public health concern worldwide. Leveraging large-scale data analytics, we assessed the risk of dementia between shingles vaccine (herpes zoster) exposure in a cohort of people with HIV (PWH) due to emerging conflicting evidence in scientific literature.

Study Objectives: To utilize national-level Veteran Affairs (VA) Informatics and Computing Infrastructure (VINCI) data and evaluate the association between shingles vaccine and dementia risk, providing insights that further biomedical research. We hypothesized that exposure to shingles vaccine (at least 2 prescriptions) is associated with a lower risk of dementia within a cohort of PWH adherent to HIV medication.

Methods: This study analyzed data from 6,284 HIV adherent persons aged ≥ 19 either having exposure to the shingles vaccine ($n=3450$) and no exposure ($n=2834$). Patient demographics and comorbidities were assessed using descriptive statistics and standardized differences. Treatment exposure was coded dynamically and Cox proportional hazards models were employed to examine dementia risk. Due to non-random treatment assignment, we utilized a propensity score matched (PSM) sample ($n=2306$ in each group) and used a time-dependent Cox proportional hazards model as well as an inverse probability treatment weighted (IPTW) Cox proportional hazards model.

Results: Non-matched results reveal those exposed to shingles vaccine were older (mean=49.25 vs. 45.49) compared to those not exposed. Non-matched adjusted hazard ratio did not reveal a significant association (HR 0.756, 95% CI 0.506–1.129); however, PSM as well as IPTW models show those persons exposed to shingles vaccine had a lower risk of dementia (HR 0.581, 95% CI 0.362–0.934 and HR 0.581, 95% CI 0.389–0.868, resp.).

Discussion: This study highlights the power of large-scale data to address public health and biomedical research questions. The decreased dementia risk associated with shingles vaccine exposure suggest further biomedical research is essential to clarify the relationship between shingles vaccine, HIV, and dementia risk, potentially improving prevention strategies and outcomes for aging and high-risk populations.

P10

Data modeling and analytics to advance the secondary use of real-world data in healthcare research

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BMC Proceedings 2025, 19(13):P10

Introduction: Health informatics focuses on strategic improvements to the quality of patient care through the study of data acquisition,

retrieval, and analytical methodologies involving real-world data [1,2]. Healthcare research with artificial intelligence and big data analytics often involves the secondary use of real-world data collected in the process of patient care. However, data modeling and analytics techniques to bridge the gap between real-world data and effective secondary uses for healthcare research are lacking [3,4].

Study Objectives: This work aimed to develop scalable methods for validating the secondary use of real-world data in healthcare research by applying principles of data modeling and analytics, including exploratory data analysis and preprocessing data. Additionally, retrospective chart reviews [5] were conducted to analyze the collection, storage, and retrieval of clinical data and to assess the interpretability and reliability of real-world data in new contexts for four clinical use cases.

Methods: To begin with, collaboration with multidisciplinary teams was crucial to ensure that the results for each use case were reliable and clinically relevant [6]. To map clinical variables to structured data elements in electronic health record systems and clinical data warehouses, exploratory data analytics and retrospective chart reviews were conducted. Clinical requirements mapped to unstructured data were categorized as incomprehensible [7]. Data analytics included descriptive statistics to summarize transformations, cluster analysis to identify groups of similar patients, and association rules to uncover patterns in patient factors. Extended entity-relational and specialized hierarchical data models were used to analyze various clinical scenarios, ensuring a thorough understanding of patient data and preventing errors in processing and decision-making downstream [8].

Results: This research provided valuable insights into the use of real-world data in healthcare research. In the first use case, a transformation learning framework was developed for an AI prediction study on end-of-life care, identifying a cohort of 15306 cancer decedents [9]. The second use case created two analytical models for a multi-institutional clinical trial, reducing the need for manual chart reviews of thousands of patients. The third established a clinical protocol to complete a retrospective review of patient eligibility criteria for seven genomics clinical trials. Finally, the fourth is a feasibility study to address forecasting hospital bed staffing using autoregressive models in the setting of community hospitals.

Discussion: This work contributes to health informatics and healthcare research by raising awareness of challenges in data curation and analytics using real-world data. A key issue was establishing a common language [10] for effective communication between all stakeholders involved in healthcare research with advancements in computational methods. This was accomplished through a systems analysis approach that included shadowing clinical experts, online meetings, reviewing the EPIC data dictionary, gathering clinical requirements, verification by clinical experts, and developing conceptual data models.

By integrating exploratory data analysis with retrospective chart reviews, a pathway was created for logically mapping the clinical research requirements to potential data elements, some of which may not exist. These uncertainties underscore the importance of validating real-world data to ensure its effectiveness for secondary use.

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P11

Accuracy of self-reported COVID-19 vaccine history compared with electronic health record data based on a U.S. longitudinal study

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BMC Proceedings 2025, **19**(13):P11

Introduction: Obtaining accurate COVID-19 vaccine history of participants is critical for clinical research yet acquiring both precise and comprehensive data remains a challenge. While EHRs are considered the gold standard, they often lack completeness, especially in the U.S., where vaccines were administered in retail pharmacies and often not incorporated into the EHR. Self-reported vaccine data can provide valuable insights, especially when EHR access is limited.

Study Objective: Our study addressed these gaps by examining the accuracy of self-reported COVID-19 vaccination dates compared to EHR data across various levels of precision—day, week, and month—as well as vaccine doses, brands, dose order, and different times during the pandemic. Utilizing data from the Innovative Support for Patients with SARS-CoV-2 Infections Registry (INSPIRE), the study evaluated the accuracy of self-reported COVID-19 vaccination dates from 2020 to 2022 and provided insights into factors influencing self-report accuracy to guide data collection and utilization in future public health efforts.

Methods: Among the 6,044 participants recruited in the INSPIRE COVID-19 study, 1,036 participants were included in the analysis meeting the criteria for having an authenticated Electronic Health Record (EHR) and reported any valid vaccine data form surveys. We compared self-reported information to that available in the EHR for the following data points: vaccine date, vaccine brand and number of doses using

rates of agreement with Cohen’s Kappa. The change in self-reported vaccine accuracy of time of vaccination and dose order was compared. **Results:** Most EHR vaccine dates were accurately reported in participants’ surveys: exact date (61.5%), week-level (76.5%), and month-level (88.1%). Participants with three EHR doses, the largest group (48.6%), showed the highest match rates across all levels. For vaccine brand, Cohen’s κ (0.951) indicated strong agreement between EHR and self-reported brands. However, 72.3% of the participants reported more doses in surveys than recorded in EHR, with weak agreement (Cohen’s $\kappa = 0.0466$). Self-reported vaccine accuracy remained consistent across different time and dose order.

Discussion: The findings demonstrate that self-reported COVID-19 vaccination data is highly accurate for both dates and vaccine brands compared to EHR records. While EHR data is considered the gold standard, it lacks comprehensive vaccine dose records and disproportionately represents individuals with higher socioeconomic status. Self-reported accuracy remained consistent over time, suggesting minimal recall bias. These results affirm the reliability of self-reported COVID vaccine information and support its inclusion in future studies to ensure more comprehensive and equitable data collection.

P12

Data science application in HIV in resource-limited settings: a systematic review

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BMC Proceedings 2025, **19**(13):P12

Introduction: Data science has advanced HIV research, but its application in resource-limited settings is underexplored. Previous reviews focused on bibliometric analyses, offering broad overviews rather than specific insights. This systematic review addresses this gap by examining data science’s role in improving HIV treatment cascade outcomes in resource-limited settings, providing targeted, actionable insights.

Objective: Understanding the current landscape of HIV data science in resource-limited settings is crucial guiding research and implementation strategies to enhance patient care and quality of life. This review summarized the evidence related to the application of data science approach in several HIV treatment and care outcomes, i.e., ART initiation, ART effectiveness/mortality, ART adherence, Viral suppression, drug interaction.

Methods: We conducted a systematic review using PubMed, Web of Science, Embase, Sociological Abstracts, IEEE, and PsycINFO to identify relevant research on HIV and data science from October 2023 to May 2024. Eligible studies included HIV treatment-related quantitative or mixed-method studies employing data science techniques between 2000 and 2023. Exclusions were descriptive studies, protocol papers, RCTs, reviews, meta-analyses, letters, book chapters, abstracts, non-English studies, or those unrelated to HIV treatment outcomes. The search combined keywords and MeSH terms like “HIV,” “AIDS,” “data science,” “machine learning,” and “resource-limited settings.” Extracted data included study settings, design, population, sampling methods, data sources, input variables, outcome measures, statistical methods, machine learning metrics, and findings. Study quality was assessed using the 14-item Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies by the National Heart, Lung, and Blood Institute, ensuring robust evaluation of included studies.

Result: Out of 10,851 records, 34 studies were included, primarily from sub-Saharan Africa (94%) using secondary data (44%). ART efficacy was examined in 13 studies, adherence in 7, and viral suppression/CD4 count in 14. Random Forest was the most common machine learning

method (38%), followed by decision trees (21%), neural networks, and logistic regression (15%). Models showed high accuracy, with Random Forest achieving 77%–99.8% and AUC scores up to 0.99. Sensitivity ranged from 60%–100%, while specificity reached 98.4% in some cases, showcasing the robust predictive power of algorithms like Random Forest, Gradient Boosted Machines, and XGBoost.

Discussion: This review highlights the growing application of data science in HIV research, particularly in resource-limited settings. With 34 studies included out of 10,851 records, the predominance of studies from sub-Saharan Africa (94%) underscores the focus on regions most affected by HIV. The widespread use of secondary data (44%) reflects the reliance on available datasets to optimize research efforts. Random Forest emerged as the leading machine learning technique, utilized in 38% of studies, likely due to its versatility, robustness, and ability to handle complex, non-linear relationships. Other commonly used techniques, such as decision trees (21%), neural networks, and logistic regression (15%), also demonstrated high predictive power. Performance metrics like accuracy (77%–99.8%) and AUC (up to 0.99) confirm the efficacy of these models in predicting ART outcomes, adherence, and viral suppression. Sensitivity (60%–100%) and specificity (up to 98.4%) further validate their clinical utility.

P13

Delivering maternal and child health data to hospitals in a secure online environment

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Issues: In developing an online data visualization platform to enhance the reporting of maternal and newborn outcomes data for hospitals in South Carolina (SC), we encountered a significant challenge: protecting the confidentiality of the data and each facility's individual reports. SC law prohibits the public sharing of identifiable hospital information without the hospital's consent, and Health Insurance Portability and Accountability Act (HIPAA) compliance rules mandate stringent data protection measures.

Interactive online platforms are invaluable for guiding planning and data-driven decision making by policymakers and clinicians. However, it is crucial to recognize the potential risks to the security and privacy of sensitive data.^[1] When transforming data into actionable intelligence, a balance must be struck between data utility and anonymity.^[2] This abstract outlines the protocols implemented to safeguard data while ensuring it remains accessible and useful for health policy and planning.

Project: The Division of Integrated Health and Policy Research (IHPR) at the Institute for Families in Society (IFS), under contract with the South Carolina Department of Health and Human Services (SCDHHS), provides maternal and newborn data to 38 birthing facilities in SC on a quarterly basis through a secure online portal. This reporting supports hospital-level quality improvement initiatives in alignment with the SC Birth Outcomes Initiatives' goals and vision to improve maternal and newborn health. IFS transforms nearly ten years of hospitals' individual data from birth, hospital, and Medicaid eligibility records into meaningful and actionable information. This process provides clinicians and decision-makers with insights into their data, allowing users to explore, identify trends more quickly, and examine relationships and patterns between measures and operations.

To comply with confidentiality protocols within this reporting environment, IFS implemented several big data solutions to protect data ingested from three summary hospital files with over 150,000 records of identifiable information. These solutions included record-level anonymization to remove protected health information, suppression of the display of small numbers, aggregation of missing/unknown into 'other' categories, application of systems regulating and restricting user-level access, encrypting data at rest, and techniques to mitigate potential unauthorized data manipulation and retrieval. Specifically, a

two-tiered approach was implemented to protect data from hacking and identification. Data access was restricted at the user-level via a third-party authentication and authorization software compliant with stringent security infrastructure requirements, and at the API-level via techniques to prevent and mitigate SQL injections (a form of database hacking).

Lessons Learned: In developing an integrated system of techniques for safeguarding hospital data from authorized and unauthorized access, IFS has furthered its understanding of how to design and support a data infrastructure that maintains the confidentiality and integrity of privileged data in an online environment. This is particularly important given the ever-increasing need for enhanced cybersecurity in an environment where both public and private entities are experiencing data breaches centered on extortion, stealing data, disrupting operations, and causing damage.

Acknowledgements

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P14

Integrating climate and health data to assess the impact of precipitation on health outcomes: a case study from Tanzania

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Background: Climate factors such as precipitation significantly influence healthcare access and health outcomes, particularly in low-resource settings.^[1] Despite the availability of rich global climate datasets and individual-level health surveys, their integration remains underexplored. This study aims to develop and present a methodological framework for integrating climate data with Demographic and Health Survey (DHS) data to assess the impact of climatic factors on health outcomes, focusing on Tanzania as a case study.

Materials and methods: We integrated a high-resolution precipitation dataset (Climate Hazards Group InfraRed Precipitation with Stations- CHIRPS) with individual-level data from the Tanzanian DHS, a nationally representative cluster sample survey. Using R, daily precipitation was linked to DHS sampling clusters via geographic coordinates. To address DHS cluster displacement (up to 2 km in urban areas, 5 km in rural areas), we created buffer zones and calculated mean precipitation within each. The outcome variable was birth location

(facility vs. home), including only the last birth for each mother. For mothers relying solely on walking, minutes to the nearest health facility was estimated using a cost-distance algorithm on a walking-only friction surface map.

Results: The sample included 5,679 most recent births from children under three at the time of the interview. Increased precipitation in the week before birth was not significantly associated with home versus facility delivery (aOR=1.002, 95% CI: 0.978–1.026). However, travel time to the nearest healthcare facility was a strong predictor of home delivery. Each additional minute of travel increased the odds of home birth by 0.7% ($p<0.0001$). Notably, self-reported travel time for mothers who relied solely on walking was only weakly correlated with travel time estimated using the cost-distance algorithm (Spearman's $\rho = 0.2052$, $p<0.0001$).

Conclusion: This methodological framework underscores the potential of integrating global climate datasets with individual-level DHS data to analyze the impact of climatic factors on health outcomes. While the initial focus was on birth location, the framework can be extended to explore other maternal and child health outcomes, such as antenatal care attendance and neonatal health indicators, as well as the coverage and timeliness of childhood vaccinations. By leveraging high-resolution precipitation data, this approach provides critical insights into how climatic variables impact healthcare access and utilization. Policymakers and public health practitioners can use these insights to develop climate-resilient healthcare strategies and improve service delivery in regions prone to extreme weather patterns.

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P15

The impact of precipitation on timeliness of childhood vaccination: insights from the 2022 Tanzania Demographic and Health Survey

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Background: Childhood vaccination is a cost-effective way to reduce morbidity and mortality from preventable diseases. Timely vaccination is essential, as delays increase infection risk and weaken immunity. Analysis of the 2015–2016 Tanzanian Demographic and Health Survey (TDHS) revealed significant vaccination delays, influenced by maternal age, socioeconomic status, education, rurality, and healthcare access. [1] However, the impact of climatic variability remains underexplored. In East Africa, rising temperatures and shifting precipitation patterns may affect vaccination timeliness, particularly in rural areas. [2] This study aims to assess the effect of precipitation in the week prior to the recommended date of the first dose of the pentavalent vaccine (Penta1), due at age 42 days, on the timeliness of its administration.

Materials and methods: We analyzed data from the 2022 Tanzania Demographic and Health Survey (TDHS) for children under three years who were alive at the time of the survey and had received the Penta1 vaccine. Children without a recorded vaccination date, with an invalid date, reported solely by maternal recall, or ineligible due to age were excluded. Precipitation data from the Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS) dataset was linked to TDHS cluster coordinates. Logistic regression was used to assess the association between precipitation in the week before the penta1 due date and delays of one week or more.

Results: A total of 4,702 children were included in the analysis. Delayed vaccination was more common among children whose mothers had lower education levels, lower income, resided in rural areas, had fewer than four antenatal care visits, or gave birth outside a health facility. Adjusted analyses showed that a 1 mm increase in precipitation in the week before the vaccination due date was associated with

a 2.7% increase in the odds of a one-week or greater delay ($p=0.024$). The effect of precipitation on vaccination delay varied by distance to the nearest health facility, with higher precipitation generally associated with increased delays across all distance categories, though these results were not statistically significant.

Conclusions: These findings suggest that precipitation may contribute to short-term childhood vaccination delays. However, the interaction between precipitation and distance to health facilities was not statistically significant. Future analyses will explore broader precipitation trends, including seasonal rainfall and extreme weather events, to assess their impact on long-term vaccination delays with greater clinical significance.

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P16

Latent class analysis of suicidal thoughts and behaviors in a national sample of high school students

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Introduction: Adolescents in the United States are experiencing suicidal thoughts and behaviors (STBs) at record high rates. Previous research and theory offer guidance on which demographic factors and adverse experiences may predict suicidality. However, the patterns of STBs experienced by adolescents and the predictive factors associated with those patterns are not well understood.

Study Objectives: Using a national dataset reporting on the health, safety, and wellbeing of high school students, we aim to identify latent classes of youth based on their reported patterns of suicidal thoughts and behaviors. Additionally, we aim to identify which characteristics and experiences are associated with those classes.

Methods: Cross-sectional survey data from the 2021 Youth Risk Behavior Surveillance System (YRBSS) was used for this study. Latent class analysis was used to group youth based on four binary indicators: persistent sadness or hopelessness, suicidal ideation, suicide planning, and suicide attempts. The SAS procedure, PROC LCA, was used for latent class analysis in SAS Version 9.4, and the best-fitting model was selected by examining indices of model fit and interpretability. A multinomial logistic regression was then performed with the latent classes of suicidal behavior as the dependent variable. A range of predictors were considered in the model, including demographic characteristics, sexual identity, self-reported mental health status, homelessness, school and neighborhood environments, substance use, and physical/sexual violence victimization. The SAS procedure PROC SURVEYLOGISTIC was used, and a multinomial distribution and GLOGIT link function were employed. A backward elimination process was used to identify confounders; those that changed the association of interest by 10% or more were retained in the final model.

Results: A 3-class model emerged as the best fit for the data (see Table 1). These classes are described as “low risk” (low endorsement of all STBs), “low suicide risk with high distress” (high endorsement of depression and suicidal ideation, low endorsement of suicide planning or attempts), and “high risk” (high endorsement of all STBs). Risk factors for the “low suicide risk with high distress” class

include being female and being bullied online (see Table 2). Risk factors for the “high risk” class include being Black and being in 9th grade. Risk factors for both risk groups include endorsing a sexual orientation other than “heterosexual,” violence victimization, poor self-reported mental health, and certain types of substance use (marijuana, prescription pain medication). Feeling close to people at school was identified as a protective factor against suicidality.

Discussion: This study identifies common clusters of suicidal thoughts and behaviors experienced by youth in the United States and pinpoints demographic and experiential risk factors associated with each latent class. Many of our findings were supported by previous research and aligned with our hypotheses; however, some were surprising. Several predictors that we expected to relate to latent class membership were not significant in our final models, including age, dating violence, school violence, feeling unsafe at school, certain types of substance use (binge drinking, cocaine, heroin, methamphetamine, injection drug use), and homelessness.

Table 1 (Abstract P16). Endorsed depression, sadness/hopelessness, suicide ideation, planned suicide and attempted suicide by class

	LC 1 High risk	LC 2 Low risk	LC 3 Low suicide risk with high distress
N (weighted %)	1505 (8.9%)	13,079 (76.3%)	2549 (14.9%)
Sadness/hopelessness	94.7%	25.4%	78.8%
Suicide ideation	99.6%	0.1%	64.1%
Planned suicide	92.1%	1.2%	38.5%
Attempted suicide	72.2%	0.9%	7.3%

Table 2 (Abstract P16). Multinomial multilevel generalized logit regression of youth sociodemographic characteristics and adverse experiences on latent classes of STBs (reference class: LC 3 low suicide risk with high distress)

	Low risk (LC 2)				High risk (LC 1)			
	OR	95% CI	p		OR	95% CI	p	
Gender								
Female vs. Male	0.81	0.67	0.98	0.03	0.74	0.56	0.98	0.04
Race/ethnicity (ref: White)								
Am Indian/Alaska Native	0.90	0.37	2.17	0.81	1.62	0.67	3.95	0.29
Asian	0.90	0.64	1.26	0.53	1.19	0.68	2.09	0.54
Black/African American	1.11	0.84	1.46	0.48	1.80	1.22	2.65	0.003
Hispanic/Latino	1.02	0.73	1.41	0.92	1.52	0.94	2.46	0.09
Multiracial—Hispanic	0.92	0.72	1.18	0.52	1.02	0.72	1.43	0.93
Multiracial—non-Hispanic	0.98	0.68	1.42	0.91	1.00	0.63	1.59	1.00
Native Hawaiian/Pacific Islander	0.62	0.17	2.24	0.47	1.63	0.24	11.07	0.62
Grade (ref: 9th grade)								
10th grade	1.02	0.80	1.31	0.86	0.83	0.59	1.15	0.26

	Low risk (LC 2)				High risk (LC 1)			
11th grade	0.89	0.69	1.13	0.33	0.47	0.34	0.67	<.0001
12th grade	1.05	0.82	1.35	0.71	0.48	0.34	0.68	<.0001
Ungraded	9.46	1.03	86.76	0.05	0.91	0.04	18.77	0.95
Sexual orientation (ref: heterosexual)								
Bisexual	0.35	0.28	0.45	<.0001	1.23	0.91	1.66	0.18
Gay or lesbian	0.39	0.26	0.57	<.0001	0.94	0.53	1.68	0.84
Some other way	0.36	0.24	0.55	<.0001	1.07	0.66	1.72	0.79
Not sure	0.45	0.32	0.62	<.0001	0.76	0.48	1.19	0.23
Don't know what this means	4.46	1.42	13.96	0.01	4.23	1.00	17.89	0.05
Bullied at school								
Yes vs. No	0.65	0.51	0.84	<.001	1.46	1.09	1.96	0.01
Bullied electronically								
Yes vs. No	0.65	0.51	0.83	<.001	1.24	0.92	1.67	0.15
Witnessed neighborhood violence								
Yes vs. No	0.75	0.60	0.94	0.01	1.32	1.02	1.72	0.04
Forced sexual intercourse								
Yes vs. No	0.65	0.47	0.89	0.01	1.75	1.27	2.41	<.001
Experienced sexual violence								
Yes vs. No	0.60	0.45	0.80	<.001	1.53	1.13	2.07	0.01
Felt close to people at school								
Yes vs. No	1.46	1.23	1.74	<.0001	0.97	0.76	1.24	0.78
Current mental health								
	0.24	0.20	0.29	<.0001	1.59	1.21	2.08	<.001
Marijuana use								
Yes vs. No	0.52	0.43	0.63	<.0001	1.91	1.46	2.50	<.0001
Prescription pain medication misuse								
Yes vs. No	0.62	0.48	0.79	<.001	1.58	1.21	2.07	<.001

P17
ClinicalConsent: AI-enhanced risk insights for clinical research participants
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Issues: The dilemma of informed consent in the clinical research setting has been extensively studied over the past several decades [3, 4, 6]. Complex medical terminologies and lengthy consent documents often hinder participants’ full understanding of the research they are consenting to [2]. This lack of comprehension can compromise the ethical foundation of clinical studies and may lead to participants making uninformed decisions. We propose ClinicalConsent, a tool that

simplifies language and incorporates interactive, on-demand Q&A to bridge the communication gap between researchers and participants [1].

Project:

The research application utilizes a secure storage service set up through Firebase, allowing for the storage of patient demographics or researcher user data based on the account type selected upon sign-up. Upon logging in, potential participants are presented with a list of studies for which their demographics identify them as suitable candidates and are geographically accessible. These studies feature AI-generated summaries derived from the extensive clinical consent forms provided by the National Institutes of Health (NIH), currently hosting over 70,000 actively recruiting studies and with the ability to add more at any time. Each study entry includes a personalized chatbot window below the summary, enabling participants to ask specific questions related to potential risks, study design (e.g., randomized controlled trials), or other relevant aspects. The chatbot, powered by a generative pre-trained transformer (GPT) and configured via the assistant's API, responds only when confident in its answers [7] and ensures all interactions are directly related to the respective clinical study (Fig. 1). For researcher accounts, the platform offers functionalities to post actively recruiting studies, enhancing the informed consent process by facilitating additional questions to provide participants with more comprehensive information related to risks. Our goal with ClinicalConsent is to empower patients to better analyze study risks, increase trust in clinical researchers, and foster an environment of shared decision-making environment in clinical research.

Lessons Learned: The development and initial implementation of ClinicalConsent highlighted the importance of participant-centered design in creating effective patient education tools. Participants appreciated the interactive features and the use of plain language, which helped to demystify esoteric medical concepts. However, challenges emerged in addressing varied literacy levels and technological proficiency among users. The experience underscored the necessity for ongoing evaluation and refinement to meet diverse patient needs. Similar research has been conducted towards improving patient comprehension in clinical research settings [5], although our database of over 70,000 NIH clinical studies with a secure login feature for patients and researchers has not yet been introduced into any other similar systems. Future iterations of ClinicalConsent will focus on recruiting participants and researchers to further build out the available clinical studies even further and to match participants with research studies in their area for which they are a match.

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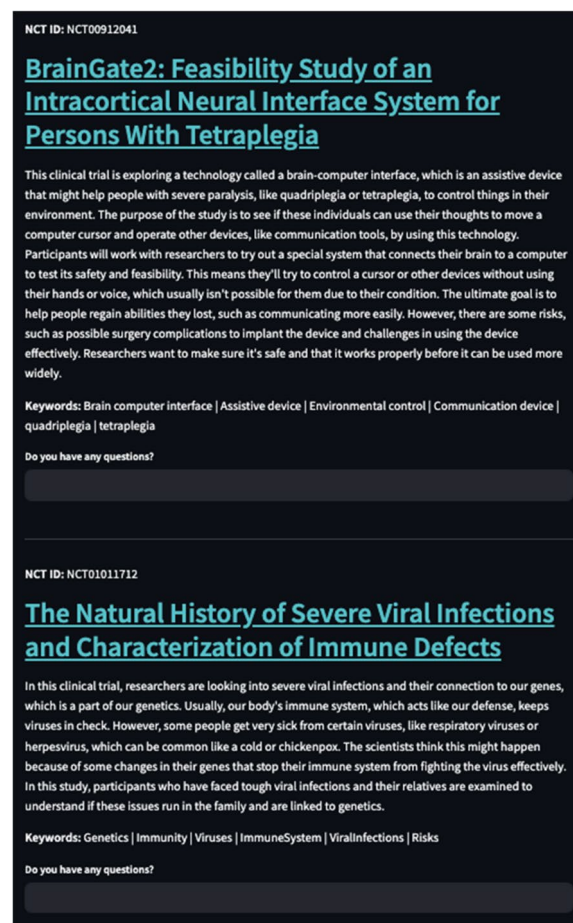


Fig. 1 (Abstract P17). Clinical trial data user interface with a generative pre-trained transformer (GPT) generated summary and chatbot capable of answering trial-specific questions for each trial

P18

Charting data construction algorithm in maternal telehealth research using national EHR data

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Introduction: The United States is amidst maternal health crisis with persistent geographic and sociodemographic disparities facing rural women and women of color. Prenatal care access is well-documented

to ensure timely detection of clinical risks and management of perinatal care planning. Disparities in access to prenatal care were expected to be mitigated through telehealth. However, identifying telehealth episodes for specific health services, such as prenatal care, can be challenging for public health researchers. Electronic Health Records data – such as the National Clinical Cohort Collaborative (N3 C) database [1]– allows researchers to identify women who received prenatal and intrapartum care [2]. Yet, constructing a detailed cohort of women who gave birth within a set time frame large-scale EHR and assessing their uptake of telehealth during varying periods of the perinatal care continuum requires a rigorous algorithm and several data assumptions.

Study Objectives: This study aims to summarize the processes of a cohort development of women who gave birth and who received labor and delivery care (i.e., intrapartum) between January 1, 2019, to December 31, 2023, within 85 health systems contributing to N3 C. We further quantified the distributions of EHR records with discontinuity of information on prenatal and intrapartum care.

Methods: The construction of the data involves a series of steps. We started by identifying pregnancy episode incidence and recording all pregnancies per person during the period of study from the N3 C Limited Datasets. Then age was restricted in all possible pregnant persons to include only those age 15–50. Unique pregnancies were identified, followed by the use of gestational concepts to determine pregnancy start dates. The pregnancy episodes data was merged with other pregnancy information to form a preliminary cohort. Concepts for telehealth and prenatal care were used to identify overlapping person identifiers within the preliminary cohort and the Limited Datasets. The event dates of care were merged with the pregnancy information and a dummy variable was created in order to identify prenatal and telehealth care within each pregnancy.

Results: During 2019–2023, 7,768,945 prenatal records were identified from the 85 contributing health systems with a total of 5,272,012 unique patients. A total of 1,113,637 childbirth deliveries were recorded for 1,060,873 individuals within the 85 contributing systems. Nearly 80% (79.88%) of patients with prenatal care records did not give birth in the contributing systems. Of those who gave birth within the contributing systems, 574,730 unique patients had gestational information and of those, 10.88% did not have prenatal care recorded within the contributing systems (Fig. 1). Of those who gave birth and had gestational information within the contributing systems, rural individuals were less likely than urban individuals to have recorded prenatal care information, and individuals of racial minorities (non-Hispanic black, Hispanic/Latino, non-Hispanic Asian, other) were more likely than white individuals to not have prenatal care records.

Discussion: Information continuity is critical for maternal health research. However, without universal nationwide information, missing information across perinatal care continuum might hinder the efficacy of EHR data to construct variables related to clinical risks during prenatal or intrapartum care, underestimating these distributions. These results suggest that leveraging electronic health data, especially when addressing maternity care access and outcome disparities, needs to consider heterogeneity of missing values across sociodemographic groups.

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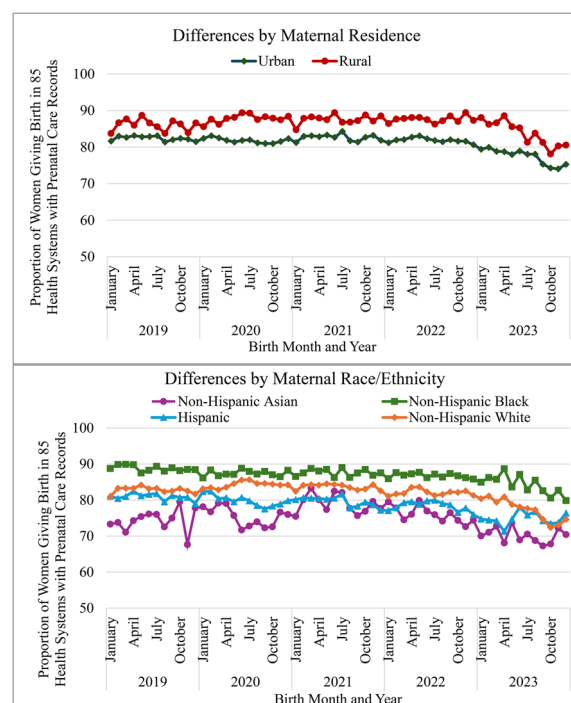


Fig. 1 (Abstract P18). Trends in the Proportion of Women Giving Birth in 85 Health Systems with Prenatal Care Records: Differences by Maternal Residence and Race/Ethnicity

P19

A big data approach to measure injection-related infections among people living with HIV in South Carolina

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BMC Proceedings 2025, 19(13):P19

Introduction: South Carolina remains heavily impacted by syndemics of HIV and injection drug use (IDU). An estimated 12% of new HIV infections are due to either IDU or IDU/MSM in South Carolina. Additionally, infective endocarditis has been on the rise among younger South Carolinians, a likely consequence of IDU. In addition to downstream harms like overdose and bloodborne pathogens (i.e. HCV co-infection), a variety of skin, soft tissue and vascular infections (SSTVIs) as well as more systemic infections (e.g. endocarditis, osteomyelitis) may follow exposures to IDU. These IDU-related infections (IRIs) can lead to severe clinical illness and hospitalization, especially among PWH who may be immunocompromised and/or have a high degree of co-morbidities. However, the epidemiology of IRIs among PWH have not been well-described in South Carolina, making it challenging to address drivers of risk and appropriately target interventions.

Study Objectives: This analysis seeks to describe the prevalence and correlates of injection-related infections (IRI) among PWH in South Carolina using health administrative data and the SC HIV Clinical Cohort since Jan 1, 2021. We hypothesize that rates of endocarditis,

osteomyelitis, sepsis/bacteremia, and SSTVIs will vary by socio-structural characteristics and HIV clinical care attributes.

Methods: Our study sample will include everyone from the SC HIV clinical cohort (regardless of HIV risk type [MSM, MSM/IDU, IDU, hetero, etc.]). First, we will construct 4 binary variables for the categories of IRI (infective endocarditis, osteomyelitis, SSTVIs, and bacteremia/sepsis) from the corresponding ICD codes (Table 1). Based on these, we will construct the primary outcome variable (any IRI) and calculate the total number of IRIs since Jan 1, 2021. Then, we will conduct bivariate logistic regression analysis to test the association between potential correlates (socio-structural characteristics and HIV care variables) and IRIs, using binary outcomes. Then, we will rerun the models using Poisson regression using IRI counts as the primary outcome. Finally, we'll conduct a sensitivity analysis with a sub-sample restricted to those identified as IDU, MSM/IDU by HIV surveillance data at the time of infection.

Results: We plan to report descriptive statistics and breakdown of the total number of IRIs by type.

Discussion: Our work seeks to shed light on significant, yet under-reported outcomes among a vulnerable population using a data science approach. Our findings will shed light on clinical outcomes and differential risk of IRIs among PWH in South Carolina and inform interventions to prevent drug-related harms. We included a wide variety of IRIs in an attempt to capture the breadth of downstream injuries and infections. A primary limitation is that in each individual case, we lack certainty that the IRI cause was directly related to IDU, however, the next steps in our work are to validate IDU exposures in the EMR using a Big Data approach. Conversely, it is likely that many IRIs took place and were managed outside of the hospital environment and never captured in electronic health records, leading to a potential underestimate in our findings.

Table 1 (Abstract P19). Descriptive results & bivariate associations with IDU-related Infections

Outcome	Variables and Source	Specifications	Description
Infective Endocarditis	ICD-9	1128	Candidal endocarditis
	ICD-9	421	Acute and subacute endocarditis
	ICD-9	424	Other diseases of the endocardium
Bacteremia/Sepsis	ICD-9	38	Septicemia
	ICD-9	4151	Pulmonary embolism and infarction
	ICD-9	4229	Septic myocarditis
	ICD-9	7855	Shock without mention of trauma
Osteomyelitis	ICD-9	7300	Acute osteomyelitis
	ICD-9	7301	Chronic osteomyelitis
	ICD-9	7302	Unspecified osteomyelitis
	ICD-9	7309	Unspecified infection of bone
Skin/Soft-tissue vascular infections (SSTVIs)	ICD-9	0400	Gas gangrene
	ICD-9	324	Intracranial and intraspinal abscess
	ICD-9	326	Late effects of intracranial abscess or pyogenic infection
	ICD-9	451	Phlebitis and thrombophlebitis
	ICD-9	5672	Other suppurative peritonitis (incl peritoneal abscess)
	ICD-9	5695	Abscess of intestine
	ICD-9	5720	Abscess of liver
	ICD-9	5901	Acute pyelonephritis

Outcome	Variables and Source	Specifications	Description
	ICD-9	681	Cellulitis and abscess of finger and toe
	ICD-9	682	Other cellulitis and abscess
	ICD-9	7071	Ulcer of lower limbs, except pressure ulcer
	ICD-9	7078	Chronic ulcer of other specified sites
	ICD-9	7079	Chronic ulcer of unspecified site
	ICD-9	7098	Other specified disorders of skin
	ICD-9	7236	Panniculitis specified as affecting neck
	ICD-9	72886	Necrotizing fasciitis
	ICD-9	7293	Panniculitis, unspecified site
	ICD-9	7854	Gangrene

P20

Computational modeling of 3D structure of the apolipoprotein ApoA-I

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Background: Investigation of apolipoproteins has the potential to discover groundbreaking new treatment methods for cancer, cardiovascular disease (CVD), and Alzheimer's disease (AD) [1,2,3]. Apolipoprotein type ApoA-I holds together high-density lipoproteins (HDLs), which are studied for their ability to decrease cholesterol level in the body and harmful side effects of administering toxic cancer drugs by targeted release to tumors [1,3]. HDLs usually have sizes ranging from 50–100 Å. Another class of apolipoproteins is ApoE, which has a variant (ApoE-2) that decreases risk of AD and a variant (ApoE-4) that increases risk of AD substantially [2].

Study Objective: To compile information about apolipoproteins using big data sourced from the RCSB Protein Data Bank (PDB), as well as to create fitted models of the apolipoproteins onto various 3D shapes that can be used to compare and evaluate them.

Methods: Structural data about the apolipoproteins' alpha carbon coordinates was extracted from PDB database. MATLAB was then used to fit these proteins to various 3D models that best described the shapes of these molecules. This data was also used to calculate various structural properties of the apolipoproteins, such as the end-to-end distance (Ete) and the average radius of gyration (R_g). The dihedral angles and the backbone angles were also calculated to construct a Ramachandran plot to determine the different secondary structural features in these proteins.

Results: It was observed that ApoA-I has a peak backbone angle of ~92° degrees and a peak backbone dihedral angle of ~52° which indicated the presence of α -helical structures (Fig. 1). The R_g of ApoA-I was approximately 38 Å which was twice that of ApoE-3 with an R_g of ~19 Å. ApoA-I was fitted into an ellipsoid and the R^2 score was close to unity (Fig. 2).

Discussion: While the average dihedral and backbone angles can be used to evaluate and compare the structures, the Ramachandran plot constructed with these angles were used to show that most of the apolipoproteins take an alpha helical secondary structure. The calculated R_g value of 38 Å explains the crucial role of ApoA-I in holding HDL particle sizes to a narrow size of 50–100 Å. Knowing that each type of apolipoprotein has a unique structure, computational modeling is potentially useful for evaluating and comparing their structures which can potentially shed light on their biological functions. R^2 values of the fitted models provided insight into the 3D shapes best representing these proteins.

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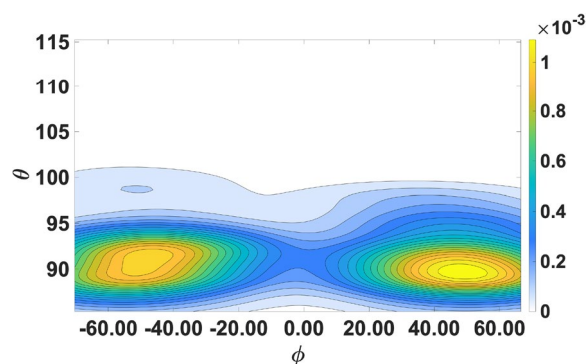


Fig. 1 (Abstract P20). Ramachandran plot of ApoA-I

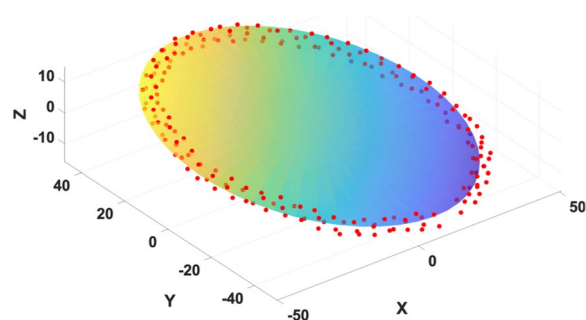


Fig. 2 (Abstract P20). The ellipsoid fitting of ApoA-I. The red markers represent the α -carbons in the backbone of ApoA-I

P21

Characterization of adults diagnosed with cancer in Haiti from January 2015 to September 2020

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Introduction: National level data related to cancer in Haiti are limited, mostly constituted by estimations derived from modelling other countries' data [1,2]. This limits Haiti's abilities to evaluate the actual cancer burden, the exact resources needed, and to implement the most appropriate prevention and control measures.

Study Objectives: This study aims to characterize adults diagnosed with cancer from January 2015 to September 2020 in health facilities with cancer diagnosis and/or treatment capabilities in Haiti.

Methods: The dataset used in this cross-sectional analysis emanates from of a project to improve cancer surveillance in Haiti, collecting paper and electronic medical record data of patients with cancer attending one of the nine main cancer centers (five hospitals, two largest private clinic, one institute, one society) from January 2015 to September 2020. Cancer cases either had a level 1 confirmation (clinical suspicion confirmed by imaging/laboratory tests) or a level 2 confirmation (histopathology) [3]. Means and percentages were used to describe demographic and clinical characteristics, and chi squared test to assess differences between women and men via Epi Info 7.

Results During the period, 4,086 adults were newly diagnosed with cancer, showing an increasing trend from 488 in 2015 to 1,011 in 2018 followed by a decrease until 2020. The majority were female (84.21%) and aged 41–60 years (49.54%) at diagnosis. Mean age at diagnosis was 52.26 ± 14.21 years overall, 52.37 ± 13.6 for females and 51.70 ± 17.04 for males. Regardless of sex, the most common cancer sites were breast (41.41%), cervix (21.11%), colon (4.87%), ovary (4.38%) and stomach (3.13%). Among females, the most frequent cancer sites were breast (48.37%), cervix (26.10%), ovary (5.20%), uterus (3.11%) and colon (2.85%). Among males, the most common cancer sites were colon (15.68%), prostate (12.27%), blood (10.71%), stomach (8.85%) and neck (6.99%). Overall, 36.98% and 37.81% of the patients were diagnosed at stage 3 and 4 respectively, 38.13% and 35.69% for female compared to 27.08% and 55.83% for male, $p < 0.0001$. Cancer diagnosis was confirmed by histopathology for 78.47% of patients, 79.25% of female vs. 74.28% of males, $p = 0.006$. Presence of symptoms was the paramount circumstance of diagnosis overall (92.20%) and across sexes (92.48% for females and 90.72% for males, $p = 0.13$).

Discussion: Trend of cancer cases in adults were increasing in the main cancer centers in Haiti, numbers in 2019 and 2020 were affected by political violence and COVID-19 lockdowns which reduced hospital attendance [4]. The most common cancers for males and females have highly effective prevention and screening methods [5]; and given that more than 90% of patients were diagnosed because of symptoms, it is of utmost importance to upgrade prevention and screening interventions at a large scale. More consideration should be given to blood and neck cancers which are not generally among the top sites in estimations available for Haiti [1,2]. An important proportion of participant were diagnosed at an advanced stage, even higher for males, showing that it is essential to improve awareness and availability of services for the population, and tailor messages to perceptions and attitudes of sociodemographic groups.

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P22

Healthcare resource allocation using Deep Reinforcement Learning to improve spatial accessibility equityShiqi Wang¹, Hanqi Li², Fahui Wang², Bing Zhou³, Yuhao Kang⁴¹School of GeoSciences, University of Edinburgh, Edinburgh, Scotland;²Department of Geography & Anthropology, Louisiana State University, Baton Rouge, Louisiana, USA; ³Department of Geography, Pennsylvania State University, Pennsylvania, USA; ⁴Department of Geography and the Environment, The University of Texas at Austin, Austin, Texas, USA**Correspondence:** Yuhao Kang (yuhao.kang@austin.utexas.edu)*BMC Proceedings 2025, 19(13):P22*

Introduction: Healthcare resource allocation often exhibits significant regional disparities that exacerbating health inequities. Traditional solutions such as Quadratic Programming (QP) have limitations in handling dynamic and multi-objective problems. We designed and implemented a Reinforcement Learning (RL) approach that offers a promising avenue by dynamically optimizing healthcare resource distribution to improve equality of spatial accessibility and adaptation to real-world complexities.

Study Objectives: This study aims to develop an RL-based framework for healthcare resource allocation by proving its feasibility in solving Maximum Accessibility Equity Problem (MAEP), accommodating diverse stakeholder objectives, and demonstrating adaptability to dynamic real-world scenarios, ultimately reducing spatial accessibility disparities and promoting equitable resource distribution.

Methods: This study utilized publicly available datasets to analyze healthcare resource optimization in two contexts: healthcare facility data from Xiantao, China, and hospital bed capacity data from Chicago, USA. A dynamic environment was developed to model the interaction between supply and demand, using the Proximal Policy Optimization (PPO) algorithm for resource reallocation. The state space represented resource distributions and population needs, while actions involved adjusting allocations to optimize accessibility. The reward function prioritized minimizing population-weighted accessibility variance while enhancing average accessibility in designated regions.

Four experimental scenarios were designed: unrestricted resource transfers, maintaining minimum resources at origin facilities, limiting transfer distances, and prioritizing average accessibility improvement in predominantly Black neighborhoods in Chicago. Evaluation metrics included accessibility variance (equality) and average accessibility in special regions. Comparisons with Quadratic Programming (QP) under identical constraints (policy, cost, and budget) validated RL's adaptability and effectiveness in dynamic, multi-objective scenarios, particularly in addressing spatial equity challenges.

Results: The experiments revealed that RL reduced accessibility variance to 20%–80% of pre-optimization levels, improving spatial equality. In underserved areas, RL increased accessibility in special regions to 140% of pre-optimization values, addressing disparities. With added constraints, such as transfer distance limits or prioritization of vulnerable groups, RL maintained equal outcomes. These results show that RL reduces accessibility variance and adapts well to multi-objective optimization. While RL may not outperform QP in single-objective tasks, it delivers stable and equitable performance across diverse experimental conditions, particularly when integrating additional objectives.

Discussion: This study demonstrates the potential of RL for healthcare resource optimization, particularly in improving spatial accessibility equity. RL reduced accessibility variance by up to 80% and increased special region accessibility by 40%, showing significant improvement compared to pre-optimization levels. While RL did not outperform QP in single-objective optimization, it proved stable and effective across various scenarios when incorporating multiple objectives. Its adaptability to real-world constraints, such as distance and policy limitations, and its scalability make it a promising approach to addressing health disparities. Additionally, the step-by-step transfer guidance in our RL design provides a clear and practical strategy for resource allocation. Looking ahead, we plan to scale up the model to tackle larger, more complex optimization problems, which will further enhance its capabilities and enable more powerful solutions to real-world healthcare challenges. This approach offers a valuable foundation for

future advancements in healthcare systems and equitable resource distribution.

P23

Demographic and socioeconomic risk factors for HIV status: analysis of the Tanzania HIV Impact Survey

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Introduction: Tanzania faces ongoing challenges in HIV prevention despite substantial testing efforts [1,2]. While risk assessment tools have shown promise in neighboring countries [3], their transferability requires validation. This study evaluates a Kenyan HIV risk scoring algorithm in Tanzania and identifies Tanzania-specific risk factors.

Study Objectives: To evaluate the applicability of a Kenyan HIV risk scoring system in Tanzania's context and identify demographic and behavioral predictors of HIV status. To provide evidence-based recommendations for developing Tanzania-specific risk assessment tools and targeted HIV testing strategies that reflect local socio-demographic and epidemiological patterns.

Methods: We analyzed data from the Tanzania HIV Impact Survey (THIS) 2016–2017** (n=24,365). The study involved two analyses. First, the Kenyan risk scoring algorithm [3] was applied to the THIS sample. The algorithm assigns weighted points across six components: ages 35–44 (8 points), primary education or less (7 points), marital status (6–17 points), multiple partners in the past 12 months (5 points), STI symptoms in the past 12 months (11 points), no prior HIV test (18 points) or prior test > 12 months ago (22 points). Cumulative scores were used to classify participants into Low, Medium, High, and Very High-risk groups, and rates of HIV infection across risk groups were compared to those in Kenyan.

Second, a survey-weighted logistic regression model examined demographic and behavioral correlates of participants' HIV status. Correlates included age categories, gender, urban/rural residence, wealth quintiles, education, multiple partners and STI symptoms in the past 12 months, marital status, and HIV testing history. To account for the complex survey design, analyses employed jackknife replication with 257 replicates.

Results: The Kenyan tool showed moderate predictive value in Tanzania: Odds of HIV infection increased across risk categories (Medium Risk: aOR=1.41, CI:1.02–1.95; High Risk: aOR=1.90, CI:1.35–2.67; Very High Risk: aOR=3.31, CI:2.43–4.50), though the variation across risk categories was substantially lower in Tanzania than in Kenya (aOR>15 for highest risk) [3].

Extended analysis revealed strong age effects, peaking at ages 45–49 (aOR=8.85). Women showed higher odds (aOR=1.43, CI:1.23–1.66), while rural residence was protective (aOR=0.57, CI:0.46–0.72). Marital status strongly predicted HIV status, with highest risks among widowed (aOR=3.32), separated (aOR=2.41), and divorced (aOR=2.32). Multiple partners (aOR=1.44), STI symptoms (aOR=1.83), and post-secondary education (aOR=0.11, CI:0.07–0.19) were significant predictors, while highest wealth quintile showed protective effects (aOR=0.69, CI:0.48–1.00).

Conclusion: Our findings reveal the potential and limitations of applying existing risk assessment tools across contexts. While the Kenyan tool maintains statistical significance in Tanzania, its reduced predictive power highlights the need for country-specific adaptations. The analysis identifies Tanzania-specific predictors that could form the foundation for a more locally relevant risk assessment tool. Particularly striking are distinct patterns related to age and marital status, and urban-rural differences. These insights provide an evidence-based framework for developing targeted HIV testing strategies and a Tanzania-specific risk-scoring system [4]. The strong associations identified could improve testing efficiency while maintaining sensitivity to local contexts. Limitations include the cross-sectional study design and potential reporting bias. Future research should focus on developing

and validating a Tanzania-specific risk scoring system incorporating these identified predictors.

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P24

Factors associated with contraceptive use among adolescent girls and young women in Lebowaqomo, Limpopo Province, South Africa

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Introduction: The use of and access to contraceptives among adolescent girls and young women (AGYW) in South Africa remains an ongoing multifactorial, social, and public health issue that needs urgent attention [1]. This accounts for the high prevalence of unplanned pregnancy, HIV/STI, unsafe abortions, and feto-maternal complications [2]. **Study objectives:** To determine the factors associated with the utilisation of contraceptives among AGYW in Lebowaqomo, Limpopo Province

Methods: An analytical, cross-sectional design was used to determine the factors associated with the utilisation of contraceptives among AGYW. A sample of 355 AGYW aged 18–24 years was considered. A semi-structured questionnaire was used to collect information on the socio-demographic, sexual, and reproductive health (SRH), and structural and behavioural factors towards contraceptives. Data were analysed using STATA version 18 (College Station, TX, USA) software.

Results: Out of 355 AGYW, 97.50% reported using contraceptives. The crude analysis showed a significant association among students ($p=0.005$; aRR=0.97, 95% CI 0.94–0.99). With regards to SRH history, age sexual of debut between 15–24 years ($p=0.020$; aRR=0.96 CI: 0.93–0.99), awareness of HIV status ($p=0.033$; aRR: 1.03, 95% CI: 1.00–1.07), those with two or more children ($p=0.011$; aRR=0.97, 95% CI 0.94–0.99) and those with one child ($p=0.013$; aRR=0.91, 95% CI: 0.93–0.98), pregnancy planning ($p=0.004$; aRR=1.06; 95% CI: 1.02–1.10) were associated with contraceptive use. There was no significant association between structural factors and contraceptive use ($p>0.05$). There was a significant association between behavioural factors and contraceptive use ($p<0.05$).

Discussion: Despite the high prevalence rate, there are gaps in SRH education and services [3]. The findings highlight the need to prioritise those with significant risk to reduce the rate of unplanned pregnancy, HIV/STI, obstetric complications, and school dropout. A multi-sectoral and transdisciplinary approach will assist in addressing contraceptive needs [4] and subsequently contribute to the public health discourse of improving SRH outcomes among AGYW [5]. This approach will promote access to SRH services and subsequently enhance pre-existing policies, standard operating procedures, and the formulation of legislation [6].

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P25

A unified EHR-based data integration approach for perinatal health policy research

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Background: The integration of maternal labor and delivery records and newborn birth certificate data is a cornerstone of perinatal health research. However, such linkages are often complex and problematic within administrative datasets. Despite the importance of a robust matching algorithm, a standardized algorithm to integrate these datasets and formalize a unified maternal-birth dyad (matched) cohort has been lacking. To fill the research gaps, we leveraged statewide population-based data and developed a unified EHR-based data integration approach designed for merging multiple EHR datasets from different sources and demonstrating a process for maternal-birth cohort construction.

Methods: Our study used two South Carolina (SC) statewide databases from January 2018–December 2022 (Fig. 1): 1) All-Payer Hospital-based Uniform Billing (UB) Database, including all hospital-based outpatient, emergency department, and inpatient records, and 2) SC Vital Statistics Records (VR) for Birth Certificates Data. Maternal-birth cohort construction involved two phases; first, hospital-based maternal childbirth hospitalization information was identified in the UB dataset using diverse sources of medical coding: International Classification of Diseases, Tenth Revision (ICD-10) diagnosis and procedure codes, Current Procedural Terminology (CPT) codes, and Diagnosis-Related Group delivery (DRG) codes. Second, we conducted a unified data integration by merging the hospital-based maternal childbirth records to the VR dataset for other birth and infant information unavailable from the hospital UB data. Data harmonization and synchronous validations were conducted to ensure accurate maternal-birth linkage and identification of unique maternal-birth records.

Results: Among the 1,117,818 hospital records for 206,337 pregnant individuals in the UB dataset, medical coding identified 247,402

childbirth records, of which 195,287 were unique childbirths or pregnancies for 169,461 birthing individuals. The second phase involved aggregating multiple birth records into single pregnancy records in the VR dataset, and then matched to maternal childbirth hospitalization record using maternal identifier, birthing month/year, and gestational age at birth. This phase identified additional 4,124 births to 2,824 birthing individuals in VR dataset only but not in the UB dataset. Using this matching algorithm, 99.2% of maternal childbirth hospitalization records were linked to the VR data for more birth and infant characteristics. However, many data management and harmonization issues arise in identifying unique childbirths in hospital and non-hospital settings and merging maternal childbirth records to birth certificate data. This study addressed the emerging issues in constructing such integrated dataset with maternal-birth dyad information and recent attempts to remedy these data harmonization issues.

Conclusions: A unified data integration approach from attributing multiple hospitalization records to each unique childbirth and from matching maternal childbirth discharges to newborn birth certificates data could leverage EHR database from multiple sources. The delivery information extraction considers a variety of medical codes comprehensively. The use of new identification index as key variable in the data merging step improves the efficiency of data integration and avoids duplicate records in the final cohort. The proposed approach is meaningful for perinatal health policy research to enable efficient cohort construction under flexible inclusion-exclusion criteria.

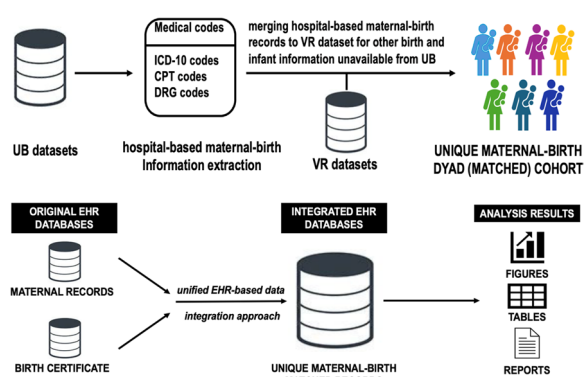


Fig. 1 (Abstract P25). Study Approach on Linkage of All-Payer Hospital-based Uniform Billing (UB) Datasets and Vital Statistics Records (VR) for Birth Certificates

P26

ML solutions for land cover classification by using hyperspectral images

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Abstract: Land Cover classification is important for tracking how land is used and changes over time. This is essential for the environment, city planning, and farming. Hyperspectral images (HSI) are especially useful because they capture more information than regular photos, giving us a lot more detail. This helps us better identify different types of land and materials, like plants, water, and buildings. Studies such as those by Ballabio et al. [1] highlight the effectiveness of Support Vector Machines (SVM) in identifying land cover types in complex terrains. This not only saves time and resources but also offers a better understanding of land use and environmental changes.

Introduction: Hyperspectral imaging (HSI) collects information across the electromagnetic spectrum, providing a detailed view of materials and image features. Unlike traditional cameras that capture images in

three color channels (red, green, and blue), hyperspectral sensors capture data in hundreds of spectral bands, allowing for the recognition of subtle differences in materials based on their spectral signatures.

One crucial application of HSI is land cover classification, which is essential for identifying and mapping different types of land cover. This includes assessing agricultural areas for soil types and crop suitability, as well as analyzing urban areas for city infrastructure planning and zoning regulations.

To enhance the accuracy and efficiency of land cover classification, machine learning (ML) models are widely used. ML models are particularly valuable in remote sensing due to their ability to process large and complex hyperspectral datasets, making them a powerful tool for extracting meaningful patterns and improving classification accuracy.

Study Objectives: The goal of this study is to leverage machine learning (ML) techniques to improve land cover classification using hyperspectral images. We aim to compare different ML models to determine which one performs best in identifying land cover types across three datasets: Pavia University (Italy), Salinas Valley (California), and Indian Pines (Indiana). By evaluating precision, recall, f1-score, and accuracy, we seek to identify the strengths and weaknesses of each approach and their suitability for different types of terrain and land cover.

Methodology: We trained and tested three machine learning models to classify different types of land cover. Logistic Regression (LR) helps sort pixels into land cover categories based on their spectral features. Support Vector Machine (SVM) uses special math functions (kernels) to separate data in a way that makes it easier to classify things like different types of plants, even when there is no straight boundary. Multi-layer Perceptron (MLP) is a deep learning model that uses activation functions to learn complex patterns from the data.

Results: The classification results varied across the three datasets. Pavia University demonstrated strong performance across all models, with MLP achieving the highest accuracy at 92%, followed by SVM at 89% and LR at 85%. Similarly, Salinas Valley, showed high accuracy, with MLP reaching 93%, SVM at 90%, and LR at 89%. However, Indian Pines, posed a greater challenge, with MLP achieving 79%, while SVM and LR struggled at 70% and 66%, respectively. This indicates that dataset complexity and spectral overlap significantly impact model performance.

Some observations include the superior performance of MLP, which consistently outperformed traditional models due to its ability to capture complex spectral patterns. SVM performed well on simpler datasets like Pavia and Salinas but struggled with Indian Pines, which contain more mixed land cover types. Logistic Regression had the lowest accuracy in all datasets, suggesting it may not effectively capture intricate hyperspectral relationships. Future improvements could include feature selection, deep convolutional networks, and hybrid ML approaches to enhance classification accuracy.

Discussion: This study confirms that Multilayer Perceptron (MLP) is the most effective model for land cover classification using hyperspectral images. MLP's deep learning structure allows it to recognize complex spectral patterns better than traditional models. However, Indian Pines proved to be the most challenging dataset, possibly due to the presence of mixed land cover types and small sample sizes for some categories.

Compared to Pavia and Salinas, where ML models performed well, Indian Pines resulted in lower accuracy, suggesting that datasets with high spectral overlap require more advanced techniques such as deep convolutional networks or additional preprocessing. Future research could explore methods like data augmentation, feature extraction, or hybrid ML models to enhance performance.

Keywords

Hyperspectral Imaging, Land Cover Classification, Remote Sensing, Machine Learning

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P27

Semaglutide: oral vs. injection and the risk of colorectal cancer using large-scale national dataJoseph Magagnoli^{1,2}, Tammy H. Cummings^{1,2}, S. Scott Sutton^{1,2}¹Department of Clinical Pharmacy and Outcomes Sciences, College of Pharmacy, University of South Carolina, Columbia, South Carolina, USA; ²Dorn Research Institute, Columbia VA Health Care System, Columbia, South Carolina, USA**Correspondence:** Joseph Magagnoli (magagnol@mailbox.sc.edu)
BMC Proceedings 2025, 19(13):P27

Introduction: Colorectal cancer (CRC) is a significant public health challenge. Leveraging large-scale data analytics, we assessed the comparative risk of CRC between oral and injectable semaglutide formulations, widely used GLP-1 receptor agonists for diabetes and obesity management. The study aimed to address critical gaps in understanding CRC risk in diverse patient populations.

Study Objectives: To utilize national-level Veterans Affairs Informatics and Computing Infrastructure (VINCI) data to evaluate the association between oral and injectable semaglutide use and CRC risk, providing insights to inform clinical decision-making and public health strategies. We hypothesized that differences in formulation could influence CRC risk due to variations in pharmacokinetics and patient characteristics.

Methods: This study analyzed data from 268,142 new-users of either injectable (n=267,258) or oral semaglutide (n=884). Patient demographics, comorbidities, and treatment characteristics were assessed using descriptive statistics and standardized differences. Kaplan-Meier survival analysis and multivariable Cox proportional hazards models were employed to examine CRC risk, adjusting for age, sex, comorbidities, and other potential confounders.

Results: Oral semaglutide users were older (66.93 vs. 64.09 years) and had a lower prevalence of BMI ≥ 30 (67% vs. 75.3%) compared to injectable users. Adjusted Cox regression (Table 1) revealed a significantly increased hazard of CRC for oral vs. injectable users (HR 3.013, 95% CI 1.25–7.28) and male sex (HR 1.666, 95% CI 1.13–2.46).

Discussion: This study highlights the power of large-scale data to address public health and biomedical research questions. The increased CRC risk associated with oral semaglutide underscores the need for further investigation into pharmacologic and patient-specific factors. These findings provide critical insights for clinicians and policymakers aiming to optimize semaglutide use while mitigating potential risks, emphasizing the role of big data in advancing precision medicine and public health.

Table 1 (Abstract P27). Hazard of Colorectal cancer: Oral(tablet) versus injection semaglutide

	HR(95%CI)
Tablet vs. injection	3.013(1.25–7.28)
Age	1.018(1.01–1.02)
Race other vs Black	0.795(0.56–1.13)
White vs. Black	0.89(0.72–1.11)
Male	1.666(1.13–2.46)
Charlson comorbidity	1.073(1.03–1.12)
Alcohol dependence	0.719(0.38–1.35)
BMI 25–29.9 vs. 18.5–24.9	0.923(0.62–1.37)
30 + vs 18.5–24.9	0.986(0.68–1.42)
Pure hypercholesterolemia	0.676(0.37–1.23)
Hypertriglyceridemia	0.873(0.71–1.07)
Hyperlipidemia	0.906(0.76–1.08)
Ischemic heart disease	1.011(0.84–1.22)
Other heart disease	0.959(0.75–1.23)
Hypertension	1.095(0.86–1.39)
Smoker	0.993(0.74–1.33)
Year	1.068(0.98–1.16)

P28

Neurocardio-trauma feedback system, An investigational study of electroencephalography and heart rate variability monitoring in emergency care clinicians for the early detection of acute stress disorderMacKenzie Meier¹, Chris Kibler¹, Hailey Sparks², Michael Campanelli³, Chris Koons³, Brandon Williams⁴, Ronald Pirallo⁵, Nicholas Boltin¹¹Department of Biomedical Engineering, University of South Carolina,Columbia, South Carolina, USA; ²Department of Computer Science,College of Charleston, Charleston, South Carolina, USA; ³University

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BMC Proceedings 2025, 19(13):P28

Introduction/Background: Acute Stress Disorder (ASD) is a psychological response to a terrifying, traumatic, or surprising event and can be a precursor to Post Traumatic Stress Disorder (PTSD) if left untreated. The current method of diagnosing ASD is anecdotal in practice, with clinical evaluation involving patient history and physical examination within days following a traumatic event. Diagnostic tools typically include screenings and questionnaires, but they require significant training and often miss initial physiological symptoms.

Objectives: The objectives of this study include designing and testing a portable EEG device that collects biopotential brainwave and heart rate activity in addition to the development of an Artificial Intelligence (AI) data-driven decision support prototype to analyze biomarkers associated with Acute Stress Disorder.

Methods: To obtain these objectives, participants from Greenville Memorial Hospital Emergency Department and Level 1 Trauma Center wore an EEG headset prior to their work shift to determine a reference signal (preliminary sample size of ~n=20). Participants then wore the EEG headset during their work shift while the timing of traumatic events and severity was recorded and biopotential data was collected and stored. De-identified data was exported and processed into various waveforms before being fed into an AI model to develop parameters to identify increased stress levels. The Acute Stress Center (ASC) has developed an application utilizing the Python “Shiny” package which will allow physicians evaluating EEG data to quickly deploy a clinical decision support system to evaluate their data. The application enables physicians to drag and drop EEG data into the application, select which waveforms are of interest to their research, and view summary statistics of their data and graphs indicating stressful events along their EEG waveform.

Results: The data received from this application includes a breakdown of the neurological responses by waveform. These include delta, theta, alpha low, alpha high, beta low, beta high, gamma low, and gamma mid waveforms. The AI model calculated individual thresholds by finding the average baseline value for the first ten minutes of data and adding one standard deviation. It then highlighted parts of the waveform indicating a stressful response in black for providers using the app to easily understand when stressful responses occurred and what happened neurologically inside the body during these events [Fig. 1]. The application also produces data statistics of each waveform including the mean, standard deviation, minimum, maximum, and quartiles of each waveform. Feature importance is also assessed and graphed to help providers understand which waveforms are the best indicators of a traumatic event [Fig. 2]. Lastly, event intensity is assessed and classifies the percent of instances as non, low, medium, and highly traumatic events.

Discussion: Overall, the application shows a promising future with helping emergency care clinicians understand the neurological response that occurs during a stressful event. The application depicts a graphical representation of waveforms for provider understanding in addition to statistics that further support and clarify variability. Additionally, the drag and drop feature allows for researchers and providers to receive a graphical representation of their own neurological data quickly. Future work on the application that is currently being studied

includes heart rate variability being evaluated for traumatic and non-traumatic events. This could help provide users with additional biopotential data to further understand stress responses during a workshop.

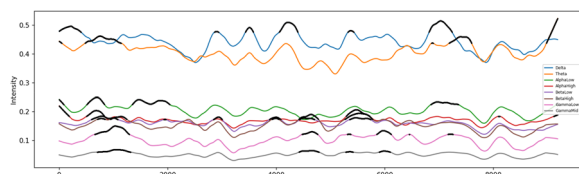


Fig. 1 (Abstract P28). Waveform analysis. This evaluation displays user data in terms of intensity vs time and separates data by waveforms. Black highlighted regions indicate a stressful event or a threshold being crossed per waveform

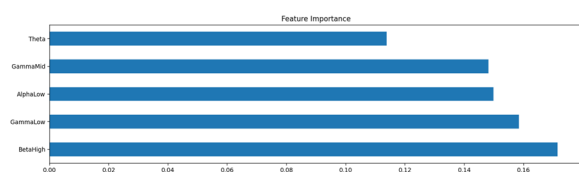


Fig. 2 (Abstract P28). Feature Importance Scale. This evaluation of user data classifies beta high waveforms as the most important predictor of a stressful event and theta as the least

P29

Leveraging machine learning algorithm to predict STI prevalence, transmission dynamics and trends in SA adults using SABSSM data: a study protocol

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Background: Sexually transmitted infections (STIs) are a significant public health concern in South Africa. South Africa has one of the highest STI burdens, with a notable impact on sexual health and overall well-being. The key STIs in the South Africa's National Strategic Plan (NSP) 2023–2028 include gonorrhoea, chlamydia, and syphilis. Compared to traditional statistical models, machine learning (ML) approaches can improve the accuracy of STI prediction.

Objective: To utilize machine learning algorithms to predict STI prevalence, transmission dynamics, and trends among South African adults.

Methods: A retrospective analysis of five cycles of population-based data from the South African National HIV Prevalence, Incidence, Behaviour, and Communication Survey (SABSSM), (2002–2017) will be conducted. The Figure below illustrates the study methodology and the use of machine learning algorithms to predict STI prevalence, transmission dynamics, and trends.

Expected Results: Integrating machine learning algorithms with SABSSM data offers a promising approach to predicting STI trends and understanding transmission dynamics in South Africa. By leveraging these advanced analytical techniques, this study aims to provide valuable insights to support public health efforts, enhance resource allocation, and guide policy development to address STI challenges effectively. The study may face limitations such as data quality issues and generalizability of ML models. The study anticipates providing a deeper understanding of STI patterns and more accurate predictions to inform public health interventions. It will provide critical insights that will support public health efforts, optimize resource

allocation, and inform policy development to tackle STI challenges more effectively.

Keywords: Machine Learning, STI Prevalence, South Africa, Transmission Dynamics, South Africa's National Strategic Plan, SABSSM

P30

Effectiveness of a digital tool in improving quality assurance of rapid HIV test outcomes in Gauteng, South Africa: evidence from the Rapid Test Continuous Quality Improvement Program

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Introduction Conventional training of quality assurance (QA) of rapid HIV testing is time consuming and difficult to implement at scale causing variability in the quality of interventions [1]. To standardize training and implementation, a digital tool such as an eLearning Management System (eLMS) will allow efficiency at greater scale and lesser cost. The use of digital health tools to collect and analyze rapid HIV testing data has become more critical in enhancing quality and reliability of test results.

Study Objective To determine pre and post changes of quality assurance outcomes between primary health care facilities and health care workers that were exposed to digital technologies and non-exposed facilities.

Methods A retrospective, quasi-experimental design will be used to assess the effectiveness of the eLearning Management System (eLMS) digital tool (Fig. 1). A total of 2526 health care workers (HCWs) who received training on QA of Rapid HIV testing administered as either, eLMS and conventional QA training will be randomized into intervention (n=782) and control groups (n=1744) respectively. Stepwise Process for Improving the Quality of Rapid Test (SPIRT) checklist was used to assess a total of 1609 primary health care facilities in Gauteng and data across five cycles (2019, 2020, 2021, 2022 and 2023) will be analyzed. The SPIRT assessments are conducted annually to monitor the progress of QA of rapid HIV testing [2]. Quantitative data will be analyzed using STATA version 18.0 (College Station, TX, USA).

Expected Results The findings of this study will identify the variables that affect the effectiveness of eLMS as a digital tool to improve QA of rapid HIV testing in PHC facilities. Furthermore, a framework for strengthening QA of rapid HIV testing in a PHC setting will be developed that will allow the government to deliver efficient and focused QA interventions at greater scale and lesser cost.

Discussions According to WHO, digital health approaches have enabled active HIV quality assurance program management. It has the potential to be used for other Point of care (POC) testing services across the continent [3]. Considering the critical role played by HIV rapid tests in improving access to HIV care and the foreseen challenges in resource-limited settings, maintaining accurate and reliable test results to ensure sustainability of quality service delivery remains crucial. This study evaluates the effectiveness of digital technologies in improving the quality of rapid HIV testing in health care facilities in South Africa in support of the National Strategic Plan (NSP) 2023–2028 [4]. The study will however use retrospective data which is used for programmatic purposes, and this pose limitations because the researcher will be restricted to the usage of the available data. Additionally, detection bias cannot be avoided because those who collected and analyzed the data will not be blinded to which group received the intervention because this is a retrospective quasi-experiment design study.

Keywords: Quality Assurance, Rapid HIV Tests, Training, digital tools, eLearning Management System.

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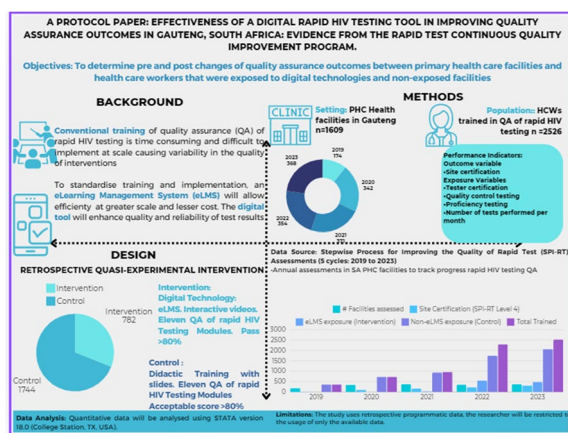


Fig. 1 (Abstract P30). Infographic summary of the abstract

P31

Differentiated service delivery model in improving HIV treatment outcomes among female sex workers in Gauteng Province, during the Covid-19 pandemic – study protocol

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Background: The Differentiated Service Delivery (DSD) model for HIV care was expanded during COVID-19 to improve access to antiretroviral therapy (ART) for people living with HIV (PLHIV), especially those facing barriers accessing healthcare [1, 2]. Female sex workers (FSWs), vulnerable to HIV, experience risks from ART interruptions [3]. The impact of DSD on FSWS, particularly in Gauteng, remains under-studied.

Aim: To assess HIV treatment outcomes following implementation of the differentiated service delivery model among FSWS in Gauteng.

Methods: The study will use secondary data from Agape Lerato, a community-based organization, covering ART initiation, adherence, viral suppression, and retention in care among FSW living with HIV. The study will use descriptive and inferential analysis to assess service accessibility, patient outcomes, and resource use [4]. It focuses on 4–6 sites in the Sedibeng district, affiliated with the Provincial Department

of Health (PDoH). The study will assess HIV treatment outcomes for FSWS, evaluating ART initiation, linkage to care, viral suppression, and adherence, while exploring how the DSD model mitigated care disruptions during the pandemic.

Discussion: The study will provide insights into the DSD model's role in improving ART adherence and viral suppression among FSWS living with HIV, informing future healthcare strategies. The DSD model may improve ART access by reducing clinic visits and offering alternative care. Evaluating ART adherence in FSW living with HIV is essential for improving their healthcare needs. Accessible, patient-centered care models are key to improving HIV outcomes.

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P32

An empirical study on the use of Large Language Models (LLMs) to translate recipes into a semi-structured data format

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Introduction: Given the importance of food for human health, recipe data, containing information about food items, ingredients, nutrients, cooking steps, and preparation time, is crucial for public health. A semi-structured multimodal representation is especially valuable for enabling machines to understand and reason about recipes, addressing the limitations of ambiguous unstructured formats. Pallagani et al. [1] introduced R3, a *Rich Recipe Representation* that incorporates multimodal data to represent recipes and meal planning data into a structured representation, which existing systems lack. R3 bridges this gap by efficiently representing information, extracting details such as macronutrients and allergens, and explaining item and meal recommendations to users [7, 8]. Because converting plain and raw text recipes into R3 is a time-intensive, tedious process, we leverage the context-aware capabilities of Large Language Models (LLMs) [5, 6, 9, 10] and use them to streamline the conversion process. This study evaluates their potential to automate recipe conversion and reduce manual effort.

Study Objectives: This study evaluates LLMs' effectiveness in automating conversion of plain text recipes into R3, thereby reducing the time and complexity needed for manual processing.

Methods: We compare three recipe conversion methods: fully manual (RC0), hybrid (RC1), and fully automated (RC2).

RC0: Following Pallagani et al.'s work [1], we use their set of 25 R3 recipes to evaluate manual conversion effectiveness.

RC1: Guided by Nagpal et al. [2], it uses ChatGPT-3.5 to generate portions of R3 (e.g., individual steps, ingredient lists). We evaluate this approach for 27 recipes.

RC2: This method provides the LLM with the expected R3 schema, an example, and a plain text entire recipe (target) to generate the full R3 representation. This is tested on 52 recipes that were previously converted.

Results: Table 1 presents our results using four metrics to evaluate R3 conversion: semantic similarity, syntactic similarity, perplexity, and JSON error count.

Semantic Similarity: Assesses if semantic meaning is preserved in the R3 s by comparing embedding vectors for original recipe text and R3 JSON strings, and calculating BERT scores [3].

Syntactic Similarity: Evaluates R3's structure by analyzing JSON keys and nesting levels of each key, and compares them with the curated set of 25 recipes from *RCO*.

JSON Decode Error Count: Evaluates LLM-generated JSON errors such as missing or extra characters like quotes, brackets, and commas.

Results show that *RCO* achieves the highest semantic accuracy but a lower syntactic score due to fewer JSON keys. *RC1* introduced extra JSON keys in the R3 s, thereby inflating syntactic scores. *RC2* proved infeasible as LLMs frequently produced invalid JSONs with multiple errors.

Discussion: Our findings suggest that while LLMs show promise, they require manual intervention, rule-based support, or fine-tuning for domain-specific tasks to effectively translate recipes into R3. Pre-trained LLMs alone are insufficient for such accurate conversion.

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Table 1 (Abstract P32). A comparison of the three R3 translation methods. Best performing methods for each metric are bolded

Config	JSON Error Counts	Syntactic Similarity	Semantic Similarity
<i>RCO</i>	0	0.584	0.763
<i>RC1</i>	0	0.778	0.733
<i>RC2</i>	26	0.376	0.758

P33

Examining the contributing factors to suicide among the black male population

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Background: Black men face a wide range of discrimination imposed by society, resulting in detrimental effects on the black community. The National Alliance on Mental Illness (NAMI, 2023) data demonstrates that suicidality among young Black or African American men aged 15–24 ranks as the third leading cause of death. This research explores the stigma surrounding mental health, social influences, and the availability of treatment, all of which contribute to the occurrence of suicide among Black men. The study examines existing research on factors that predict mental health issues and obstacles that hinder Black men from receiving necessary care. This study aims to diminish the stigma associated with mental health, increase awareness, and provide support tailored to the unique needs of Black men. By addressing the stigma, social factors, and accessibility to treatment, we can reduce disparities and foster a welcoming and empathetic community that promotes positive mental health outcomes.

Brief Project Description: This project looks at the challenges Black men face that negatively affect their mental health. One major challenge is the lack of community support and understanding, which makes it hard to talk openly about mental health. Another issue is the limited access to fair and affordable healthcare, as Black men are often left out of mental health programs and policies. These problems create a cycle where many Black men don't get the help they need to improve their well-being.

The findings show how important it is to address these challenges, especially since Black men have some of the highest suicide rates. This project calls for programs and policies to be implemented to better support Black men and address their mental health needs. By tackling the challenges they face, such as a lack of community support and limited access to fair healthcare, these initiatives can ensure Black men receive the help they need to improve their well-being.

Lessons Learned: After completing this study, I became more aware of the many health disparities marginalized communities face, not just in mental health but also in physical health. It highlighted how systemic barriers affect access to proper care and resources, leaving these communities at a disadvantage. One significant lesson I learned is the lack of representation, especially for men, in research. This underrepresentation has a direct impact on their health outcomes. If men are not studied and their specific needs are not understood, specialists cannot determine the best ways to support them.

This gap in research reinforces existing disparities, making it even harder for men, particularly Black men, to receive the care they need. It shows me how important spreading awareness is so that health professionals understand the challenges that need to be addressed. By increasing awareness, we can ensure that programs and policies are better informed and focused on creating solutions that consider the unique struggles faced by marginalized communities.

These lessons have deepened my commitment to advocating for health equity. They have also emphasized the importance of representation in research and policymaking, as these are key steps in improving health outcomes for marginalized communities and ensuring they receive the support they deserve.

P34

Willingness to pay for routine childhood vaccination in Northern Tanzania: a contingent valuation method

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Background: Tanzania has one of the highest coverages of routine immunization in Sub-Saharan Africa with 71.8% of children aged 12–23 months having received basic antigens. However, rural-urban disparities exist, and several regions reported substantially lower vaccination rates, including 56% in Mwanza Region and 32% in Shinyanga Region [1, 2].

Study objective: This study examined mothers' willingness to pay (WTP) for routine childhood vaccination using contingent valuation methods (CVM). WTP reflects the net value of a vaccination to

the mother and may be used to assess the relative importance of diverse facilitators and barriers, including access constraints and vaccine hesitancy, on vaccination uptake.

Methods: A cross-sectional cluster sample survey of 400 mothers from 4 rural districts in Northern Tanzania was conducted from June–July 2024. Mothers were presented with CVM questions in a double-bounded dichotomous choice format to elicit their maximum WTP for routine vaccination. Respondents were asked “if you were asked to pay TSH1000 each time your child receives a vaccination, would you get your child vaccinated?” Responses determined the next bidding amount of either TSH500 (if “no”) or TSH2000 (if “yes”), with open-ended WTP elicited below TSH500 and above TSH2000. Bivariable and multivariable linear regression models were estimated to characterize the relationship between WTP and various sociodemographic and access characteristics. 1 USD equaled TSH2685 at the time of the study. To characterize households’ relative economic position, an asset index was constructed using principal components analysis.

Results: 15% of respondents were not willing to pay anything for routine childhood vaccination. The mean (median) WTP was TSH1555 (TSH2665); the maximum WTP was TSH30000. In multivariable analyses, mothers who had post-primary education had a higher WTP compared to mothers with less than primary education (coef: TSH2157.593, $p=0.016$); those who have no insurance had lower WTP (coef: -TSH554.87, $p=0.048$). WTP was strongly associated with households’ economic position ($p=0.003$). In bivariable analyses, WTP was also associated with travel and waiting time ($p=0.016$), and vaccine hesitancy ($p=0.042$), but these were not statistically significant after adjusting for covariates.

Implications: Low rates of childhood vaccinations in the face of high rates of WTP among mothers point to system-level barriers that must be addressed for mothers to consistently follow through on their intention to vaccinate their children. Studies may also consider examining client satisfaction regarding vaccination to assess system-level barriers such as waiting time, provider attitude, provider scheduled sensitiveness, etc. Targeted interventions towards improving knowledge on the benefits and misconceptions on RV will improve uptake.

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P35

Does today’s internet access effectively bridge the digital divide for urban patients? A preliminary study associated with healthcare inequality among pregnant mothers with substance use disorder in the United States

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Background: This study investigates the relationship between disparities in digital access and infant outcomes among mothers with substance use disorders (SUD). Maternal SUD during pregnancy poses a

significant public health challenge, adversely impacting both maternal and infant health [1]. Despite the widespread availability of the internet, the digital divide remains a major concern, particularly in healthcare. This divide hampers access to critical information and resources, especially for vulnerable populations like pregnant mothers in rural areas with underdeveloped IT infrastructure, as well as in urban areas where patients may lack adequate healthcare access and information despite existing IT infrastructure [2,3,4].

Materials and methods: The data for this study were obtained from five sources: the Healthcare Cost and Utilization Project’s State Inpatient Datasets (HCUP-SID), the American Hospital Association (AHA) Annual Survey, the AHA IT Survey, Census Data, and County Health Rankings. Our dataset comprises 248,150 birth records from New Jersey, spanning three years from 2018 to 2020. Of these records, 7.24% are associated with preterm births, and 3.55% are labeled as low birth weight (LBW) with variables linked to maternal demographics, infant outcomes, and regional digital access. Comprehensive feature importance analysis was performed using the Boruta method, followed by stepwise logistic regression to identify key predictors. Class imbalance was addressed using the Synthetic Minority Over-sampling Technique (SMOTE), and statistical analyses were conducted for deeper feature insights.

Results: This study identifies key predictors of preterm birth and LBW among mothers with substance use disorders using the Boruta method and stepwise logistic regression. Telehealth services, including consultations, stroke care, and remote monitoring, significantly reduced adverse outcomes. Areas with limited digital access, such as low internet and cellular plan availability, exhibited higher rates of preterm and LBW births. Income inequality emerged as a critical factor, with higher inequality increasing the likelihood of adverse outcomes. Private insurance and higher household income reduced risks, underscoring the importance of addressing socioeconomic and digital disparities to improve maternal and infant health.

Conclusions: Our preliminary findings reveal significant associations between limited internet access and adverse infant outcomes, highlighting the urgent need for targeted interventions to bridge the digital divide and improve maternal and infant health in communities affected by substance use disorders. The study reveals a notable difference in preterm birth rates between mothers with SUD and those without, indicating mothers with SUD have a higher incidence of preterm births. Furthermore, the predominantly urban nature of the mom population in this study highlights that even in urban environments, there are significant challenges existing around the digital divide problem.

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P36

Big data analytics for cardiovascular risk prediction: insights into feature importance

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Introduction: Effective risk assessment tools are urgently needed, as cardiovascular disease is one of the leading causes of morbidity and death worldwide [1]. This research harnesses big data analytics through logistic regression and decision tree methodologies to uncover significant predictors. We seek to provide valuable insights for improving clinical practices and public health interventions in cardiovascular risk management.

Study Objective: Our study aims to identify key predictors of cardiovascular disease risk using logistic regression and decision tree models. It analyzes medical and demographic factors, evaluates feature importance, and compares model performance. The goal is to use big data to generate actionable insights to enhance risk prediction, clinical decision-making, and public health strategies.

Research Method: This study employs logistic regression and decision tree models to predict cardiovascular disease (CVD) risk using a Kaggle dataset.[2]. The methodology begins with data preparation and selecting relevant clinical and demographic features. Based on these predictors, the logistic regression model estimates the probability of CVD, providing statistical significance [3].

In contrast, the decision tree model captures non-linear relationships by recursively splitting the data based on information gain, resulting in a hierarchical structure of decision rules [4]. We evaluate the model performance using accuracy, F1 score, precision and recall, alongside feature importance scores derived from the respective models. This comparative analysis highlights the strengths of each model, with logistic Regression offering clear interpretability and decision trees revealing complex interactions among predictors, thereby enhancing risk assessment and diagnostic prioritization in clinical practice.

Results: The findings highlight the significant role of machine learning models in predicting cardiovascular risk (see Fig. 1). In Logistic Regression, Chest Pain Type and ST Depression were the most critical predictors, with importance scores of approximately 0.95 and 0.82, respectively. The Decision Tree model also recognized these predictors with lower scores of 0.24 and 0.21. Moderately important features included Sex, Number of Vessels Colored by Fluoroscopy, and Thallium Stress Test Results. Logistic Regression achieved a superior accuracy of 0.97 compared to 0.74 for the Decision Tree, emphasizing the importance of model selection in enhancing cardiovascular risk assessment (see Fig. 2).

Discussion: This study highlights the pivotal role of big data analytics in advancing cardiovascular risk prediction. The findings reveal that logistic regression surpasses decision tree models' accuracy, underscoring the importance of selecting appropriate models for clinical use. Significant predictors, including Chest Pain Type and ST Depression, emerged as key indicators, corroborating existing research emphasizing their importance in cardiovascular evaluations [4]. The study's implications extend to clinical practice, where these predictive models can aid in the early detection of high-risk patients, improving patient outcomes. Additionally, insights into feature importance can inform targeted interventions and resource allocation in public health initiatives, demonstrating the critical need for ongoing advancements in data-driven approaches to cardiovascular disease management [5].

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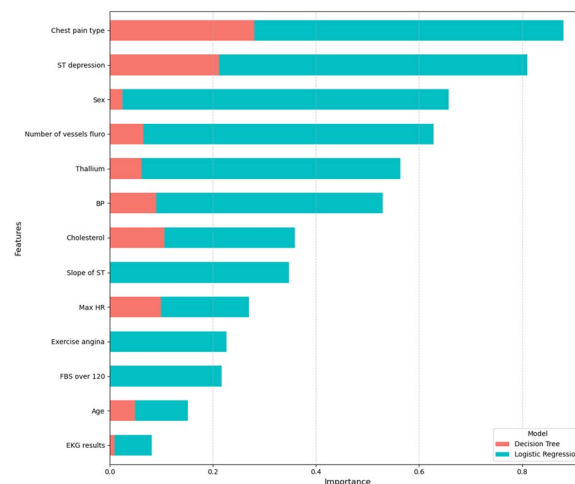


Fig. 1 (Abstract P36). Feature Importance: Decision Tree vs Logistic Regression

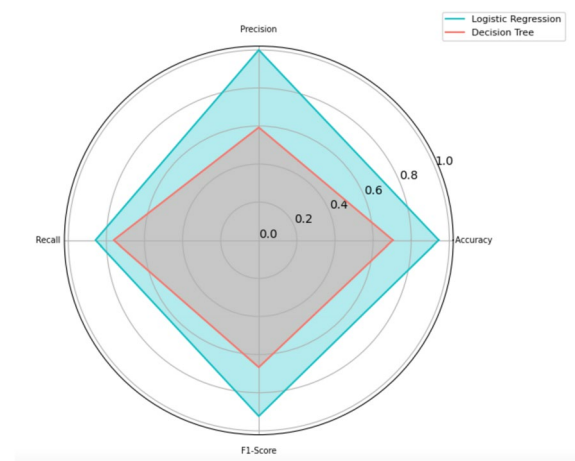


Fig. 2 (Abstract P36). Model Performance: Decision Tree vs Logistic Regression

P37

Optimizing HIV treatment outcomes through data triangulation: aligning treatment and consumption metrics

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Background: Nigeria's HIV program has made remarkable progress in expanding access to antiretroviral therapy (ART). However, persistent discrepancies between reported HIV treatment data and antiretroviral drug consumption data undermine program efficiency and effectiveness. These misalignments raise concerns regarding data accuracy, supply chain management, and resource allocation.

Objectives:

1. Identify and quantify the inconsistencies between reported HIV treatment data and antiretroviral drug consumption data across healthcare facilities in Nigeria.

2. Analyze potential causes of misalignment, including data reporting errors, supply chain inefficiencies, and systemic challenges in health facilities.
3. Utilize Patient Biometric Systems (PBS) and advanced data analytics tools (STATA & Tableau) to reconcile treatment and consumption data.
4. Assess the impact of biometric tracking and data analytics interventions on reducing discrepancies in ART treatment and drug consumption.
5. Develop recommendations for strengthening data management systems, improving supply chain transparency, and optimizing ART resource distribution.

Aims:

1. To improve the accuracy and reliability of HIV treatment and antiretroviral drug consumption data through data triangulation.
2. To ensure that ART patient numbers reported align with actual drug utilization, reducing cases of over- or under-reporting.
3. To promote the adoption of digital health solutions, including biometric tracking and real-time analytics, for improved program monitoring and decision-making.
4. To support sustainable, data-driven approaches in Nigeria's HIV program, ensuring efficient resource allocation and enhanced treatment outcomes.
5. To advocate for policy and system-level improvements that strengthen HIV data reporting mechanisms and supply chain management across different states.

Methods: A data triangulation approach was employed to align treatment and consumption metrics. Key interventions included the use of Patient Biometric Systems (PBS) for tracking ART patients and leveraging data analytics tools such as STATA and Tableau to analyze discrepancies at the facility level. Data were sourced from the National Data Repository (NDR), Logistics Management Information System (LMIS), and health facility reports to compare patient treatment numbers with antiretroviral drug consumption trends.

Results: Preliminary findings indicate that while ART coverage has expanded, discrepancies persist in over 20% of facilities, where reported treatment figures do not correlate with actual drug consumption. These inconsistencies suggest challenges in data reporting, supply chain inefficiencies, and potential cases of over- or under-reporting of ART numbers. Implementing PBS has improved data reliability in select facilities, reducing discrepancies by 30%. However, full alignment remains a challenge due to variations in data reporting mechanisms across different states.

Conclusions: Addressing discrepancies between HIV treatment data and drug consumption is critical for improving program efficiency, ensuring accurate reporting, and optimizing resource allocation. Strengthening data management systems, expanding the use of biometric tracking, and integrating real-time analytics can enhance alignment between treatment and consumption data. Further investment in digital health solutions and capacity-building efforts is essential to achieve data-driven decision-making and sustained progress in Nigeria's HIV response.

Keywords: HIV treatment, antiretroviral therapy, data triangulation, supply chain management, patient biometric systems, Nigeria, STATA, Tableau.

P38

Did the ACIP policy recommendation on adolescent vaccine bundling influence same-day administration of Tdap-HPV vaccines? Evidence from NIS-Teen 2011–2022

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Background: The Advisory Committee on Immunization Practices (ACIP) recommended in 2014 that adolescent vaccines, including HPV and Tdap, be administered concomitantly, a practice known as vaccine bundling.[1] Bundling adolescent vaccines presents a pragmatic opportunity to reduce missed vaccination opportunities, streamline care delivery and leverage the high acceptance of Tdap to increase HPV uptake.[2] However, the extent to which the 2014 ACIP recommendation has influenced healthcare providers'practices of adolescent vaccine bundling remains largely unknown, creating a critical knowledge gap in understanding the policy's real-world impact.

Objectives: This study examined trends in Tdap-HPV vaccine bundling and assessed the impact of the 2014 ACIP policy change recommending the bundling of adolescent vaccines on healthcare providers'practice of same-day administration of Tdap and HPV vaccines.

Methods: This retrospective cohort study analyzed provider-verified immunization data from the 2011–2022 National Immunization Survey-Teen (NIS-Teen) for 160,643 adolescents who received the Tdap vaccine at ages 9–14 years. Participants were categorized as pre-ACIP Policy (Tdap administered before 2014) or post-ACIP Policy (Tdap administered in 2014 or later). Tdap-HPV bundling, defined as receiving both vaccines on the same day, was coded as 1 (bundled) or 0 (unbundled). Segmented regression with state fixed effects and covariates (e.g., age, sex, maternal demographics, insurance, and poverty level) assessed changes in bundling trends and levels before and after the policy. Marginal effects quantified the predicted probability of bundling for each year relative to the policy recommendation.

Results: The proportion of Tdap-HPV bundling increased from 2.8% (95%CI: 2.0–4.0%) in 2006 to 44.9% (95%CI: 29.7–61.1%) in 2022. For male adolescents, it increased from 0.1% (95%CI: 0.0–0.3%) in 2008 to 43.6% (95% CI: 24.3–65.0%) in 2022; and for female adolescents, it increased from 5.8% (95% CI: 4.1–8.2%) in 2008 to 46.3% (95% CI: 24.4–69.7%) in 2022. Pre-ACIP Policy, the proportion of Tdap-HPV bundling increased by 3.34 percentage points per year ($p<0.001$). Following the ACIP policy, the annual rate of increase in Tdap-HPV bundling accelerated by an additional 0.99 percentage points per year ($p<0.001$; Table 1). Fig. 1 shows the marginal effects of Tdap-HPV vaccine bundling across the pre- and post-ACIP policy period.

Discussion: The ACIP policy did not result in an immediate significant increase in Tdap-HPV bundling. However, the policy had a significant impact on the rate of increase in bundling, suggesting that its influence grew over time. This builds on prior studies, such as Zhu et al.[3], which documented Tdap-HPV bundling increasing from 22.9% in 2014 to 39.1% in 2018, and Doke et al.[4] and Moss et al.,[5] which observed upward trends in vaccination bundling. However, these studies primarily focused on descriptive trends over limited timeframes, often within specific populations, without assessing the ACIP policy's impact. Our study provides critical evidence of the ACIP policy's direct influence on national vaccine bundling practices, filling a significant gap in the literature. These findings underscore the importance of policy-driven interventions in shaping provider behaviors and optimizing vaccine bundling strategies.

Keywords

Adolescent vaccination, HPV, Tdap, vaccine bundling

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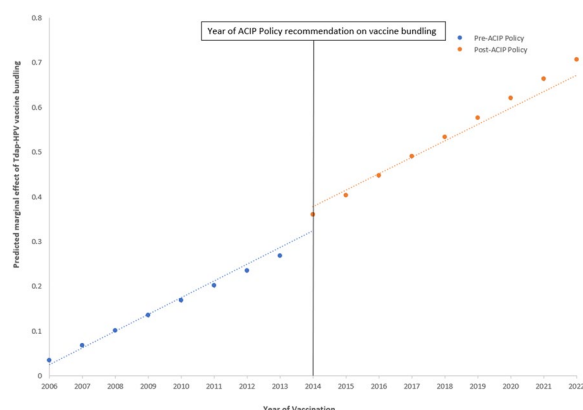
Table 1 (Abstract P38). Segmented regression analysis outputs of Tdap-HPV bundling and ACIP Policy change adjusting for study covariates and state fixed-effect

Tdap-HPV Bundling	Coefficient	std. err	t	P > t	[95% conf. interval]	
Time ^a	0.0334	0.0013	24.21	< 0.001*	0.0306	0.0361
ACIP Policy ^b	0.0099	0.0079	1.25	0.212	−0.0056	0.02534
Time*ACIP Policy ^c	0.0099	0.0023	4.22	< 0.001*	0.0053	0.0145
Constant	0.35092	0.0181	19.39	< 0.001*	0.3154	0.3863

^a Time represents periods of vaccination relative to 2014, and ranged from −8 to +8

^b ACIP Policy—categorized as 1(period from 2014 and afterwards) and 0 (period before 2014)

* Statistically significant

**Fig. 1 (Abstract P38).** Marginal effects of Tdap-HPV vaccine bundling pre- and post-ACIP policy recommending adolescent vaccine bundling

P39

Pre-exposure prophylaxis outcomes following implementation of innovative interventions during the Covid-19 lockdown period among women in the North-West Province, South Africa: a retrospective analysis

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Background: Oral pre-exposure prophylaxis (PrEP) has been shown to be an effective HIV prevention tool among key populations including young women [1–2]. However, the risk of HIV acquisition is only reduced when PrEP uptake and adherence are high [3]. The uptake and adherence to PrEP may have been lower in Sub-Saharan African countries, including South Africa during the Covid-19 lockdown. The emergence of the Covid-19 pandemic has disrupted access to health care including HIV prevention and treatment services [4,5].

Aim: This study assesses changes in PrEP initiation, uptake, and adherence before and during the Covid-19 lockdown period among young women in the North-West Province of South Africa

Methods: This retrospective analysis evaluated programme data collected by TB HIV Care on HIV PrEP initiation, uptake, and adherence from December 2018 – January 2021 among women aged 16 to 35 years in the three sub-districts (Matlosana, Maquasi Hills, and JB Marks) of the Dr Kenneth Kaunda District were initiated into PrEP. The women were offered oral PrEP during the above-mentioned period through a mobile clinic vehicle. We assessed changes in PrEP initiation, uptake, and adherence before (Dec 2018– Feb 2020) and during (Mar 2020–Jan 2021) the Covid-19 lockdown period. The initiation was defined as the number of people commencing PrEP; uptake as the proportion of clients receiving HIV testing services and offered PrEP for 1-month return; and adherence as the number of people remaining between months 2 and 4. Data was captured on a Microsoft Excel Sheet and analysed using STATA (College Station, TX, USA) software version 23. Descriptive statistics were used to compute the frequencies and percentages of categorical variables.

Preliminary Results: Innovative PrEP interventions such as social media platforms were utilized for PrEP continuity during the Covid-19 lockdown period. A total of 1339 young women who engage in transactional sex were counselled and offered PrEP. Out of these, 475 were initiated on PrEP. There was a decrease from 67% in the proportion of those tested and initiated on PrEP before Covid-19 compared to 38% during the lockdown period. The 1-month return of PrEP also decreased from 62% compared to 38% during Covid-19. Furthermore, adherence was extremely impacted with a 21% decrease during the lockdown period than before 79% the lockdown period (Fig. 1).

Conclusion: The current study showed a decrease in PrEP initiation, uptake and adherence during the Covid-19 lockdown period despite social media platforms that were used to ensure continuity of services. More innovative interventions are needed for the improvement of PrEP outcomes to achieve HIV control and prepare for future pandemics.

Acknowledgments

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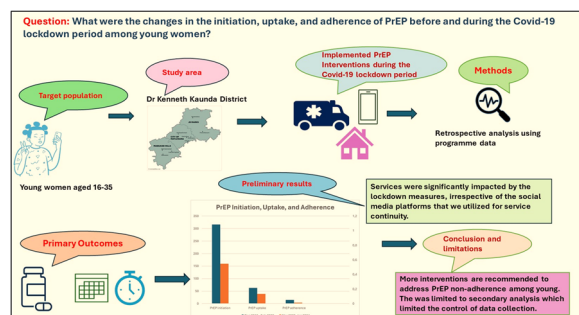


Fig. 1 (Abstract P39). Overview of the study

P40

Using deep learning to disentangle artifacts in multimodal datasets

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Introduction: The potential benefits of multimodal biosignal recordings and data fusion for patient monitoring and preventive self-monitoring are enormous. However, confounders complicate assessing biological rhythms from multimodal biosignal recordings. Investigators need to reject, correct, or control for confounders acting as artifacts (e.g., eye movement contamination in the electroencephalogram, or EEG). The availability of multimodal datasets and the rise of deep learning open new opportunities in automatically learning and disentangling artifacts from signals of interest.

Study Objectives: We aimed to use deep-learning to remove EOG artifacts in multimodal EEG/eye-tracking (ET) datasets.

Methods: We used an open-access multimodal EEG/ET dataset (EEGeyeNet [1]) and a recurrent neural network (a two-layer Long Short-term Memory model followed by a 0.5 dropout and a fully-connected layer) to learn the part of the EEG predictable from ET data and remove that predicted signals from the EEG. We used the *Large Grid* task from EEGeyeNet. In this task, 30 participants looked at a series of dots appearing at 25 predetermined positions on the screen (810 trials/participant). EEG and ET were recorded simultaneously using a high-density 129-channel net and video-infrared eye-tracking. We then compared the performance of this approach with the classical approach relying on independent component analysis (ICA). We trained and tested the model within subjects.

Results: We assessed the evoked EEG time series before and after EOG artifact reduction, once using the proposed deep learning model, and once using ICA. As shown in Fig. 1, the proposed deep learning model is more conservative in EOG reduction than ICA and distorts the signal less in the pre-stimulus period.

Discussion: In this study, we learned the part of EEG predictable from ET and demonstrated that, even with a single recording, it is possible to learn this relationship with a satisfying degree of precision. This relationship is made up of various contributions, including a component due to the artifact generated by the movement of the eye and the neural activity systematically correlated with the eye movement (e.g., motor control, change in visual input). The approach considered in this study cannot disentangle these different components. In future work, we plan to address this limitation by designing a generative model of how EOG artifacts are generated from eye movements relying on biophysical knowledge. In combination with the approach presented

here, the EOG and the neural component associated with eye movements could then be disentangled and allow more precise analyses of neural activity.

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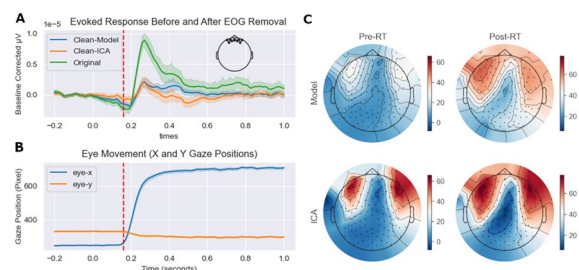


Fig. 1 (Abstract P40). A-B: Evoked response for looking at a dot on the right side of the screen before and after EOG removal (A), and the average eye movement during those trials (B). The vertical red dashed line represents the average reaction time (RT) across participants. C: Spatial projection of the predicted EOG signal for the pre-RT period and the post-RT period, as predicted by the model and ICA

P41

Association between physical activity and depression symptoms among anemic and non-anemic individuals: findings from the 2007-2018 National Health and Nutrition Examination Survey

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Background: In United States (US), the prevalence of anemia has increased from 4% to 6.5% between 1999-2020.[1] Several studies have reported an association between depression symptoms and anemia [1-4]. Strong evidence supports the benefits of physical activity in reducing depression symptoms across diverse populations [5-7]. Although studies elsewhere have explored the role of physical activity in alleviating depression symptoms in anemic individuals [8, 9], there is a need to explore this association and identify modifiable risk factors to prevent depression symptoms among anemic individuals in the US. The study investigated the association between moderate to vigorous physical activity (MVPA) and depression symptoms among anemic and non-anemic individuals in the US.

Materials and Methods: We used data from the 2007-2018 National Health and Nutrition Examination Survey (NHANES). Our analytical sample included 29,169 non-pregnant individuals who were 18 years or above. Depression symptoms were assessed using the Patient Health Questionnaire (PHQ-9) and the total PHQ score was classified as no depression symptoms (0), minimal to mild depression symptoms (1-9), and moderate to severe depression symptoms (10-27). Moderate and vigorous physical activity per week were summated for work, leisure, and transportation (where, one minute of vigorous activity = two minutes of moderate activity) to create MVPA. MVPA was categorized into – No MVPA, Insufficient MVPA (0-150 minutes/week), and Sufficient MVPA (>150 MVPA min/week). Anemia status was classified with a cut-off of 13 gram/dL and 12 gram/dL of hemoglobin for men and women respectively. Adjusted multinomial logistic regression was performed to determine the associations between MVPA status and

depression symptoms in anemic and non-anemic groups, after controlling for demographic, socio-economic, health risk, health behavior and health status variables.

Results: Bivariate analysis showed females, non-Hispanic whites, those aged 40–59, and with functional disabilities were significantly more likely to report depression symptoms in both anemic and non-anemic groups ($p < 0.05$). Adjusted multinomial regression results showed that for both anemic and non-anemic groups, individuals with sufficient MVPA had significantly lower odds of moderate to severe depression symptoms when compared to those with no MVPA [anemic – aOR 0.38, 95% CI 0.15–0.96; non-anemic – aOR 0.62, 95% CI 0.47–0.82].

Conclusions: Our study reported a protective association of recommended physical activity against depression symptoms in both anemic and non-anemic groups. Given that anemic individuals are more prone to depression symptoms, additional recommendations for sufficient physical activity should be designed for anemic individuals in addition to treating anemia to prevent depressive symptoms.

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P42

Machine learning applications on Computed Tomography Angiography (CTA) data for diagnosis of Peripheral Artery Diseases (PAD)

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BMC Proceedings 2025, **19(13)**:P42

Background: Characterized by the narrowing of the large arteries supplying the limbs, Peripheral Artery Diseases (PAD) is a growing health issue for the rapidly aging US population. The incidence of lower extremity amputation for PAD in South Carolina is significantly higher than the national average, associated with high levels of morbidity and mortality, up to 40%–50% within 1 year of initial diagnosis and treatment [1]. It is difficult to predict outcomes for individual PAD patients as a validated metric to quantify calcification of the lower extremities, comparable to the widely accepted Agatston score for coronary artery calcification, has not been developed. Furthermore, due to individual variability in disease progression and response to treatment, there is a significant unmet need for prognostic tools to risk-stratify PAD patients for surgical intervention according to

expected outcomes. CT angiography, a type of CT scan, is used to visualize arteries and diagnose arterial diseases by identifying blockages, narrowing, or aneurysms [2].

Objective: The objective of this project is to develop a computer aided diagnostic technique (CAD) utilizing machine learning application for automated diagnosis of PAD.

Methods & Results: To address the challenge of PAD diagnosis, we have proposed a meta-learning dense neural network (DNN) to aid surgeons in the analysis of Computed Tomography Angiography (CTA) images of the lower extremities, specifically to segment and track objects across multiple sequential images for identifying and quantifying vessel calcification [3]. For our training, we also have utilized synthetic image data generation techniques for transfer learning. Doing so, evaluation of our proposed DNN architecture and segmentation models will have significant pixel accuracy, mean intersection over union (mIoU), Dice coefficient, and precision/recall/F1-score's.

Discussion: Our proposed machine learning architecture can effectively predict outcomes for individual PAD patients as a validated metric to quantify calcification of the lower extremities. It can further be extended to other arterial diseases detected utilizing CTA imaging.

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P43

Effectiveness of Tenofovir-induced tool in improving early screening of acute kidney injury among female sex workers in Northwest Province: protocol for a quasi-experimental study

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BMC Proceedings 2025, **19(13)**:P43

Background: Tenofovir disoproxil fumarate (TDF) is a joint antiretroviral therapy (ART) used in Sub-Saharan Africa (SSA), but it has been linked to renal toxicities, including acute kidney injury (AKI). Current nephropathy tests, which rely on increased urinary creatinine levels, are not sensitive enough for early AKI detection and require laboratory facilities. This is a significant issue in SSA, where AKI leads to reduced ART adherence and increased hospitalisation rates. Female sex workers (FSWs) are disproportionately affected by HIV, yet research on AKI among this population is scarce.

Aim: To assess the effectiveness of the TDF-induced AKI tool in improving screening, early diagnosis, and management of AKI among FSWs in the North West Province of South Africa.

Methods: This quasi-experimental study will assess the effectiveness of a TDF-induced AKI screening tool among FSWs in the North West Province, South Africa (Fig. 1).

Public health implication: The results are expected to guide the development of scalable AKI POC screening tools for early detection, improve healthcare models and diagnostic frameworks and inform policy, benefiting broader healthcare systems in resource-limited settings.

Limitations of the study: The limitations will be the small sample size, limited access to FSWs in one province, and migration of the FSWs to another province for baseline follow-up and the correct use of the AKI POC.

Expected Outcome: This study emphasizes the need for better diagnostic tools for early AKI detection, especially in resource-limited settings. The introduction of the ALERT-AKI test offers a promising solution to address renal complications in HIV-positive individuals, and the findings will inform scalable AKI screening tools, improving healthcare models and ART adherence across sub-Saharan Africa.

Keywords: Antiretroviral therapy, TDF-induced Acute kidney injury, Female Sex Workers, Diagnostic kits, Point-of-care screening, HIV; Quasi-experimental design, South Africa

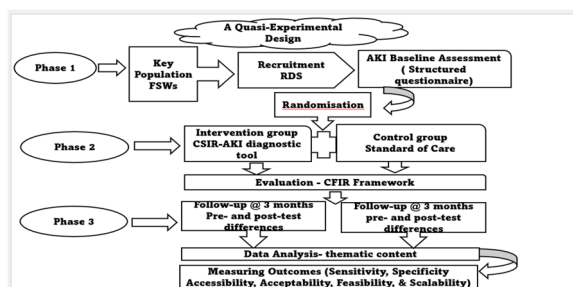


Fig. 1 (Abstract P43). Summary of Methods

P44

Assessment of the Routine Health Information Management System in monitoring HIV prevention and treatment cascade in South Africa

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Introductions/Background: There is a need to prioritize timely and high-quality data on HIV prevention and treatment captured on the routine health information system (RHIMS) ahead of the United Nations' 2030 Agenda [1]. The RHIMS has great potential for improving HIV care outcome monitoring. However, the data captured on the RHIMS has gaps, including the lack of key population (KP) inclusivity and the disaggregation thereof. The HIV prevalence among general population in South Africa is estimated at 12.7% [2], and it is significantly higher among KP, with 55% among PWID, 54.6% among TG women, 49.7% among FSW, and 28.6% among MSM [3–5]. To date, there has been no descriptive epidemiological analysis on the nature and structure of RHIMS data for the purpose of including KP data disaggregation.

Methods: Exploratory-descriptive approach was used to analyze de-identified HIV routine program data from the National Department of Health. A total of 297657 which were captured on the system were analysed using exploratory data analysis. A standard data extraction tool in Excel will be employed to retrieve deidentified data from the HIV database. Data analysis was done using Python 3.11.4 programming language.

Preliminary results: In terms of the first 95 cascade, 297657 clients that tested for HIV, and 9133 (3.1%) tested positive for HIV. For the second 95 cascade, 2391 (80.6%) were initiated on ART and for those on ART 1246 (88.5%) had viral load testing done and 923 (74.1%) were virally suppressed. The viral load completion rate was at 90.2% with the suppression rate at 79.5%. High viral load suppression was observed among females within the age category 45–54 years at 88.0%, followed by those within the age group of 55–64 years at 84.0%. Most of the clients were taking Dolutegravir (DTG)-based regimen, with higher proportion among females at 99.7%, compared to males at 99.1%.

Discussion: As countries strive to reach the 95–95–95 HIV targets and end the HIV/AIDS epidemic as a public health threat by 2030, the evolution of RHIMS is essential to uncover epidemiologic patterns and service gaps, enabling differentiated HIV care and improved reporting. To address this gap, we explored opportunities for integrating unique identifiers for KPs and other critical variables into national data collection and reporting tools using exploratory data analysis. Our study found that the system can generate HIV cascades disaggregated by gender and children under 15 and adults who are 15 years and above.

Conclusions: Monitoring and evaluation systems must be robust enough to accurately report on the programmatic coverage of HIV services, including those for key populations. Our study findings will contribute to the development of a framework for the inclusion of KPs unique identifier code on the RHIMS.

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P45

Burnout and empathy among community-engaged medical students

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Introduction: Medical students experience high rates of burnout while progressing through their training. Prior research has demonstrated that these high rates of burnout paired with continuing medical studies correlate with a decline in empathy. However, empathy is a key component of effective and compassionate patient care, so it is essential that medical training supports programs that foster or maintain empathy. Community engagement is one possible avenue for increased empathy among medical students.

Study Objectives: This study aimed to assess the impact of participation in Root Cause, a monthly student-led community health fair in Greenville, South Carolina, on burnout and empathy among medical students, highlighting the role of volunteerism in student well-being and the importance of community engagement opportunities for medical trainees.

Methods: University of South Carolina School of Medicine Greenville (USCSOMG) student-volunteers at Root Cause events completed a voluntary survey prior to each event which assessed demographics, burnout via the Maslach Burnout Inventory (MBI), and empathy via the Toronto Empathy Questionnaire (TEQ). The MBI is a validated tool for assessing burnout across three dimensions: emotional exhaustion, depersonalization, and personal accomplishment. The TEQ is a 16-point self-reported evaluation of empathetic tendencies. One hundred and five surveys collected from 2022 to 2024 were analyzed to determine the relationships between attendance, burnout, and empathy. Statistical tests included Pearson correlations and unpaired t-tests.

Results: During the 2022-2023 academic year, no statistically significant correlation was found between Root Cause attendance and burnout. However, in 2023-2024, increased Root Cause attendance correlated with decreased levels of burnout among students ($r = -0.3077$, 95% CI $[-0.05608, -0.5226]$, $p = 0.0177$). Furthermore, in 2023-2024, students who completed event surveys at multiple Root Cause events had a lower mean burnout score than students who completed only one event survey (mean difference $= -3.783$ (95% CI $[0.006643, -7.632]$, $p = 0.0540$). Root Cause attendance and level of empathy among student volunteers did not have an association in either academic year.

Discussion: We found that increased community engagement through health care events could lead to decreased levels of burnout among medical students. No change in empathy is noted, which may be attributed to existing high levels of empathy in medical student volunteers and/or a maintenance of empathy levels among these students. These findings highlight the importance of monitoring burnout and empathy in medical students engaged in community outreach activities such as Root Cause. Assessing the efficacy of these community engagement programs through evaluations including the MBI and TEQ underscores the value in further development of programs that promote well-being to foster effective, compassionate future physicians. As medical schools grapple with high levels of burnout experienced by their students, establishing avenues for students to volunteer in the community may prove to be beneficial to student wellness and lead to positive impacts on future patient care.

P46

Impacts of Covid-19 Pandemic on HIV Treatment Outcomes Among Men Who Have Sex with Men in South Africa Using Routine Programme Data: An Interrupted Time Series Analysis

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Background: South Africa is the longest-standing epicentre of the HIV epidemic with approximately 8 million people living with HIV (PLHIV). Key populations including men who have sex with men (MSM) living with HIV have been exposed to increased rates of isolation prior to the Covid-19 pandemic. The pandemic may have worsened the isolation of MSM from healthcare systems, leading to poor HIV outcomes. This study aims to assess the impact of the Covid-19 pandemic on the HIV treatment outcomes among MSM in selected provinces of South Africa.

Materials and methods: This study used de-identified routinely collected data among MSM in the Key Population Men who have Sex with Men (KP MSM) Prevention Programme from 01 January 2018 to 31 December 2022. The data was collected in five districts across three provinces of South Africa (i.e., Gauteng, Mpumalanga, and Kwa-Zulu Natal) during the Covid-19 lockdown, wherein 96992 MSM were

reached. The programme is led by an implementation partner of the National Department of Health (NDoH) in selected Provinces of South Africa. Descriptive statistics was conducted to describe the characteristics of the study population. An interrupted time series analysis was applied to quantify the differences in the utilization of HIV treatment services and outcomes among MSM using STATA (College Station, TX, USA) software version 18.

Results: Differentiated service delivery (DSD) models were implemented to mitigate the impact of the Covid-19 pandemic. These included facility- and community-based DSD models. After the intervention, the influence of time becomes more pronounced, leading to an increase in the outcome by 0.001572 units for each additional time unit. An interrupted time series analysis revealed that the relationship between the reactive HIV test and time significantly changes after the intervention ($p < 0.001$), resulting in a steeper upward trend in the reactive HIV test post-intervention. See the segmented regression analysis with 95%CI in Fig. 1 below.

Conclusions: The results are not conclusive. Future studies should investigate the consistent implementation of various DSD models on HIV treatment outcomes among key populations over time. This can contribute towards reaching HIV epidemic control and future pandemic preparedness by ensuring the continuity of HIV services.

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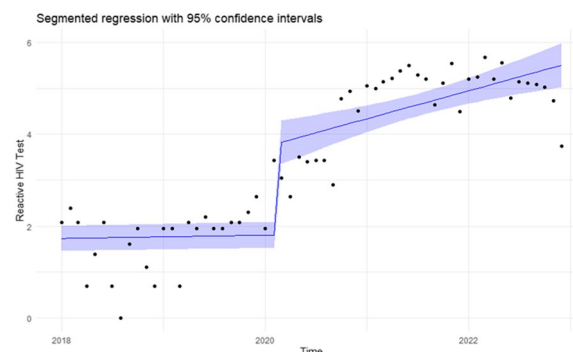


Fig. 1 (Abstract P46). Segmented regression analysis with 95%CI for reactive HIV test among men who have sex with men

P47

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<https://pubmed.ncbi.nlm.nih.gov/38921826/>.

P48

Geographical patterns of COVID-19 vaccine inequality by race and ethnicity: evidence from Jefferson County, Kentucky

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BMC Proceedings 2025, **19**(13):P48

Introduction: Two-dose COVID-19 vaccines protect residents from severe outcomes, i.e., hospitalization and mortality [1]. Racial disparities in COVID-19 vaccine uptake are attributed to vaccine attitudes and behavior [2]. However, evidence on spatial heterogeneity in vaccination uptake in respect to social determinants of health (SDOH) across spatial units can help generate tailored vaccination promotional programs.

Study Objectives: Report COVID-19 vaccination rates (VRs) in the 2nd quarter (5/31/2021) and 6th quarter (5/31/2022) among Jefferson County residents by race and ethnicity across regions (West, Southwest, South, Central, Inner East, Outer East) in Kentucky to identify spatial-temporal vaccine inequality in consideration of respective regional SDOH factors, i.e., income, education, employment, and health insurance.

Methods: COVID-19 vaccine counts were obtained from the Kentucky Immunization Registry. Reporting of different races for unique recipients across doses 1–4 was observed. To quantify the spatial-temporal vaccine inequality, a consistent race was assigned to each recipient across doses. Three methods were established: race reported at first dose, latest dose, and most frequent across doses. Dose-specific VRs were computed using the US 2020 Census zip code-level population data. Reported dose one race was assigned for subsequent doses, as it led to the least VR overestimations. SDOH factors were obtained from the American Community Survey. Regions were formed by grouping neighboring zip codes. VRs for the 2nd quarter (Q2) and 6th quarter (Q6) by race and ethnicity are mapped in Figures 1–2, respectively. Racial density maps are also included in Figures 1–2.

Results: VR increased by 19.2% between Q6 (59.8%) and Q2 (40.6%) across Jefferson County. Across regions, Whites and non-Hispanics had the highest VRs compared to other races, except in the West by Q2; Asians were highest in the West. Compared to Q2, notable changes in VRs can be observed among Whites in the West (28.7%), Blacks in the Inner East (26.1%) and Outer East (22.5%), Asians in the Southwest (22.2%) and Inner East (18.6%), Multiracial in the West (24.3%), Other Races in the West (22.39%), Southwest (24.5%), and South (25.9%), and Hispanics in the Southwest (24.7%) and South (23.2%).

Discussion: Compared to Jefferson County's overall SDOH (% Bachelor Degree [BD]: 20.7, % of Graduate Degree [GD]: 14.1, Average Family Income [INC]: \$107,580.6, % White: 63.7, % Black: 21.0), Inner East (BD: 33.4, GD: 29.5, INC: \$152,846.2, % White: 84.9, % Black: 5.3) and Outer East (BD: 30.2, GD: 21.8, INC: \$148,487.7, % White: 75.3, % Black: 9.8) are the affluent regions. West (BD: 6.2, GD: 3.0, INC: \$47,441.9, % White: 15.5, % Black: 77.9) is the least affluent region. VRs of races are highly associated with the regional rate of higher education and average family income (Figures 1–2). VR is generally lower in Black individuals who showed a higher COVID-19 vaccine hesitancy than other races [3, 4]. Dissection of VRs of races by regional affluence indicates that higher education and family income are key drivers behind vaccine uptake among Black individuals (Figures 1–2). Furthermore, White individuals in the Black-majority region (West) had higher VRs. Campaigns in Black-majority, least-advantaged communities need to address vaccine mistrust.

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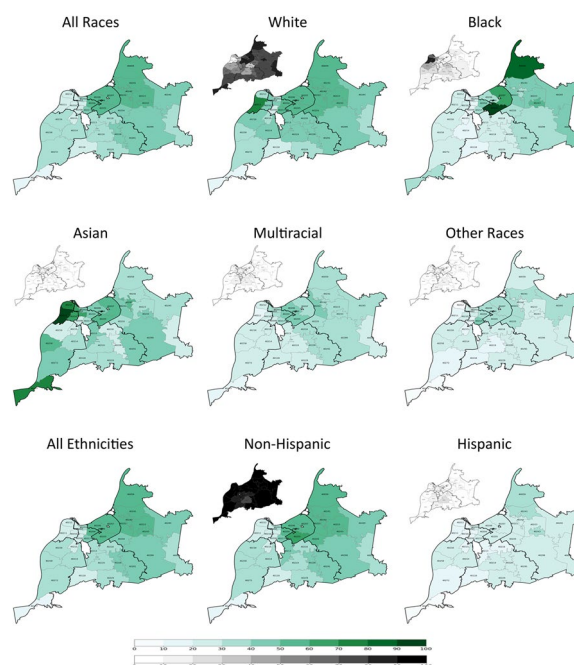


Fig. 1 (Abstract P48). Two-dose COVID-19 vaccination rates at the end of the second quarter of the vaccination campaign on 5/31/2021 by race, ethnicity, and ZIP code (population density maps located at the top left corners)

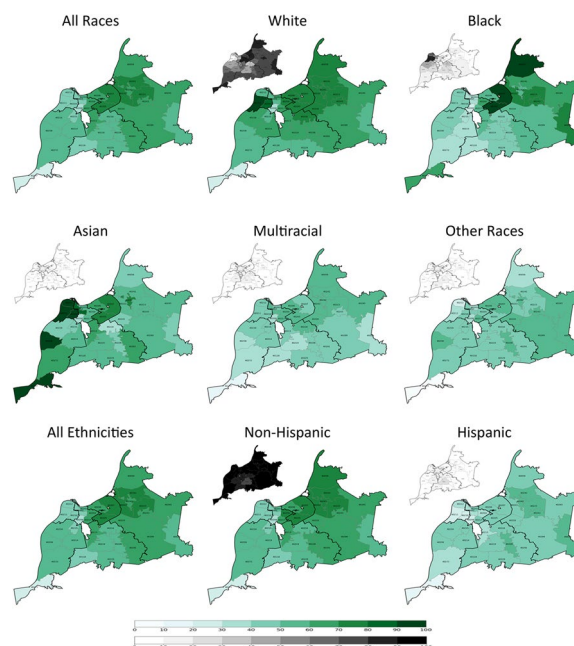


Fig. 2 (Abstract P48). Two-dose COVID-19 vaccination rates at the end of the sixth quarter of the vaccination campaign on 5/31/2022 by race, ethnicity, and ZIP code (population density maps located at the top left corners)

P49

Mining Reddit for insights: analyzing sentiments and topics on opioid harm reduction using unsupervised machine learningJahnvi Kari¹, Aashi Sethiya², George Shaw, Jr.³¹Ardrey Kell High School, Charlotte, North Carolina, USA; ²School of Data Science, University of North Carolina at Charlotte, Charlotte, North Carolina, USA; ³Department of Health Management and Policy, School of Data Science, University of North Carolina at Charlotte, Charlotte, North Carolina, USA**Correspondence:** George Shaw, Jr (gshaw11@charlotte.edu)*BMC Proceedings 2025, 19(13):P49*

Introduction: Opioid addiction is a chronic, relapse-prone disease characterized by compulsive substance use due to its prolonged effects on the brain [1,2]. Despite advancements, stigmatization remains a significant barrier for individuals who use drugs [3]. Increasingly, communities and individuals are turning to social networking sites (SNS) for support [4].

Study Objectives: Few studies have examined Reddit data at the intersection of harm reduction information and sentiment analysis regarding opioid use disorder (OUD) [5]. This study aims to fill this gap by examining users' sentiments, latent topics, and SNS topical interactions regarding harm reduction information associated with OUD [6,7].

Methods: The dataset was curated from the subreddits *r/news*, *r/AskReddit*, *r/IAmA*, and *r/AddictionAdvice*. Data were collected from December 2010 to March 2024 using the Reddit API in Python, with query terms such as "drugs," "opioids," and "naloxone." Comprehensive pre-processing was performed to eliminate unrelated content, stop words, and special characters. Sentiment analysis was conducted using the Linguistic Inquiry Word Count (LIWC) tool, identifying positive and negative sentiments, with neutral sentiments removed. BERTopic Topic Modeling was applied to the positive and negative sentiments using sentence transformers (all-MiniLM-L6-v2), UMAP, HDBSCAN, and c-TF-IDF parameters to identify latent topics. To examine topic coherence, two researchers conducted descriptive content analysis to label the top 20 topics from each sentiment document cluster [8]. Cluster mapping and similarity matrices were used to visually display cluster associations.

Results: A total of 16,051 comments from Reddit were identified. Using the LIWC analysis tool, 4,678 comments (29%) were identified as positive and 6,060 (38%) with negative sentiments. Before removing neutral sentiments, 192 individual topics were identified. Several iterations of BERTopic-based topic modeling were conducted to assess model variability, resulting in 79 positive topics and 72 negative topics. Sixty-five percent of the negative sentiments sampled were related to opioid harm reductions and drug effects. Conversely, only 45% of the sentiment topics were related to opioid harm reduction strategy and other drug-related topics.

Discussion: Despite strong moral and political divides, Reddit users largely expressed that legal and healthcare systems for opioid addiction are difficult to navigate and often ineffective, inadvertently encouraging opioid misuse [9]. Positive topics were related to general harm reduction information and drug testing, while negative topics focused on the drug effects associated with opioid use, particularly amphetamines. Two latent topics among the negative sentiments were drug criminalization and insurance fraud. These findings highlight the usefulness of Reddit for tracking user perspectives on opioid misuse and harm reduction information, as well as the applicability of computational models for social science research [10]. Future research should examine the high costs and biases in OUD treatment from a public policy perspective using a computational framework. Additionally, incorporating generative AI models with human oversight when examining topic coherence is recommended.

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P50

Uncertainty in computational flow cytometry: improving cell-type classificationFNU Shrishti¹, Jacob Sander², Jansen N Seheult⁴, Brian Jalaian^{2,3}, Achraf Cohen¹¹Mathematics and Statistics, University of West Florida, Pensacola, Florida, USA; ²Intelligent Systems and Robotics, University of West Florida, Pensacola, Florida, USA; ³Computer Science, University of West Florida, Pensacola, Florida, USA; ⁴Department of Laboratory Medicine and Pathology, Mayo Clinic, Rochester, Minnesota, USA**Correspondence:** Achraf Cohen (acohen@uwf.edu)*BMC Proceedings 2025, 19(13):P50*

Introduction: Supervised machine learning is increasingly used for flow cytometric gating but faces challenges with labeling errors, novel immunophenotypes, and out-of-distribution (OOD) recognition. Advances in uncertainty quantification (UQ) enhance prediction reliability and enable OOD detection [1]. Inspired by Sander et al.'s work on robust open-set recognition [2], we apply UQ to improve flow cytometry classification, gating, and immunophenotyping by identifying OOD events.

Study objectives: This study develops a metamodel-based UQ framework to enhance prediction confidence and interpretability in natural killer (NK) cell immunophenotyping for pediatric solid cancer datasets. It also addresses ambiguous cases to enable more informed and reliable decision-making in biomedical research.

Methods: This study analyzed publicly available NK-focused mass cytometry data from the TRICICL study [3], which evaluated NK cell immunity defects in pediatric solid cancer patients. K-means clustering assigned cluster labels to cell subsets, eliminating the need for manual labeling. The preprocessing pipeline incorporated standardization and dimensionality reduction via Uniform Manifold Approximation and Projection (UMAP) for visualization. Base models, including multinomial logistic regression, random forest, and XGBoost, were trained on the cluster-labeled data. A metamodel-based Uncertainty Quantification (UQ) framework was implemented, where a secondary binary classifier estimated confidence in base model predictions. The metamodel received the original features alongside the base model's predicted outputs to generate certainty scores, identifying predictions likely to be incorrect.

Results: K-means clustering identified cell populations, with the optimal cluster number determined using the Elbow (k=3) and

Silhouette ($k=9$) methods; $k=9$ was selected for better separation. Performance metrics for the three machine learning models are shown in Table 1. XGBoost achieved the highest accuracy, precision, recall, and F1 score. UQ was conducted using a binomial logistic regression meta-model to determine the probability of the baseline model incorrectly classifying a cell's cluster label. The meta-model's receiver operating characteristic (ROC) area under the curve (AUC) was 0.9439, indicating strong discrimination between correct and incorrect labels—a UQ threshold of 0.87 distinguished correct (18,383) and incorrect (621) predictions.

Conclusions: This study demonstrates the feasibility of a UQ meta-model framework to identify incorrect cell label classifications made by a supervised ML model and to flag OOD events for manual review. The UQ threshold balanced model interpretability with the need for additional review of uncertain predictions. Additional work will compare UQ metamodel approaches with other UQ approaches, such as ensemble methods and variational layers to improve OOD detection for clinically relevant tasks that rely on flow cytometric data.

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Table 1 (Abstract P50). Performance Metrics of Machine Learning Models

Baseline Model	Accuracy	Macro Average			Weighted Average		
		Precision	Recall	F1-score	Precision	Recall	F1-score
Logistic Regression	96%	89%	91%	90%	96%	96%	96%
Random Forest	96%	90%	91%	90%	96%	96%	96%
XGBoost	97%	92%	92%	93%	97%	97%	97%

P51

Determinants of antiretroviral therapy adherence among farm workers living with HIV in a selected sub-district in Limpopo Province, South Africa

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Introduction: Despite global efforts to improve access to antiretroviral therapy (ART) for all people living with human immunodeficiency virus (PLHIV), sub-optimal adherence to ART remains a concern [1]. While ART adherence has focused on various population groups, farm workers, who are essential for the food security industry are often underserved. Adherence to recommended ART dosages determines the effectiveness of long-term ART and consequently optimal health outcomes such as viral load suppression [2,3].

Study Objectives: To determine the factors associated with ART adherence among farm workers living with HIV.

Method: We used a descriptive-analytical cross-sectional study on a random sample of 445 farm workers aged 18 and above working in a selected subdistrict in Limpopo Province. A semi-structured questionnaire was used to gather information on ART adherence, socio-demographic factors, clinical knowledge, behavioural and perceptions of farm workers. Clinical data were extracted from clinic folders and Three Interlinked Electronic Register for Tuberculosis and HIV (TIER. Net) version 1.13. Data analysis was performed using STATA version 17. Logistic regression age categorised:18–35, 36–49, +50 years. A p -value < 0.05 was considered as statistically significant.

Results: The average age of the population was 43.5 years. A proportion of 74.2% ($n=330$) farm workers were adherent on ART. The following variables were associated with adherence to ART on univariate model: gender ($p=0.042$), marital status ($p=0.038$), type of employment ($p=0.011$), stable income ($p=0.027$), ability to take medication on duty ($p=0.035$), source of information ($p=0.008$). A higher proportion of ART adherence was observed among females +50 years (115(77.2%), $p=0.017$), farmers aged 36–49 year old in a relationship (150(80.2%), $p=0.009$), 36–49 years with access to sick leave (170(90.9%), $p=0.021$), 36–49 years with ability to take medication on duty (179(96.2%), $p=0.021$), 18–35 years who are Christians (102(93.6%), $p=0.042$).

Discussion: ART adherence was associated with gender, marital status, type of employment, stable income source, source of information and taking medication at work. Strategies to improve ART adherence should implement context appropriate interventions to support adherence within different population groups.

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P52

Factors associated with sexually transmitted infection literacy among men who have sex with men and transgender women in Soweto: a machine learning approach

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Background: Sexually transmitted infections (STIs) are a big health problem. Human immunodeficiency virus (HIV), viral hepatitis, and other STIs account for 2.3 million deaths annually. Machine learning algorithms are used to report STI literacy [1]. Key populations are a vulnerable group who can benefit from STI literacy which is essential for promoting informed and empowered sexual choices.

Objective: To identify factors associated with STI literacy among MSM and transgender men and women in Soweto using machine learning algorithms.

Methodology: A retrospective observational study involving 1240 MSM and transgender men and women from Soweto was conducted using secondary data collected through self-administered questionnaires by Best Health Solutions. The collected data were analysed using supervised machine learning algorithms including XGBoost,

AdaBoost, Naive Bayes, Decision tree, Random Forest, Logistic regression, and K-Nearest Neighbors [2]. Data were split into training (80%) and testing (20%) sets [3]. The outcome variable was based on the question: "Have you ever received counselling or education about STI diagnosis and results interpretation?" This variable was a binary factor with "Yes" and "No" representing STI literacy. We employed repeated 10-fold cross-validation with 100 repetitions to ensure reliable performance estimates [4]. Metrics such as AUC (ROC), accuracy, sensitivity (recall), and precision were used to evaluate the model's classification ability [5]. The final model was assessed based on its accuracy in distinguishing between individuals with and without STI literacy. Variable importance was calculated to highlight the most influential predictors [6].

Preliminary Results: Based on the number of people who responded yes to the question "Have you ever received counselling or education about STI diagnosis and results interpretation", an STI literacy rate of 68.4% was determined among MSM and transgender individuals in Soweto. The random forest model produced the best metrics with a precision of 0.995 (99.5%). The area under the curve (AUC) was 0.988, indicating that the model is very effective in distinguishing between individuals with STI literacy and those without. Overall, the fitted random forest model appeared to be the most effective in identifying the factors associated with STI literacy.

Discussion: This study used advanced machine-learning techniques to establish the STI literacy rate among MSM and transgender men and women in Soweto. Barrier methods used, understanding that STIs can be asymptomatic, confidence in STI knowledge, STI testing in the past, marital status, familiarity with STIs, STI testing in the past, and sexual orientation were found to be important in predicting STI literacy [8]. The random forest model was developed to predict STI literacy based on these factors, enabling informed decision-making and improving sexual health outcomes. The findings may help to address critical knowledge gaps and misconceptions about STIs, offering a strong foundation for developing tailored educational interventions [7]. By addressing these gaps, public health initiatives can become more effective in reducing the spread of STIs and improving overall health outcomes for vulnerable populations.

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P53

Telemental health service utilization among children in foster care in a southeastern state: the roles of child race/ethnicity, urban/rurality, and mental health diagnoses

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Background: Foster children in rural areas face disproportionate barriers to accessing mental health services, including provider shortages, transportation challenges, and other social determinants of health [1]. Despite the widespread adoption of telemental health (TMH), limited data is available on TMH use among foster children in rural areas, making it difficult to address their significant barriers to mental health care. We aimed to (1) demonstrate the overall rates of TMH use among foster children with mental health diagnosis in a southeastern state; and (2) assess the variations in the TMH use among foster children with a primary mental health diagnosis by their race/ethnicity, urban/rurality, and mental health diagnoses.

Methods: We analyzed statewide integrated administrative data, which included foster care and service records from the Department of Social Services, mental health service data from the Department of Mental Health, and Medicaid claims data from the Department of Health and Human Services. We conducted descriptive and bivariate analyses (e.g., chi-square, Fisher's exact test, t-test) to compare the rates of TMH use overall and by child sociodemographic and clinical factors. Multivariable logistic regressions were used to examine whether race/ethnicity, urban/rurality, and mental health diagnoses were associated with their use of TMH. We tested the interaction between the place of residence and race on use of TMH services.

Results: Among 640 children who received mental health services, 264 (41.2%) used in-person service only, 34 (3.4%) used TMH services only, and 342 (34.1%) utilized both. Children diagnosed with depression ($p < 0.01$), conduct disorder ($p < 0.01$), or ADHD ($p < 0.01$) were more likely to use TMH services than those without these conditions. Bivariate analysis reveals that TMH use varies by race ($p = 0.04$) among foster children living in rural areas, showing children of other races were more likely to use TMH services compared to Non-Hispanic (NH) Whites (Table 1). After adjusting for sociodemographic and clinical factors, rural-urban differences in TMH use varied by race/ethnicity ($p = 0.01$). While rural and urban foster children of Black and White identity had similar rates of TMH use, foster children identified as other races – such as Asian, American Indian/Alaskan Native, Hispanics, and Native Hawaiian/Other Pacific Islander, who reside in rural areas had higher odds ($OR = 6.56$; $p = 0.02$) of using TMH services than their peers in urban areas (Table 2).

Conclusions: Tailored strategies might be necessary to engage underrepresented racial/ethnic groups in TMH services, especially in urban underserved communities. Children diagnosed with depression, conduct disorder, or ADHD are more likely to use TMH services than those without these conditions, suggesting the potential of TMH services in meeting the mental health needs of foster children with these conditions.

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Table 1 (Abstract P53). Bivariate analysis of TMH status by children's race in rural areas

Race	Any TMH n (%)	No TMH (in-person) n (%)	Fisher Exact Test p-value
NH Black	16 (57)	12 (43)	0.96
NH White	38 (58)	28 (42)	reference
Other	13 (87)	2 (13)	0.04

Table 2 (Abstract P53). Adjusted Odds Ratios for TMH use among children in foster care for the interaction between race and region

	Odds Ratio	95% CI
NH Black vs Other in rural region	0.17	0.03 0.97
NH Black vs NH White in rural region	1.05	0.40 2.74
Other vs NH White in rural region	6.12	1.21 31.12
Rural vs Urban in Other	6.56	1.33 32.29
Rural vs Urban in NH Black	0.89	0.38 2.10
Rural vs Urban in NH White	0.86	0.47 1.56

P54

An adaptation of the principal component analysis technique for identifying significant predictors of hereditary disease

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Introduction: We propose a modification to the Principal Component Analysis (PCA) [1] which can be effective for identifying significant predictors of hereditary diseases (such as heart disease and diabetes) [2]. Our objective is to develop an efficient method for identifying a small subset of significant predictors from a collection of large, often strongly correlated predictors.

Methods: PCA identifies a set of basis vectors along which there is lot of variation in the values of the predictors. These basis vectors are in fact the eigen vectors of the covariance matrix, the main diagonal of which has the variance measures for each predictor. The off-diagonal terms of the covariance matrix are the covariances between the predictors. The principal axes are often used as significant predictors of the output. However, linear combinations of predictors that have high variances along that axis are not adequate for linear regression, because large variances along an axis do not imply a large covariance with the output. By replacing each main diagonal terms with the covariance between the output and the corresponding predictor, the covariance matrix can be kept symmetric and generate eigenvectors which have a large covariance with the output. A challenge emerges when diagonal terms become negative. We address this by changing the sign of negative diagonal terms and multiplying the corresponding row and column by -1 . In addition, only the predictors (and not the target) are standardized to keep the eigenvalues positive.

Results: After testing this approach on healthcare data sets [3,4,5] and comparing it with a brute force approach (which checks the improvement in the adjusted R^2 for every remaining predictor), we found that

in the cholesterol dataset, ratio and HDL emerged as dominant predictors, with minimal improvement from additional variables. For diabetes prediction, BMI and S5 (log of serum triglycerides) were identified as key initial variables. In blood pressure modeling, Pulse and Weight served as primary predictors. For Parkinson's disease, jitter and harmonics-to-noise ratio (HNR) were significant predictors. The significant predictors were inferred by calculating a weighted sum of the eigenvalues from the eigenvectors in which the variables appeared.

Conclusions and future work: Our modified PCA approach demonstrated promising results across diverse datasets. By effectively identifying significant predictors, we achieved higher initial R^2 values with fewer variables. Though validation on larger datasets and optimization of eigenvalue thresholds remain necessary, the technique showed potential for improving model building efficiency across different domains.

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P55

Disparities in mental health and substance use disorder related postpartum readmissions

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Introduction: Postpartum readmission (PPR) occurs in 1–2.6% of all U.S. birth hospitalizations and represents a significant, often preventable burden—particularly for psychiatric conditions and substance use disorders. Many mothers face barriers to accessing mental health and substance use treatment during the perinatal period, with rural communities disproportionately affected by access-related disparities. Addressing this public health challenge and reducing inequities in PPR is essential to advancing equitable maternal care.

Study Objective: This study aimed to investigate the intersections of residential rurality and race/ethnicity on all-cause, mental health disorders and substance use disorders (SUD) PPR throughout 1-year post-birth.

Methods: A statewide retrospective cohort study using birth certificates linked to all-payer hospital data for mothers giving birth in South Carolina from January 2018 through December 2021. Our main exposures were maternal race/ethnicity and rural-urban residence. To allow for administrative censoring, our main outcome of PPR was defined as an inpatient readmission at 4 time points: within 42, 90, 180 and 365 days after initial birth hospitalization discharge (hereafter, post-discharge) using *International Classification of Disease, Tenth Revision, Clinical Modification (ICD-10-CM)* diagnosis and procedure codes for delivery-related diagnosis and procedures. We report hazard ratios (HR) and 95% confidence intervals (CI) from Cox proportional hazards models, adjusted for prenatal characteristics (e.g., age, education) and clinical factors (e.g., length of stay, mode of birth).

Results: Of 195,311 mothers, 28,617 (14.6%) resided in rural residential areas and 166,694 (85.3%) resided in urban residential areas. Among the overall sample, 0.6% of mothers had mental health and SUD related PPR within 42 days post-discharge, to 2.3% within 365 days post-discharge. Using urban residence as the reference group, the adjusted hazard ratios (HRs) for rural mothers were: all-cause postpartum readmissions (PPR) HR 1.2 (95% CI: 1.1–1.3), mental health-related PPR HR 1.1 (95% CI: 0.9–1.2), and substance use disorder (SUD)-related PPR HR 1.1 (95% CI: 1.0–1.3). Considering the interaction between maternal residence and race/ethnicity, with urban residence and non-Hispanic (NH) White as the reference group, all-cause PPR was lower for NH Black rural mothers (HR 0.9, 95% CI: 0.8–1.0) and Hispanic rural mothers (HR 0.5, 95% CI: 0.3–0.9). For mental health-related PPR, HRs were 0.9 (95% CI: 0.8–1.1) for NH Black rural mothers and 0.5 (95% CI: 0.1–1.9) for Hispanic rural mothers. For SUD-related PPR, HRs were 1.0 (95% CI: 0.8–1.3) for NH Black rural mothers and 0.6 (95% CI: 0.1–4.4) for Hispanic rural mothers.

Discussion: Rural mothers face a slightly higher risk of all-cause postpartum readmissions compared to urban counterparts, with less pronounced differences for mental health and substance use disorder (SUD)-related readmissions. The interaction between maternal residence and race/ethnicity shows that rural non-Hispanic Black and Hispanic mothers have lower risks of all-cause readmissions than urban non-Hispanic White mothers, though patterns for mental health and SUD-related readmissions vary and are less consistent. These findings highlight the complex interplay of geographic and racial/ethnic factors in postpartum health outcomes, emphasizing the need for tailored interventions to address disparities.

P56

Exploring state-level variations in maternal telehealth policies in the United States

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Background: Telehealth – telecommunication via cellular devices, computers, video call, or virtual patient portal – has promises to improve medical care and patient education [1]. The benefits of telehealth are well-documented [2 including increased flexibility, improved scheduling convenience, reduced geographic barriers to specialty care, remote monitoring of vital signs and health information, and support for clinical decision-making by providers. However, telehealth uptake have been limited and varies substantially across geography.⁴

Objective: This study synthesized literature on state-level telehealth policies to identify states with reimbursement policies for each telehealth modality: video, audio-only, asynchronous (or store-and-forward), and remote-patient monitoring.

Methods: This review study aims to synthesize state-level telehealth reimbursement policies in the U.S. from 2013 to 2023, with a focus on Medicaid coverage and reimbursement variations for synchronous (live video or audio-only), asynchronous (store-and-forward), and remote patient monitoring services. A comprehensive literature search were conducted in PubMed, Embase, Web of Science, and CINAHL using relevant search terms related to telehealth, reimbursement policies, Medicaid telehealth coverage, and modality-specific reimbursement (synchronous, video, phone/audio, asynchronous, and remote monitoring). The search was limited to U.S.-based studies published in English between January 1, 2013 and December 31, 2024. To ensure a thorough policy review, we also conducted a gray literature search, including state government reports, Medicaid policy documents, and insurance industry publications.

Results: Since 2013, many states have begun providing reimbursement for at least one of the 4 policies (Fig. 1), the expansion has been

gradual but remained steady since 2020. The most drastic of increases were audio-only telehealth reimbursement, from 0 states in 2013–2023 to 47 states after 2020, the COVID 19 year. The number of states that implemented telehealth reimbursement for remote-patient monitoring increased from 6 states in 2013 to 38 states in 2023. There has been a huge increase in the number of states that have implemented more than 1 policy.

Conclusion: Although state policy changes can promote greater access to healthcare through telehealth, significant disparities may persist due to substantial variations in telehealth regulations across states.

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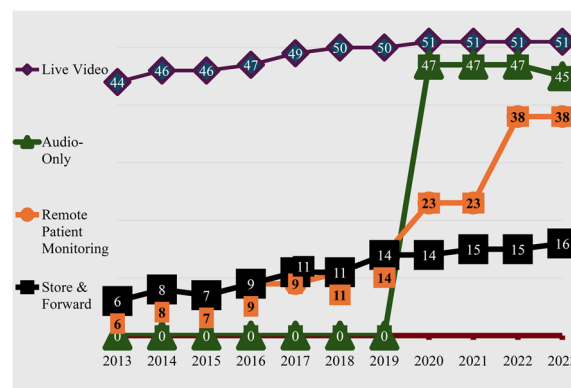


Fig. 1 (Abstract P56). Number of States with Medicaid Reimbursement for Telehealth by Modality, 2013–2023

P57

Daily associations between physical activity and cognitive function in older adults: an ecological momentary assessment study

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Introduction: Maintaining cognitive health is vital for the independence and well-being of older adults. Current research lacks ecological validity and temporal insight, as activity patterns and cognitive function fluctuate daily. This study utilizes smartphone and sensor technology to capture real-world, time-sensitive data, exploring the role of daily activity patterns in predicting cognitive outcomes.

Objectives: This study utilized Ecological Momentary Assessment (EMA) to investigate daily associations between physical activity and cognitive function in older adults over a two-week period. Participants' physical activity was continuously monitored using activity trackers, while self-reported emotions and basic information were collected through standardized forms.

Methods: This study enrolled 102 older adults (mean age = 68.31 years, 35% male) with neuropsychological disease risk factors but without dementia or mild cognitive impairment at enrollment. Inclusion criteria included independent ambulation and either a family history of Alzheimer's disease, subjective cognitive decline or memory complaints, or being overweight or obese (BMI > 25). Participants completed a 14-day protocol between January 2021 and December 2022. Daily surveys recorded bedtime and wake time, followed by cognitive assessments using the Symbol Search Test administered through an NIH-funded smartphone app (M2 C2). Cognitive tests were conducted each morning and at four semi-random times throughout the day. Machine learning models, including Decision Tree Regression and Support Vector Regression (SVR), were developed to predict cognitive outcomes. Predictors included sleep duration and quality, emotional states, activity levels, location, and screen time, with control variables such as age, sex, education, race, marital status, and living alone.

Results: The best-performing approach combined a linear mixed-effects model and Support Vector Regression (SVR). Step 1 used the mixed-effects model to adjust for control variables, generating residuals representing unexplained variance. Step 2 incorporated these residuals and additional predictors into the SVR model. The SVR achieved optimal performance based on MSE and R-squared metrics. Positive coefficients, such as for sleep quality and positive emotions, indicated improved Symbol Search Test scores, while negative coefficients, like stress and screen time, were associated with lower scores. Covariates with p-values < 0.05 demonstrated statistically significant relationships with cognitive performance.

Discussion: This study underscores the importance of demographic factors, including age, sex, education, and marital status, in influencing cognitive function among older adults. Emotional states, sleep duration, and being at home were also strongly associated with daily cognitive performance, highlighting the multifaceted nature of cognitive health. These findings emphasize the need for real-world, time-sensitive data to capture the dynamic interplay of daily factors on cognition. The use of smartphone and sensor technology provides a valuable approach for understanding these associations and offers potential for future applications in monitoring and intervention. Future research should investigate the role of physical activity patterns in cognitive health. Leveraging wearable devices to analyze activity data collected during semi-random intervals could reveal distinct movement patterns. By categorizing these patterns into meaningful variables, researchers can explore how variations in activity levels contribute to cognitive outcomes, paving the way for targeted interventions to support cognitive health in aging populations.

P58

A multilevel temporal analysis disparities in COVID-19 vaccine uptake in children: evidence from Louisville, Kentucky

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Introduction: Children's COVID-19 vaccination can limit COVID-19 transmission [1]. In December 2020, Pfizer and Moderna vaccines received emergency use authorization (EUA) for adults(2). The EUA

expanded to 12–15 and 5–11-year-old children in May and October 2021, respectively [2]. Despite these efforts, Black and White children had a lower-than-average vaccination rate [3].

Study Objectives: We aimed to analyze the temporal COVID-19 vaccination trends in children by combinations of sex, race, and ethnicity. We devised analytical methods to tackle the challenges of using immunization registry data, including missing demographics, discrepancies in reporting demographics across doses, and potential overrepresentation of some demographic groups.

Methods: We used 12/2020-05/2022 Kentucky Immunization Registry (KYIR) data from Louisville to calculate the number of vaccinated individuals by age, sex, race, and ethnicity, and census data to estimate the vaccine-eligible population. We imputed missing demographic information in the KYIR data using the pertinent information reported for previous or subsequent doses. Observing discrepancies in reporting race across doses, we devised an algorithm for assigning individuals a single race across doses. Still, vaccination rates for some Multiracial and Some Other Races were substantially overestimated. We addressed it by devising an algorithm to reallocate the estimated excess number of vaccinated Multiracial and Some Other Races to other races. In addition, we randomly distributed those who reported Unknown Race to specific races proportionate to their respective population share. We kept the COVID-19 vaccine recipients who were under 16 when receiving the first dose. Ultimately, we computed the cumulative COVID-19 vaccination by sex, race, ethnicity, race-sex, and ethnicity-sex.

Results: As of 5/31/2022, the overall COVID-19 doses one and two vaccination rates were 63.9% and 56.1% at 12–15 years and 36.7% and 30.6% at 5–11 years. During this period, 12–15- and 5–11-year-old White and Other Race children had the highest doses one and two vaccination rates, and Black children had the lowest (Fig. 1). Hispanic had also lower rates than non-Hispanic in both age groups (Fig. 1). Considering children of all races and ethnicities across sexes on 5/31/2022, Black boys had the lowest doses one and two vaccination rates of 44.6% and 37.5% at 12–15 years and 22.3% and 16.9% at 5–11 years. In addition, Hispanic boys in both age groups were less vaccinated than Hispanic girls (Table 1, Fig. 1).

Discussion: By 5/31/2022, Black children of 12–15 years and 5–11 years in Louisville had the lowest COVID-19 doses one and two vaccine rates, significantly lower than those for White and Asian residents. In addition, Hispanic children in both age groups had lower vaccination rates than non-Hispanic. This finding strongly aligns with national studies [3], highlighting persistent racial inequalities and socio-economic marginalization. Moreover, COVID-19 doses one and two vaccination rates were lower in males than in females for Whites and Blacks of both age groups. We found the largest female-male difference in doses one and two vaccination rates among White and Other Race children 12–15 years and 5–11 years. However, comparing all race-sex and ethnicity-sex groups, Black boys had the lowest doses of one and two vaccination rates among both age groups.

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Table 1 (Abstract P58). Vaccination rate differences by race and sex on 5/31/2022, the end of the study period, for children in Jefferson County, Kentucky

Dose Number	Race					Ethnicity	
	White	Black	Asian	Other	Multi Racial	Hispanic	Non-Hispanic
1	63.9	70.5	45.9	60.1	71.5	70.1	55.8
2	56.1	62.9	38.8	54.4	55.2	63.3	45.8
1	0.0	+ 6.6	−18.0	−3.8	+ 7.6	+ 6.2	−8.1
2	0.0	+ 6.8	−17.3	−1.7	−0.9	+ 7.2	−10.3
1	65.5	73.1	47.2	59.9	69.7	69.7	55.4
2	57.8	65.6	40.0	54.3	54.9	63.2	45.8
1	−3.1	−5.0	−2.6	+ 0.1	+ 3.6	+ 0.8	−2.5
2	−3.4	−5.2	−2.5	+ 0.9	+ 0.7	+ 0.3	−0.1
1	36.7	43.4	22.4	43.4	50.9	38.4	27.6
2	30.6	35.5	17.2	36.7	37.0	32.9	19.1
1	0.0	+ 6.7	−14.3	+ 6.7	+ 14.2	+ 1.7	−9.1
2	0.0	+ 5.6	−13.4	+ 6.1	+ 6.0	+ 2.9	−11.5
1	36.9	41.4	22.5	43.3	53.2	38.9	28.5
2	30.8	35.6	17.4	35.8	39.4	33.5	19.6
1	−0.4	−0.1	−0.2	+ 0.2	−4.7	−1.1	−1.7
2	−1	−0.1	−0.5	+ 1.8	−4.9	−1.1	−0.9

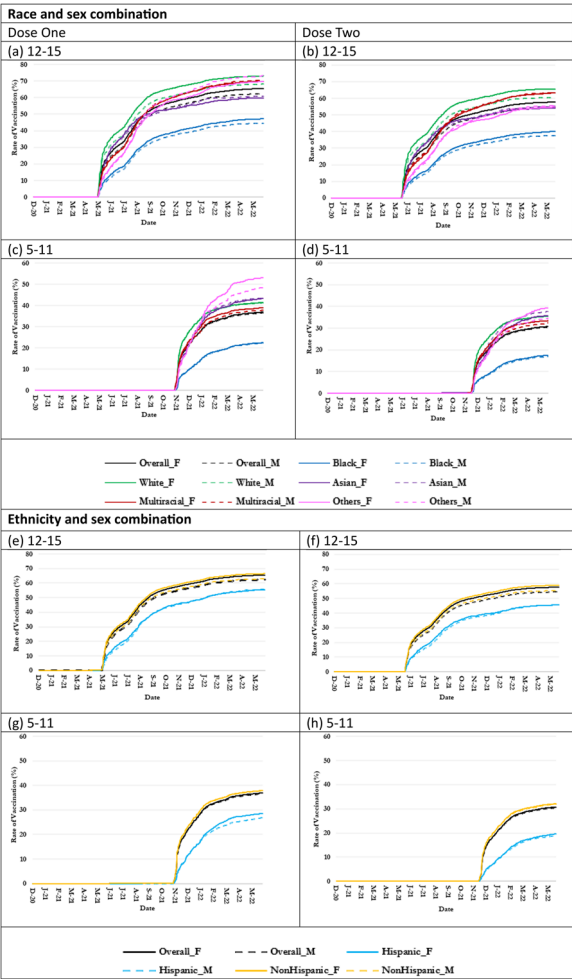


Fig. 1 (Abstract P58). Trends in one and two-dose COVID-19 vaccination rates in 5–15 years old population by race/ethnicity and sex in Jefferson County, Kentucky

P59
Resilience, stressors, and intersectionality: understanding alcohol and drug use disparities among sexual minority populations
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Background: Substance use disorders, particularly alcohol and drug misuse, disproportionately affect sexual minority populations (SMPs), driven by minority stress factors such as discrimination, stigma, and structural inequities [1, 2]. Despite growing awareness, existing research often overlooks the nuanced interplay between resilience and stressors within intersectional identities, including race/ethnicity, gender, sexual orientation, income, and education [3]. This study addresses these gaps using Multilevel Analysis of Individual Heterogeneity and Discriminatory Accuracy (MAIHDA) [4, 5] to explore resilience and stressor profiles with substance use outcomes in SMPs.

Methods: We analyzed cross-sectional data from the Generation Study (N=1518) [6]. Substance use outcomes included alcohol use (Alcohol Use Disorders Identification Test [AUDIT], Cronbach's alpha=0.79) and drug use (Drug Use Disorders Identification Test [DUDIT], Cronbach's alpha=0.86). Resilience factors (i.e., social support, life satisfaction, social well-being, and community connectedness) and stressors (i.e., everyday discrimination, perceived stigma, healthcare stereotype threat, and internalized homophobia) were assessed. Intersectional strata were defined by education (less than high school vs. at least high school), birth sex (male vs. female), sexual orientation (gay, lesbian, bisexual, other SMPs), income (low, medium, high), and race/ethnicity (White, Black, Hispanic). Descriptive statistics, Bayesian hierarchical modeling, and MAIHDA quantified variance due to intersectional strata and evaluated predictor-outcome relationships. Multiple imputations addressed missing data and ensured robust analysis by merging sparse strata.

Results: Alcohol use was positively associated with everyday discrimination and negatively associated with perceived stigma. Drug use was more strongly linked to everyday discrimination and inversely related to social well-being. Resilience factors like social support and community connectedness showed protective effects, varying across intersectional strata. Bayesian modeling identified everyday discrimination as the strongest predictor for alcohol ($\beta=0.10$, 95% CI [0.01, 0.18]) and drug use ($\beta=0.23$, 95% CI [0.15, 0.32]) (Table 1). MAIHDA revealed significant between-strata heterogeneity, with intersectional strata accounting for 1.47% (AUDIT) and 3.86% (DUDIT) of total variance. Including predictors reduced DUDIT strata-level variance by 65%, emphasizing the importance of stressors and resilience factors. Negative associations were observed for perceived stigma, highlighting its complex and varying role across the models (Fig. 1a, 1b).

Conclusions: This study underscores the complex interplay between resilience, stressors, and intersectional identities in substance use behaviors among SMPs. Consistent with minority stress theory [7], findings reveal the adverse impact of stressors and the protective role of resilience factors. MAIHDA enhanced understanding of heterogeneity across strata, providing actionable insights for reducing substance use disparities through targeted, equity-focused interventions [8–10].

Acknowledgements
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Table 1 (Abstract P59). Bayesian Regression Coefficients for Stressor and Resilience Factors Predicting Alcohol Use (AUDIT) and Drug Use (DUDIT) Among Sexual Minority Groups

Predictor	Estimate	Est. Error	95% CI (Lower)	95% CI (Upper)	Rhat	Bulk ESS	Tail ESS
ALCOHOL USE AS THE OUTCOME (AUDIT)							
Connectedness	0.02	0.05	−0.08	0.11	1.0	11,209	6138
Social Well-Being	0.02	0.04	−0.05	0.08	1.0	10,301	6961
Life Satisfaction	0.03	0.02	−0.01	0.07	1.0	10,304	6517
Social Support	−0.0	0.02	−0.05	0.04	1.0	10,639	6198
Everyday Discrimination	0.10	0.04	0.01	0.18	1.0	10,111	6240
Perceived Stigma	−0.06	0.03	−0.12	0.0	1.0	10,760	5672
Healthcare Threat	0.01	0.03	−0.05	0.06	1.0	11,481	5983
Internalized Stigma	−0.04	0.04	−0.11	0.04	1.0	11,831	6024
Connectedness	0.09	0.05	−0.0	0.18	1.0	10,742	6382
Social Well-Being	−0.17	0.04	−0.23	−0.10	1.0	9660	5296
Life Satisfaction	−0.01	0.02	−0.05	0.03	1.0	7872	5683
Social Support	0.01	0.02	−0.03	0.05	1.0	10,568	5696
Everyday Discrimination	0.23	0.04	0.15	0.32	1.0	9700	6046
Perceived Stigma	−0.11	0.03	−0.17	−0.05	1.0	11,429	5953
Healthcare Threat	−0.05	0.03	−0.10	−0.00	1.0	11,437	6222
Internalized Stigma	0.03	0.04	−0.03	0.10	1.0	12,347	5923

Notes:

Estimate: Posterior mean of the regression coefficient, indicating the central tendency of the posterior distribution

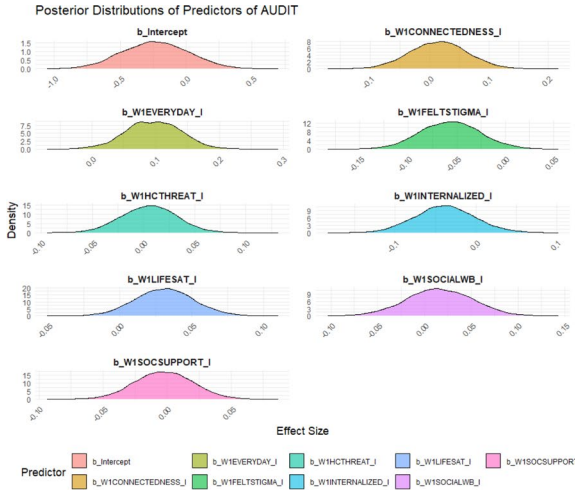
Est. Error: Standard deviation of the posterior distribution, reflecting uncertainty around the estimate

Credible Interval (I-95% CI, u-95% CI): 95% credible interval showing the range within which the parameter likely falls with 95% probability. Indicates statistical or practical relevance (whether it includes 0)

Rhat: Convergence diagnostic; values near 1 indicate good convergence

Bulk ESS & Tail ESS: Effective sample sizes assessing posterior sample quality. Bulk ESS reflects mixing in the bulk, while Tail ESS assesses reliability in the tails

a. Posterior Distribution of Predictors of AUDIT



b. Posterior Distribution of Predictors of DUDIT

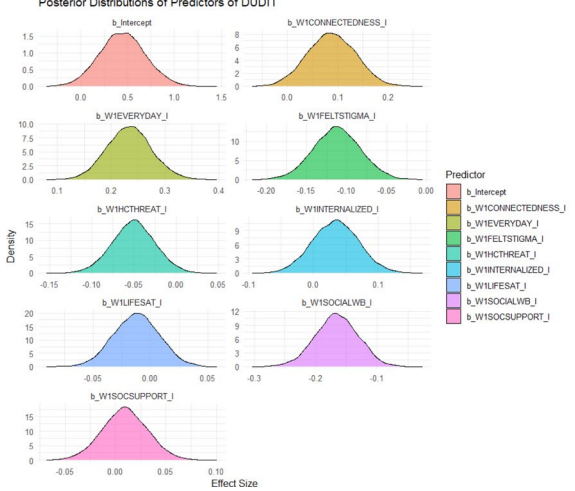


Fig. 1 (Abstract P59). a. Posterior Distribution of Predictors of AUDIT. b. Posterior Distribution of Predictors of DUDIT. Notes for Fig. a, b: key predictors include resilience factors (social support [W1SOCIETY_I], life satisfaction [W1LIFESAT_I], social well-being [W1SOCIALWB_I], and community connectedness [W1 CONNECTEDNESS_I]) and stressors (everyday discrimination [W1EVERYDAY_I], perceived stigma [W1 FELTSTIGMA_I], healthcare stereotype threat [W1HCTHREAT_I], and internalized homophobia [W1INTERNALIZED_I])

P60

Wearable devices as tools for health interactions: a social media analysis of public attitudes and challenges

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Study Objectives: The rapid advancement of wearable technologies that track users' activity, fitness, and physical levels has made health-related interactions popular and easily accessible in our daily lives. Meanwhile, users' attitudes toward using the product have largely been overlooked, despite having significant mediating effects [1]. As wearable devices increasingly collect and analyze different types of personal data, it has become more important to study how users perceive the information generated by these products to understand how they benefit people's health. Using sentiment analysis, this study aims to examine public attitudes and perceived challenges toward data tracking and the outputs of these products.

Methods: This study adopted content analysis on different social media platforms. Using the social media analysis software Quid Monitor, a two-year span of data from December 6, 2022, to December 6, 2024, was extracted from popular domains, including Reddit, X, YouTube, Bluesky, and Facebook. To avoid biases from commercial posts that generally contain positive sentiments and highlight functionalities, this study targeted analyzing replies and comments on these social media. Search terms included "Wearable," "Smartwatch," "Fitness tracker," and top smartwatch brand names interconnected with perception keywords such as "Trust," "Accuracy," "Correct," "Valid," "Reliable," and "Right."

Results: A total of 207,779 mentions were extracted from the social media platforms. The sentiment analysis showed that 44.3% of the mentions were positive (104,057 mentions), 25% were negative (58,779 mentions), and 30.7% were neutral (72,153 mentions). The top indexed terms included "Right" with 48,106 mentions, "Accurate" with 41,967 mentions, "Accuracy" with 11,557 mentions, and "More accurate" with 9,877 mentions. These terms occurred at the top in all three sentiment types. Fig. 1 shows the overview of users' sentiment toward smartwatches.

Discussion: Wearable devices play a significant role in fitness tracking and health monitoring, as users rely on them heavily for daily health insights. And users appreciate features like heart rate monitoring, GPS tracking, and integration with fitness apps. Despite that, fitness tracking devices are widely discussed for their accuracy and functionality. The technology still faces several limitations in accurately tracking those common health metrics. The functional limitations have appeared as one of the most significant predictors that negatively impact the continuous usage of smartwatches [2]. Thus, users are advised to use these devices as trend indicators for their health monitoring rather than absolute measures or to consider professional medical advice for health-related concerns. These challenges highlight the need for continuous improvement in wearable technology to enhance accuracy and reliability.

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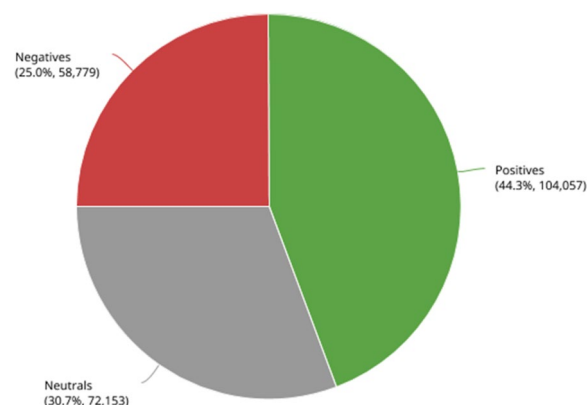


Fig. 1 (Abstract P60). Sentiment Overview of Users' Smartwatch Attitude

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